

APPENDIX A

THE REAL NUMBERS

In this Appendix we will show how the rational and the real numbers can be represented faithfully in set theory as the natural numbers are; that is, we will identify some characteristic, set theoretic properties of these systems and we will prove from the axioms of **ZDC** the existence and uniqueness (up to isomorphism) of structured sets with these properties. The proofs are quite simple as far as set theory goes, but they use ideas from algebra and analysis, which we will present in outline.

There are two, standard representations of the real numbers in set theory, Cantor's *equivalence classes of Cauchy sequences* and Dedekind's *Dedekind cuts*. Here we will employ a somewhat novel construction which interweaves between these two and (incidentally) establishes their equivalence. The basic notion that we need is that of a quotient of a set A by an equivalence relation on A , described in Problem **x4.5**.

Recall that a **determining surjection** for an equivalence relation $\sim$ on a set A is any surjection

$$\pi : A \twoheadrightarrow B$$

such that for all $x, y \in A$,

$$x \sim y \iff \pi(x) = \pi(y).$$

When this holds, we call B a **quotient of A by $\sim$** . The *canonical surjection* of $\sim$ is the mapping

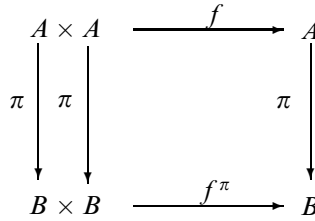
$$x \mapsto [x/\sim] \quad (x \in A)$$

with quotient the set of equivalence classes $\llbracket A/\sim \rrbracket$, but there may be others, more illuminating of the situation, e.g., those described in Problems **x4.6**, **x4.7** and **x4.8**. Determining surjections are especially useful in the study of congruence relations.

A.1. Definition. Suppose $\sim$ is an equivalence relation on A and

$$f : A \times A \rightarrow A$$

The material in this Appendix can be read after Chapter **10**. It assumes some theoretical knowledge of the Calculus, and it is not a prerequisite for understanding the remainder of these Notes.

FIGURE A.1. Congruence for f .

is a binary function. We call $\sim$ a **congruence for f** if for all $x, x', y, y' \in A$,

$$x \sim x' \ \& \ y \sim y' \implies f(x, y) \sim f(x', y').$$

Similarly, $\sim$ is a **congruence for a binary relation $P \subseteq A \times A$** , if for all x, x', y, y' ,

$$x \sim x' \ \& \ y \sim y' \implies [x P y \iff x' P y'].$$

We can obviously define the notion of congruence for functions and relations of any number of arguments, in the same way.

The next theorem deals with one of the simplest and most basic algebraic constructions.

A.2. Theorem. *Let $\pi : A \twoheadrightarrow B$ be a determining surjection of some equivalence relation $\sim$, so that for all x, y in A , $x \sim y \iff \pi(x) = \pi(y)$.*

(1) *If $\sim$ is a congruence for a function $f : A \times A \rightarrow A$, then there exists exactly one function $f^\pi : B \times B \rightarrow B$ on the quotient B such that the diagram in Figure A.1 commutes, i.e.,*

$$f^\pi(\pi(x), \pi(y)) = \pi(f(x, y)) \quad (x, y \in A). \quad (\text{A-1})$$

(2) *If $\sim$ is a congruence for a relation $P \subseteq A \times A$, then there exists exactly one relation $P^\pi \subseteq B \times B$ on the quotient B such that*

$$\pi(x) P^\pi \pi(y) \iff x P y \quad (x, y \in A).$$

PROOF. The form of (A-1) makes it clear that at most one function can satisfy it, so it is enough to show that at least one function does. Put

$$f^\pi =_{\text{df}} \{((\pi(x), \pi(y)), \pi(z)) \mid x, y, z \in A \ \& \ f(x, y) = z\}.$$

To verify that the set of pairs f^π is a function, we must check that

$$((u, v), w), ((u, v), w') \in f^\pi \implies w = w'. \quad (\text{A-2})$$

From the hypothesis of (A-2) and the definition of f^π , there exist $x, y, z \in A$ such that

$$u = \pi(x), v = \pi(y), \quad w = \pi(z), \quad f(x, y) = z$$

and also $x', y', z' \in A$, so that

$$u = \pi(x'), v = \pi(y'), \quad w' = \pi(z'), \quad f(x', y') = z'.$$

It follows that

$$\pi(x) = \pi(x'), \pi(y) = \pi(y'), \pi(f(x, y)) = \pi(f(x', y'))$$

since $\sim$ is determined by π and it is a congruence for f , and the last of these equalities yields the desired $w = w'$.

The characteristic property (A-1) of f^π follows immediately from its definition and the proof of Part (2) is similar. $\dashv$

A.3. Exercise. Prove Part (2) of A.2.

The axiomatic characterizations of the rationals and the reals are based on the notion of an *ordered field*, which codifies the basic properties of addition, multiplication and ordering in these number systems.

A.4. Definition. A **field** is a structured set $(F, 0, 1, +, \cdot)$ of objects with the following properties.

(F1) $0, 1 \in F$, $0 \neq 1$, and $+, \cdot$ are binary functions on F .

(F2) The addition function $+$ satisfies the identities

1. $(x + y) + z = x + (y + z)$,
2. $x + y = y + x$,
3. $x + 0 = x$,

and for every x , there exists some x' , such that $x + x' = 0$.

(F3) The multiplication function $\cdot$ satisfies the identities

1. $(x \cdot y) \cdot z = x \cdot (y \cdot z)$,
2. $x \cdot y = y \cdot x$,
3. $x \cdot 1 = x$,

and for every $x \neq 0$, there exists some x'' , such that $x \cdot x'' = 1$.

(F4) Addition and multiplication together satisfy the identity

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

A.5. Lemma. Every field F has the following properties:

(1) For each x there exists exactly one x' such that $x + x' = 0$, and we denote it $-x$; for every $x \neq 0$ there exists exactly one x'' such that $x \cdot x'' = 1$ and we denote it by x^{-1} .

(2) $x \cdot 0 = 0$.

(3) $x \cdot y = 0 \implies x = 0 \vee y = 0$.

(4) $(-x) \cdot y = -(x \cdot y)$.

PROOF. (1) If $x + x' = 0$ and $x + y = 0$, then from the axioms

$$\begin{aligned} y &= y + 0 = 0 + y = (x + x') + y = x + (x' + y) \\ &= x + (y + x') = (x + y) + x' \\ &= 0 + x' = x' + 0 = x'. \end{aligned}$$

The proof about x^{-1} is similar.

(2) $x \cdot 0 = x \cdot (0 + 0) = x \cdot 0 + x \cdot 0$, and therefore

$$\begin{aligned} 0 &= x \cdot 0 + -(x \cdot 0) = (x \cdot 0 + x \cdot 0) + -(x \cdot 0) \\ &= x \cdot 0 + ((x \cdot 0) + -(x \cdot 0)) = x \cdot 0 + 0 = x \cdot 0. \end{aligned}$$

(3) If $y \neq 0$, then some y^{-1} exists such that $y \cdot y^{-1} = 1$, so that

$$x = x \cdot 1 = x \cdot (y \cdot y^{-1}) = (x \cdot y) \cdot y^{-1} = 0 \cdot y^{-1} = y^{-1} \cdot 0 = 0.$$

(4) $x \cdot y + (-x) \cdot y = y \cdot x + y \cdot (-x) = y \cdot (x + (-x)) = y \cdot 0 = 0$, and

(1) implies that $(-x) \cdot y = -(x \cdot y)$. +

We gave this proof in full as an example, justifying each step from the field axioms. In the future we will cut corners, skip details or (more often) use identities which obviously hold in every field without proof or explicit mention.

A.6. Exercise. Every field F satisfies the identity

$$(x + y)^2 = x^2 + 2xy + y^2,$$

where $2 = 1 + 1$. (Give the proof in full detail.)

A.7. Exercise. The doubleton $\{0, 1\}$ of the first two natural numbers is a field, with the obvious operations, and in this field $1 + 1 = 0$. It follows that the field axioms do not imply $1 + 1 \neq 0$ and we must be a bit careful!

A.8. Definition. An **ordered field** is a structured set

$$(F, 0, 1, +, \cdot, \leq)$$

where $(F, 0, 1, +, \cdot)$ is a field, the binary relation $\leq$ is a linear ordering of F and the following conditions hold for all $x, y, z \in F$:

$$\begin{aligned} x \leq y &\implies x + z \leq y + z, \\ z > 0 \ \& \ x \leq y &\implies z \cdot x \leq z \cdot y, \end{aligned}$$

where $z > 0$ naturally abbreviates $0 \leq z \ \& \ z \neq 0$.

A.9. Exercise. In every ordered field,

$$z > 0 \ \& \ x < y \implies z \cdot x < z \cdot y.$$

A.10. Lemma. Every element x in an ordered field F satisfies the inequality $x \cdot x = x^2 \geq 0$, so that $0 < 1$ and for all x , $x > 0 \implies x + 1 > 0$.

PROOF. If $x = 0$, then $x^2 = 0 \geq 0$, and if $x > 0$, then $x \cdot x \geq x \cdot 0 = 0$, so that the only interesting case is when $x < 0$. Adding $-x$ to both sides of this inequality, we get $0 < -x$, so that we can multiply $x < 0$ by $-x$ and we get $(-x) \cdot x < (-x) \cdot 0$, i.e., $-(x^2) < 0$ from the preceding Lemma, and adding x^2 to this inequality we get $0 < x^2$. The conclusion $0 < 1$ follows because $0 \neq 1$ and $1 = 1^2$, and the last claim holds because $0 < x \implies 1 < x + 1$, so that $0 < x + 1$ by the transitivity of $\leq$. +

The Lemma makes it clear that we will not find in ordered fields the anomaly $1 + 1 = 0$ of Exercise A.7. Something much stronger is true.

A.11. Lemma. *Suppose F is an ordered field and set*

$$\mathbb{N}_F = \bigcap \{X \subseteq F \mid 0 \in X \ \& \ (\forall x)[x \in X \implies x + 1 \in X]\};$$

it follows that $(\mathbb{N}_F, 0, (x \mapsto x + 1))$ is a Peano system.

PROOF. By $(x \mapsto x + 1)$ we mean the function S which associates with each $x \in \mathbb{N}_F$ the element $x + 1$ of F , which is also a member of $\mathbb{N}_F$ by the definition. The first three axioms of Peano are obvious and the fourth ($x + 1 \neq 0$) holds because by the definition,

$$\mathbb{N}_F \subseteq \{x \in F \mid 0 \leq x\},$$

and by the Lemma, $x \geq 0 \implies x + 1 \geq 1 > 0$. The Induction Axiom follows immediately from the definition of $\mathbb{N}_F$ as an intersection. $\dashv$

A.12. Exercise. *Suppose F is an ordered field, $\mathbb{N} = \mathbb{N}_F$ is the set of its natural numbers and $+\mathbb{N}$, $\cdot\mathbb{N}$, $\leq\mathbb{N}$ the addition, multiplication and the wellordering of $\mathbb{N}_F$ as these are defined in Chapter 5. Prove that these functions and the relation $\leq$ coincide with the respective objects in F , e.g.,*

$$(\forall x, y \in \mathbb{N})[x +_{\mathbb{N}} y = x + y].$$

The members of $\mathbb{N}_F$ are the **natural numbers of F** . The (rational) **integers** of F comprise the natural numbers and their negations,

$$\mathbb{Z}_F = \mathbb{N}_F \cup \{-x \mid x \in \mathbb{N}_F\},$$

and it is easy to check that they are exactly all differences $u - v$, with $u, v \in \mathbb{N}_F$.

The basic idea for the axiomatic characterization of the rationals is that they are an ordered field, and that every fraction is a quotient of integers, i.e.,

$$\frac{u - v}{m} = m^{-1} \cdot (u - v),$$

where m, u and v are natural numbers and $m \neq 0$. This simple observation yields not only the axioms for the rationals, but also proofs of their existence and uniqueness.

A.13. Definition. A **system of rational numbers** is any ordered field F which satisfies the condition

$$(\forall x \in F)(\exists m, u, v \in \mathbb{N}_F)[m \neq 0 \ \& \ x = m^{-1} \cdot (u - v)].$$

A.14. Theorem (Uniqueness of the rational numbers). *For any two systems of rational numbers F_1, F_2 there exists exactly one bijection*

$$\pi : F_1 \xrightarrow{\sim} F_2$$

which is an isomorphism, i.e.,

1. $\pi(0_1) = 0_2, \pi(1_1) = 1_2$.
2. $\pi(x +_1 y) = \pi(x) +_2 \pi(y), \pi(x \cdot_1 y) = \pi(x) \cdot_2 \pi(y)$.
3. $x \leq_1 y \iff \pi(x) \leq_2 \pi(y)$.

In stating this theorem we decorated the various objects with the superscripts 1 or 2 to clarify the field to which they belong, e.g., $+_1$ is addition in F_1 and 0_2 is the zero element of F_2 . This is awkward and unnecessary, because it is always obvious which superscript is needed: e.g., the identity $\pi(0) = 0$ cannot mean anything else but $\pi(0_1) = 0_2$, since π is a function with domain F_1 and image F_2 . In the proof and in the future we will follow the general algebraic practice by which all the zero elements are 0, all additions are $+$, etc. We will also begin to skip the $\cdot$ of multiplication,

$$xy =_{\text{df}} x \cdot y.$$

PROOF. By the uniqueness of the natural numbers and **A.11**, we know that there exists a “canonical” isomorphism

$$\rho : \mathbb{N}_1 \xrightarrow{\sim} \mathbb{N}_2,$$

where $\mathbb{N}_1$ and $\mathbb{N}_2$ are the sets of natural numbers in F_1 and F_2 , respectively. We set

$$\pi = \{(m^{-1}(u - v), \rho(m)^{-1}(\rho(u) - \rho(v))) \mid m, u, v \in \mathbb{N}_1, m \neq 0\},$$

so that $\pi \subseteq F_1 \times F_2$ and it is enough to show first that π is a function, then that it is a bijection, and finally that it is an isomorphism, as we defined this in the formulation of the theorem.

To verify first that π is a function, we must show that if

$$m_1^{-1}(u_1 - v_1) = m_2^{-1}(u_2 - v_2), \quad (\text{A-3})$$

then

$$\rho(m_1)^{-1}(\rho(u_1) - \rho(v_1)) = \rho(m_2)^{-1}(\rho(u_2) - \rho(v_2)). \quad (\text{A-4})$$

The field axioms imply easily that (A-3) and (A-4) are respectively equivalent to

$$\begin{aligned} m_2 u_1 + m_1 v_2 &= m_1 u_2 + m_2 v_1, \\ \rho(m_2)\rho(u_1) + \rho(m_1)\rho(v_2) &= \rho(m_1)\rho(u_2) + \rho(m_2)\rho(v_1), \end{aligned}$$

and the first of these yields immediately

$$\rho(m_2 u_1 + m_1 v_2) = \rho(m_1 u_2 + m_2 v_1)$$

which in turn implies the second, because ρ is an isomorphism of $\mathbb{N}_1$ with $\mathbb{N}_2$ and it respects addition and multiplication by **Problem x5.4**.

The same simple method can be used directly to prove the additional conclusions, that π is one-to-one and finally an isomorphism. $\dashv$

A.15. Exercise. *Work out in detail the proofs of*

$$\begin{aligned} \pi(x + y) &= \pi(x) + \pi(y), \\ x \leq y &\iff \pi(x) \leq \pi(y). \end{aligned}$$

A.16. Theorem (Existence of the rational numbers). *There exists a system of rational numbers.*

PROOF. If we have the rationals, we can define the set

$$A = \{(m, u, v) \mid m, u, v \in \mathbb{N} \text{ \& } m \neq 0\}$$

of triples of natural numbers, and on it the relation

$$(m, u, v) \sim (m', u', v') \iff_{\text{df}} m'u + mv' = mu' + m'v,$$

which (quite obviously) satisfies

$$(m, u, v) \sim (m', u', v') \iff \frac{u - v}{m} = \frac{u' - v'}{m'}.$$

This means that $\sim$ is an equivalence relation determined by the surjection

$$\pi : A \twoheadrightarrow \mathbb{Q}, \quad \pi(m, u, v) = \frac{u - v}{m}. \quad (\text{A-5})$$

We do not have the rationals yet, but we have A and $\sim$: the idea for the proof is to define the rationals as a quotient of A by $\sim$ so that (A-5) holds. First we must show that

(1) $\sim$ is an equivalence relation. As an example, we verify that $\sim$ is transitive. From its definition, if

$$(m_1, u_1, v_1) \sim (m_2, u_2, v_2) \text{ \& } (m_2, u_2, v_2) \sim (m_3, u_3, v_3),$$

then the identities

$$m_2u_1 + m_1v_2 = m_1u_2 + m_2v_1,$$

$$m_3u_2 + m_2v_3 = m_2u_3 + m_3v_2$$

hold in the natural numbers, and if we multiply the first of these by m_3 and the second by m_1 and then add them, we get

$$\begin{aligned} m_3m_2u_1 + m_3m_1v_2 + m_1m_3u_2 + m_1m_2v_3 \\ = m_3m_1u_2 + m_3m_2v_1 + m_1m_2u_3 + m_1m_3v_2. \end{aligned}$$

Subtract now $m_3m_1v_2$ and $m_1m_3u_2$ from the two sides and divide by m_2 , which gives

$$m_3u_1 + m_1v_3 = m_1u_3 + m_3v_1,$$

i.e., $(m_1, u_1, v_1) \sim (m_3, u_3, v_3)$. Reflexivity and symmetry are proved in the same way.

(2) *Definition of the rationals.* Since $\sim$ is an equivalence relation, there exists a surjection

$$\pi : A \twoheadrightarrow \mathbb{Q}$$

onto some set $\mathbb{Q}$ which determines it, so that

$$(m_1, u_1, v_1) \sim (m_2, u_2, v_2) \iff \pi(m_1, u_1, v_1) = \pi(m_2, u_2, v_2).$$

This $\mathbb{Q}$ is the set of rationals in the system under construction, and it remains only to specify 0 and 1, to define addition, multiplication and the ordering, and finally to prove that the axioms for the rationals hold. To help follow the argument we will start right away using the notation

$$\frac{u - v}{m} =_{\text{df}} \pi(m, u, v)$$

as an abbreviation, i.e., without defining separately “subtraction” or “division”.

The zero and the one are defined in the obvious way,

$$0 = \frac{0-0}{1} =_{\text{df}} \pi(1, 0, 0), \quad 1 = \frac{1-0}{1} =_{\text{df}} \pi(1, 1, 0).$$

(3) *Addition of rationals.* With the representation of rationals as quotients of a difference of numbers by a number which we are using, the classical formula for addition of fractions takes the form

$$\frac{u_1 - v_1}{m_1} + \frac{u_2 - v_2}{m_2} = \frac{(m_2 u_1 + m_1 u_2) - (m_2 v_1 + m_1 v_2)}{m_1 m_2}.$$

So we define first on the set A the binary function f_+ which corresponds to this formula,

$$f_+((m_1, u_1, v_1), (m_2, u_2, v_2)) = (m_1 m_2, (m_2 u_1 + m_1 u_2), (m_2 v_1 + m_1 v_2)).$$

With a bit of arithmetic we can prove that for all $x, y, x', y' \in A$,

$$x \sim x' \ \& \ y \sim y' \implies f_+(x, y) \sim f_+(x', y'),$$

i.e., $\sim$ is a congruence for f_+ . It follows by **A.2** that there exists a (unique) function

$$+ : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$$

which satisfies the identity

$$\pi(x) + \pi(y) = \pi(f_+(x, y)) \quad (\pi(x), \pi(y) \in \mathbb{Q}).$$

Verification of the axioms (F2) for addition needs a bit more of arithmetic, but at least the condition for 0 is obvious:

$$\pi(m, u, v) + \pi(1, 0, 0) = \pi(m \cdot 1, 1 \cdot u + m \cdot 0, 1 \cdot v + m \cdot 0) = \pi(m, u, v).$$

(4) *Multiplication of rationals.* Following the same method, we define first the function $f \cdot : A \times A \rightarrow A$ which corresponds to multiplication when we represent rationals by triples of natural numbers,

$$f \cdot ((m_1, u_1, v_1), (m_2, u_2, v_2)) =_{\text{df}} (m_1 m_2, u_1 u_2 + v_1 v_2, u_1 v_2 + u_2 v_1),$$

we verify next that $\sim$ is a congruence for $f \cdot$ and we define the multiplication operation on fractions $\cdot$ by **A.2** so that it satisfies the identity

$$\pi(x) \cdot \pi(y) = \pi(f \cdot (x, y)) \quad (\pi(x), \pi(y) \in \mathbb{Q}).$$

Verification of axioms (F3) and (F4) requires just a few computations.

(5) *Ordering of the rationals.* The critical equivalence in this case is

$$\frac{u_1 - v_1}{m_1} \leq \frac{u_2 - v_2}{m_2} \iff m_1 v_2 + m_2 u_1 \leq m_2 v_1 + m_1 u_2.$$

We first define the relation $P \subseteq A \times A$ by

$$(m_1, u_1, v_1) P (m_2, u_2, v_2) \iff_{\text{df}} m_1 v_2 + m_2 u_1 \leq m_2 v_1 + m_1 u_2,$$

we verify that $\sim$ is a congruence for P and using **A.2** we define $\leq$ on the quotient $\mathbb{Q}$ so that

$$\pi(x) \leq \pi(y) \iff x P y \quad (\pi(x), \pi(y) \in \mathbb{Q}).$$

It follows easily that $\leq$ is a linear ordering, and that the structured set

$$(\mathbb{Q}, 0, 1, +, \cdot, \leq)$$

is an ordered field.

It remains to verify that $\mathbb{Q}$ is a system of rational numbers.

Lemma 1. *For each natural number k , $\pi(1, k, 0) \in \mathbb{N}_{\mathbb{Q}}$, i.e., the rational $\pi(1, k, 0)$ belongs to the set of natural numbers of the ordered field $\mathbb{Q}$.*

Proof. By induction on k , $\pi(1, 0, 0) = 0$ (by definition) and (easily) by the definition of rational addition

$$\pi(1, Sk, 0) = \pi(1, k, 0) + 1,$$

so that $\pi(1, k, 0) \in \mathbb{N}_{\mathbb{Q}} \implies \pi(1, Sk, 0) \in \mathbb{N}_{\mathbb{Q}}$. + (Lemma 1)

Lemma 2. *For all $(m, u, v) \in A$,*

$$\pi(m, u, v) = \pi(1, m, 0)^{-1}(\pi(1, u, 0) - \pi(1, v, 0)), \quad (\text{A-6})$$

where $^{-1}$ and $-$ are the multiplicative and additive inverse (partial) functions of the field $\mathbb{Q}$.

Proof. Having proved already that $\mathbb{Q}$ is a field, we know that (A-6) is equivalent to

$$\pi(1, m, 0)\pi(m, u, v) + \pi(1, v, 0) = \pi(1, u, 0),$$

and the latter identity is easy to verify with a direct computation. + (Lemma 2)

The two Lemmas together show that the structured set $(\mathbb{Q}, 0, 1, +, \cdot, \leq)$ satisfies the characteristic property of the rationals and this completes the proof. +

As we did with the natural numbers, we now fix a specific system of rational numbers $(\mathbb{Q}, 0, 1, +, \cdot, \leq)$ whose elements we will henceforth call **rational numbers**. This is convenient, it helps avoid awkward expressions like “members of any system of rational numbers” and the like. However, it is important to emphasize (once more) that the significant mathematical fact is the existence and uniqueness up to isomorphism of one such system: it was precisely the corresponding mathematical facts about the natural numbers that we have used in the proofs of this Appendix, not the specific identity of “the natural numbers”.

A.17. Exercise. *The set $\mathbb{Q}$ of rationals is countable.*

A.18. Exercise. *In the proof of Part (1) of the theorem we “subtracted” the same number from an identity and then “divided” an identity by the same number. Justify these steps by verifying the following two properties of the natural*

numbers:

$$\begin{aligned}x + y &= x + z \implies y = z, \\c \neq 0 \&c \cdot x = c \cdot y \implies x = y.\end{aligned}$$

A.19. Exercise. For every ordered field F , there exists exactly one **imbedding** of the rationals in F , i.e., an injection

$$\pi : \mathbb{Q} \hookrightarrow F$$

which satisfies the identities

$$\begin{aligned}\pi(0) &= 0, \quad \pi(1) = 1, \\ \pi(x + y) &= \pi(x) + \pi(y), \quad \pi(xy) = \pi(x)\pi(y), \\ x \leq y &\iff \pi(x) \leq \pi(y).\end{aligned}$$

It follows that the image $\pi[\mathbb{Q}] \subseteq F$ of π is a system of rational numbers (with the 0 and 1 of F and the restrictions of the operations and the ordering of F).

There is a beautiful theorem of Cantor which characterizes the ordering of the rationals independently of their algebraic structure. For it, we need first some definitions.

A.20. Definition. Suppose $\leq$ is a linear ordering on a set A and $B \subseteq A$. We call B **dense in A** if

$$(\forall x, y \in A)[x < y \implies (\exists b \in B)[x < b \&b < y]].$$

A linear ordering $\leq$ is **dense in itself** if its field (A) is dense in A .

A.21. Exercise. The ordering of every ordered field is dense in itself and has no minimum or maximum element.

A.22. Theorem (Cantor). Any two countable, linear, dense in themselves orderings $(A, \leq_A)$ and $(B, \leq_B)$ without minimum or maximum element are **similar**, i.e., there exists an order-preserving correspondence $f : A \hookrightarrow B$.

In particular, every countable, linear, dense in itself ordering is similar with the ordering $\leq_{\mathbb{Q}}$ of the rational numbers.

PROOF. The hypothesis gives us enumerations without repetitions

$$A = \{a_0, a_1, \dots\}, \quad B = \{b_0, b_1, \dots\}$$

of A and B . We will define by recursion a sequence

$$f_0, f_1, \dots,$$

with the following properties, for every $n \in \mathbb{N}$.

(1) f_n is a *finite, partial function* from A to B , i.e.,

$$\text{Function}(f_n) \& f_n \subseteq A \times B,$$

and f_n is finite as a set of ordered pairs.

(2) f_n is monotone and one-to-one on its domain, i.e.,

$$x, y \in \text{Domain}(f_n) \& x <_A y \implies f_n(x) <_B f_n(y).$$

(3) $f_n \subseteq f_{n+1}$.

$$(4) \{a_0, a_1, \dots, a_n\} \subseteq \text{Domain}(f_n).$$

$$(5) \{b_0, b_1, \dots, b_n\} \subseteq \text{Image}(f_n).$$

If we can succeed in this, then the union $f =_{\text{df}} \bigcup_{n=0}^{n=\infty} f_n$ is (easily) a function by (3), it is one-to-one and monotone by (2) and

$$\text{Domain}(f) = A, \quad \text{Image}(f) = B$$

by (4) and (5).

At the basis of the recursive definition we start with

$$p_0 = \{(a_0, b_0)\},$$

so that all the conditions of the result hold trivially.

Suppose now that we have already defined f_n and enumerate its finite domain of definition and image in increasing order:

$$\text{Domain}(f_n) = D_n = \{x_0 <_A x_1 < \dots <_A x_m\},$$

$$\text{Image}(f_n) = I_n = \{y_0 <_B y_1 <_B \dots <_B y_m\}.$$

Since f_n is monotone, we have

$$f_n(x_i) = y_i \quad (i = 0, \dots, m).$$

We construct the next f_{n+1} in two steps, i.e., first we will define a partial function $f'_{n+1} \supseteq f_n$ which satisfies (1) – (4) and then define $f_{n+1} \supseteq f'_{n+1}$ which satisfies all (1) – (5).

STEP 1. If $a_{n+1} \in \text{Domain}(f_n)$, set $f'_{n+1} = f_n$. Otherwise there are three cases.

CASE 1a. $a_{n+1} <_A x_0$. In this case we find some $y' \in B$ satisfying $y' <_B y_0$ (which exists because B has no minimum) and set

$$f'_{n+1} = f_n \cup \{(a_{n+1}, y')\}.$$

CASE 1b. $a_{n+1} >_A x_m$. In this case we find some $y' \in B$ satisfying $y' >_B y_m$ (which exists because B has no maximum) and we set

$$f'_{n+1} = f_n \cup \{(a_{n+1}, y')\}.$$

CASE 1c. For some i , $x_i <_A a_{n+1} <_A x_{i+1}$. In this case we find some $y' \in B$ satisfying $y_i <_B y' <_B y_{i+1}$ (which exists because B is dense in itself) and we set

$$f'_{n+1} = f_n \cup \{(a_{n+1}, y')\}.$$

In all cases, the proof that f'_{n+1} satisfies (1) – (4) is simple.

In STEP 2 of the construction we consider the element b_{n+1} of B and we distinguish again cases: first if $b_{n+1} \in \text{Image}(f'_{n+1})$ (in which case we set $f_{n+1} = f'_{n+1}$) and if not, then three cases again, in symmetry with STEP 1. We skip the details. $\dashv$

The fundamental intuition about the real numbers is that on the one hand they are an ordered field, so that their arithmetic and ordering satisfy the same laws as the rationals, and on the other, they are in one-to-one correspondence with the points of the “complete” geometric line so that there are

no “gaps” between them. In formulating the property of completeness we follow Dedekind.

A.23. Definition. A linear ordering $\leq$ on a set A is **complete** if every non-empty subset of A which has an upper bound has a least upper bound. A **system of real numbers** is any **complete ordered field**, i.e., any ordered field in which the ordering is complete.

A.24. Exercise. *The ordering of the rationals is not complete, because the set*

$$X = \{r \mid r^2 < 2\}$$

is bounded from above but has no least upper bound.

A.25. Lemma. *Every complete, ordered field F has the archimedean property*

$$(\forall x \in F)(\exists n \in \mathbb{N}_F)[x < n],$$

i.e., the set $\mathbb{N} = \mathbb{N}_F$ of its natural numbers is not bounded from above.

PROOF. Assume towards a contradiction that the set $\mathbb{N}$ has an upper bound, so that it has a least upper bound $x = \sup \mathbb{N}$ by the completeness property. The element $x - 1$ is not an upper bound of $\mathbb{N}$ because $x - 1 < x$, so there must exist some $n \in \mathbb{N}$, $x - 1 < n$: but this implies $x < n + 1$ which contradicts the assumption that x is an upper bound of $\mathbb{N}$. $\dashv$

A.26. Exercise. *In every complete, ordered field F ,*

$$\varepsilon > 0 \implies (\exists n \in \mathbb{N}) \left[\frac{1}{n+1} < \varepsilon \right].$$

A.27. Exercise (Density of the rationals). *In every complete, ordered field F ,*

$$x < y \implies (\exists r \in \mathbb{Q})[x < r \text{ \& } r < y],$$

where $\mathbb{Q} = \mathbb{Q}_F$ is the set of rationals in F .

We now aim to show that *there exists a complete ordered field* and that *any two complete, ordered fields are isomorphic*. Since the completeness property is geometric (or topological), these proofs of existence and uniqueness depend on geometric ideas. Specifically, we will need some basic definitions and results from the *theory of limits* which is studied in Calculus. We will review these here, briefly but without limiting ourselves to the absolute minimum list of theorems necessary to prove the existence and uniqueness of the reals: we have included several Lemmas and Exercises because they support the proposition that the notion of a complete, ordered field represents faithfully our geometric intuitions about the real numbers.

Since we will be manipulating (infinite) sequences a great deal, we will use consistently the familiar, simple notation which avoids the introduction of a distinct name for each sequence and treats the variable as an index, so that a sequence $(n \mapsto a_n)$ is denoted by $\langle a_n \rangle$ or $\langle a_0, a_1, \dots \rangle$. For example, the sequence

$$\left\langle \frac{1}{n+1} \right\rangle = \left\langle \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \dots \right\rangle$$

is the function $f : \mathbb{N} \rightarrow \mathbb{Q}$ defined by the formula

$$f(n) = \frac{1}{n+1}.$$

The **absolute value** function is defined in every ordered field in the usual way,

$$|x| =_{\text{df}} \max\{x, -x\} = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

A.28. Definition. Suppose $(F, 0, 1, +, \cdot, \leq)$ is an ordered field.

(1) A sequence $\langle x_n \rangle$ of elements of F **converges to** $x \in F$ or **has limit** x , if

$$(\forall \varepsilon \in F, \varepsilon > 0)(\exists K \in \mathbb{N})(\forall n \in \mathbb{N})[n \geq K \implies |x - x_n| < \varepsilon].$$

We will use the notation

$$x_n \rightarrow x \iff_{\text{df}} \langle x_n \rangle \text{ converges to } x.$$

(2) A sequence $\langle x_n \rangle$ has the property of Cauchy, or (simply) **is Cauchy** if

$$(\forall \varepsilon \in F, \varepsilon > 0)(\exists K \in \mathbb{N})(\forall n, m \in \mathbb{N})[n, m \geq K \implies |x_n - x_m| < \varepsilon].$$

A.29. Definition. For all $a < b$ in an ordered field F , the set

$$(a, b) =_{\text{df}} \{x \in F \mid a < x < b\} \tag{A-7}$$

is the **open interval** with endpoints a and b . A set $G \subseteq F$ is **open** if it is a (possibly empty) union of open intervals, equivalently

$$x \in G \implies (\exists a < b)[x \in (a, b) \subseteq G].$$

We will also use the standard notations for closed and half-open intervals, e.g.,

$$(a, b] = \{x \in F \mid a < x \leq b\},$$

and also for the half-intervals with $-\infty$ or ∞ as one or both of their endpoints, e.g.,

$$(-\infty, b) = \{x \in F \mid x < b\}, \quad [a, \infty) = \{x \in F \mid a \leq x\}.$$

A.30. Exercise. Prove that the family of open sets in an ordered field is a topology and that the definition of limits for sequences in **A.28** is equivalent to the topological definition of limits given in **10.42**.

These definitions are notorious for the difficulty of understanding what they mean and learning how to use them. We emphasize that here we study them in the context of an arbitrary ordered field which need not be complete, for example, the rationals. It is useful to formulate conditions equivalent to convergence and the property of Cauchy, on the basis of the following notion.

A.31. Definition. A sequence $\langle x_n \rangle$ in an ordered field **settles** in an open interval (a, b) , if after a certain stage all its terms belong to some closed subinterval $[a', b'] \subseteq (a, b)$:

$$\langle x_n \rangle \rightsquigarrow (a, b) \iff_{\text{df}} (\exists K, a', b')(\forall n \geq K)[a < a' \leq x_n \leq b' < b].$$

Notice that if $\langle x_n \rangle \rightsquigarrow (a, b)$, then all its terms after a certain stage belong to (a, b) ,

$$\langle x_n \rangle \rightsquigarrow (a, b) \implies (\exists K)(\forall n \geq K)[a < x_n < b];$$

this weaker property is all we need for many applications of the definition of $\langle x_n \rangle \rightsquigarrow (a, b)$.

A.32. Exercise. $\langle x_n \rangle \rightsquigarrow (a, b)$ if and only if there exists some $\delta > 0$ such that $a + \delta < b - \delta$ and the set $\{n \in \mathbb{N} \mid x_n \notin [a + \delta, b - \delta]\}$ is finite.

A.33. Exercise. For all open intervals I, J , $\langle x_n \rangle \rightsquigarrow I \ \& \ I \subseteq J \implies \langle x_n \rangle \rightsquigarrow J$.

A.34. Exercise. For all open intervals I, J ,

$$\langle x_n \rangle \rightsquigarrow I \ \& \ \langle x_n \rangle \rightsquigarrow J \implies \langle x_n \rangle \rightsquigarrow I \cap J \implies I \cap J \neq \emptyset.$$

A.35. Exercise. Every sequence $\langle x_n \rangle$ which settles in some open interval (a, b) is **bounded**, i.e., $(\exists w)(\forall n)[x_n \leq w]$.

The next Lemma makes it possible in many cases to avoid the so-called “method of epsilons”, which is illustrated by its proof.

A.36. Lemma. For every sequence in any ordered field F :

(1) $\langle x_n \rangle$ converges to x if and only if $\langle x_n \rangle$ settles in every open interval which contains x ,

$$x_n \rightarrow x \iff (\forall a, b \in F)[a < x < b \implies \langle x_n \rangle \rightsquigarrow (a, b)].$$

(2) $\langle x_n \rangle$ is Cauchy if and only if for every $\varepsilon > 0$, there exists an open interval (a, b) in which $\langle x_n \rangle$ settles and such that $(b - a) \leq \varepsilon$:

$$\langle x_n \rangle \text{ is Cauchy} \iff (\forall \varepsilon > 0)(\exists a, b)[a < b \leq a + \varepsilon \ \& \ \langle x_n \rangle \rightsquigarrow (a, b)]$$

PROOF. (1) If $x_n \rightarrow x$ and $a < x < b$, then the definition of convergence with

$$\varepsilon = \frac{\min(x - a, b - x)}{2}$$

supplies a number K such that

$$n \geq K \implies |x - x_n| < \frac{\min(x - a, b - x)}{2},$$

which with a bit of inequality massaging implies that

$$n \geq K \implies a < a' = \frac{a + x}{2} < x_n < \frac{x + b}{2} = b' < b,$$

so that $\langle x_n \rangle \rightsquigarrow (a, b)$. For the proof in the other direction: for every $\varepsilon > 0$, $\langle x_n \rangle \rightsquigarrow (x - \varepsilon, x + \varepsilon)$, so that for some K ,

$$n \geq K \implies |x - x_n| < \varepsilon.$$

(2) If $\langle x_n \rangle$ is Cauchy, then for every $\varepsilon > 0$ there exists some K such that $n, m \geq K \implies |x_n - x_m| < \frac{\varepsilon}{4}$, which immediately implies that

$$\langle x_n \rangle \rightsquigarrow (x_K - \frac{\varepsilon}{2}, x_K + \frac{\varepsilon}{2}).$$

and this interval has length ε . In the other direction, for every $\varepsilon > 0$ there exists some (a, b) with $(b - a) \leq \varepsilon$, so that $\langle x_n \rangle \rightsquigarrow (a, b)$, and so for some K ,

$$n, m \geq K \implies [a < x_n < b \ \& \ a < x_m < b \implies |x_n - x_m| < b - a \leq \varepsilon],$$

which means that $\langle x_n \rangle$ is Cauchy. $\dashv$

A.37. Corollary. *If $\langle x_n \rangle$ converges to some x , then it is Cauchy.*

PROOF. For every $\varepsilon > 0$, $\langle x_n \rangle \rightsquigarrow (x - \frac{\varepsilon}{2}, x + \frac{\varepsilon}{2})$ by (1) of **A.36**, so it is Cauchy by (2) of **A.36**. $\dashv$

A.38. Exercise. $x_n \rightarrow x \ \& \ x_n \rightarrow y \implies x = y$. This allows us to introduce the classical notation

$$x = \lim_n x_n \iff_{\text{df}} x_n \rightarrow x.$$

A.39. Lemma. *If $\langle x_n \rangle$ is Cauchy in a complete, ordered field F , then $\langle x_n \rangle$ has a limit, i.e., $x_n \rightarrow x$ with some x .*

PROOF. Let

$$X =_{\text{df}} \{u \in F \mid (\exists v)[u < v \ \& \ \langle x_n \rangle \rightsquigarrow (u, v)]\}.$$

Since $\langle x_n \rangle$ is Cauchy, there exists some (c, d) such that $\langle x_n \rangle \rightsquigarrow (c, d)$, and hence

$$u \in X \implies u \leq d,$$

since $d < u \implies \langle x_n \rangle \rightsquigarrow (c, d) \cap (u, v) = \emptyset$ which is not possible; hence X is bounded from above and it must have a least upper bound

$$x = \sup X.$$

We will show that

$$a < x < b \implies \langle x_n \rangle \rightsquigarrow (a, b)$$

which implies $x_n \rightarrow x$ by **A.36**. Using the hypothesis $a < x < b$ and the fact that $\langle x_n \rangle$ is Cauchy, we find first some (u, v) such that

$$v - u < \min(x - a, b - x), \quad \langle x_n \rangle \rightsquigarrow (u, v). \quad (\text{A-8})$$

By definition, $u \in X$, and hence (1) $u \leq x$, because x is an upper bound of X . On the other hand, v is also an upper bound of X (because the assumption $\langle x_n \rangle \rightsquigarrow (u', v')$ with $v < u'$ implies that $\langle x_n \rangle$ settles in two disjoint intervals), and hence (2) $x \leq v$, since x is the least upper bound of X . Now (1) and (2) together yield $u \leq x \leq v$, which together with (A-8) implies $a < u \leq x \leq v < b$, so that $\langle x_n \rangle \rightsquigarrow (a, b)$. $\dashv$

The next two basic theorems relate completeness as we defined it (following Dedekind) with the notion of completeness historically associated with the name of Cantor.

A.40. Theorem (The nested interval property). *Suppose that every Cauchy sequence in an ordered field F converges, and that*

$$[x_0, y_0] \supseteq [x_1, y_1] \supseteq \cdots \quad (\text{A-9})$$

is a nested sequence of closed intervals such that

$$\lim_n (y_n - x_n) = 0; \quad (\text{A-10})$$

it follows that the intersection $\bigcap_n [x_n, y_n]$ is a singleton

$$\bigcap_n [x_n, y_n] = \{w\},$$

and its only member is the common limit of the sequences $\langle x_n \rangle$ and $\langle y_n \rangle$,

$$w = \lim_n x_n = \lim_n y_n.$$

PROOF. The basic observation is that for every number K and every $\delta > 0$, $\langle x_n \rangle \rightsquigarrow [x_K - \delta, y_K + \delta]$ by (A-9). Now (A-10) implies that $\langle x_n \rangle$ is Cauchy and hence $x_n \rightarrow x$ for some x , using the hypothesis. Moreover, $x < x_K \implies \langle x_n \rangle \rightsquigarrow (x - 1, x_K)$ by **A.36**, which contradicts the basic properties of the relation $\rightsquigarrow$ since $(x - 1, x_K) \cap [x_K, y_K] = \emptyset$, hence, $x_K \leq x$, for every K . By a similar argument $x \leq y_K$, for every K , so that in the end $x \in \bigcap_n [x_n, y_n]$. Symmetrically, $\langle y_n \rangle$ converges, $y = \lim_n y_n \in \bigcap_n [x_n, y_n]$, and for every n , $|x - y| \leq (y_n - x_n)$ which implies $x = y$ by (A-10). $\dashv$

A.41. Theorem. *An ordered field F is complete if and only if it has the archimedean property (A.25) and every Cauchy sequence in F has a limit.*

PROOF. One direction is known from Lemmas **A.25** and **A.39**, so it is enough to show that if F has the archimedean property and every Cauchy sequence in F converges, then F is complete.

Suppose then that X is a non-empty, bounded from above set in F , so that there exists some point $x_0 \in X$ and some upper bound y_0 of X . Beginning with $[x_0, y_0]$, we define by recursion a sequence of closed intervals $[x_n, y_n]$ which satisfy the following conditions:

1. $x_n \leq x_{n+1} < y_{n+1} \leq y_n$,
2. $(y_n - x_n) = 2^{-n}(y_0 - x_0)$,
3. $[x_n, y_n] \cap X \neq \emptyset$,
4. $(\forall x \in X)[x \leq y_n]$.

In detail, to define $[x_{n+1}, y_{n+1}]$ we distinguish two cases: if $w = \frac{1}{2}(x_n + y_n)$ is an upper bound of X , we set $[x_{n+1}, y_{n+1}] = [x_n, w]$, otherwise $[x_{n+1}, y_{n+1}] = [w, y_{n+1}]$. Proof that $[x_{n+1}, y_{n+1}]$ satisfies (1) – (4) is trivial.

Lemma. $\lim_n (y_n - x_n) = 0$.

Proof. The archimedean property implies that for every $\varepsilon > 0$, there exists some natural number $K > 0$ such that

$$\frac{y_0 - x_0}{\varepsilon} < K \leq 2^K,$$

where the inequality $K \leq 2^K$ is verified easily (by induction!). It follows that for every $n \geq K$,

$$(y_n - x_n) = 2^{-n}(y_0 - x_0) \leq 2^{-K}(y_0 - x_0) < \varepsilon, \quad \dashv (\text{Lemma})$$

Now **A.40** implies that

$$\bigcap [x_n, y_n] = \{w\}$$

where $w = \lim_n x_n = \lim_n y_n$, and it is enough to verify that this common limit w is the least upper bound of X . We compute:

$$\begin{aligned} w < t &\implies \langle y_n \rangle \rightsquigarrow (w - 1, t) && \text{because } \lim_n y_n = w, \\ &\implies y_n < t && \text{for some } n, \\ &\implies t \notin X && \text{because } y_n \text{ is an upper bound of } X, \end{aligned}$$

so that w is an upper bound of X . Also,

$$\begin{aligned} t < w &\implies \langle x_n \rangle \rightsquigarrow (t, w + 1) && \text{because } \lim_n x_n = w, \\ &\implies t < x_n && \text{for some } n, \\ &\implies (\exists x \in X)[t < x] && \text{by the definition of } x_n, \end{aligned}$$

so that there is no upper bound of X smaller than w . $\dashv$

It is worth pointing out here that there exist *Cauchy complete* ordered fields which are not archimedean, Problem **xA.2**.

Following Cantor (up to a point), we will construct a complete, ordered field as a quotient of the Cauchy sequences on the rationals by the following, natural equivalence relation.

A.42. Definition. We call two sequences of rationals $\langle x_n \rangle$ and $\langle y_n \rangle$ **asymptotically equivalent** if their difference converges to 0, in symbols,

$$\langle x_n \rangle \approx \langle y_n \rangle \iff_{\text{df}} (x_n - y_n) \rightarrow 0.$$

A.43. Theorem. (1) *Two Cauchy sequences $\langle x_n \rangle$ and $\langle y_n \rangle$ are asymptotically equivalent if and only if they settle in the same open intervals:*

$$\langle x_n \rangle \approx \langle y_n \rangle \iff (\forall a < b)[\langle x_n \rangle \rightsquigarrow (a, b) \iff \langle y_n \rangle \rightsquigarrow (a, b)].$$

(2) *If $\langle x_n \rangle$ and $\langle y_n \rangle$ are Cauchy, then*

$$\begin{aligned} \langle x_n \rangle \not\approx \langle y_n \rangle &\implies (\text{there exist open intervals } I, J) \\ &[I \cap J = \emptyset \ \& \ \langle x_n \rangle \rightsquigarrow I \ \& \ \langle y_n \rangle \rightsquigarrow J]. \end{aligned}$$

(3) *The relation $\approx$ is an equivalence relation on the set*

$$\mathcal{C}(F) =_{\text{df}} \{ \langle x_n \rangle \mid \langle x_n \rangle \text{ is Cauchy on } F \}.$$

PROOF. (1) Suppose first that $\langle x_n \rangle \approx \langle y_n \rangle$ and $\langle x_n \rangle \rightsquigarrow (a, b)$, so that for some K_0 and some $\delta > 0$,

$$n \geq K_0 \implies a + \delta \leq x_n \leq b - \delta.$$

Using $\langle x_n \rangle \approx \langle y_n \rangle$, choose K_1 such that

$$\begin{aligned} n \geq K_1 &\implies |x_n - y_n| < \frac{\delta}{2} \\ &\implies x_n - \frac{\delta}{2} < y_n < x_n + \frac{\delta}{2}. \end{aligned}$$

From these two implications we get easily that

$$n \geq \max(K_0, K_1) \implies a + \frac{\delta}{2} \leq y_n \leq b - \frac{\delta}{2}.$$

so that $\langle y_n \rangle \rightsquigarrow (a, b)$. In the other direction, for every $\varepsilon > 0$ there exists an open interval (a, b) with $b - a \leq \varepsilon$ such that $\langle x_n \rangle \rightsquigarrow (a, b)$ by (2) of **A.36**, so by the hypothesis, we also have

$$n \geq K \implies [a < x_n < b \ \& \ a < y_n < b] \implies |y_n - x_n| < \varepsilon.$$

(2) Directly from the definition of convergence,

$$\neg[(x_n - y_n) \rightarrow 0] \implies (\exists \varepsilon > 0)(\forall K)(\exists n, m \geq K)[|x_n - y_m| \geq \varepsilon]. \quad (\text{A-11})$$

By **A.36**, there exist open intervals I and J of length $< \frac{\varepsilon}{2}$ (with this ε) such that $\langle x_n \rangle \rightsquigarrow I$ and $\langle y_n \rangle \rightsquigarrow J$. If some $z \in I \cap J$ existed, then for K sufficiently large so that

$$n, m \geq K \implies x_n \in I \ \& \ y_m \in J,$$

we would have

$$n, m \geq K \implies |x_n - y_m| \leq |x_n - z| + |z - y_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which contradicts (A-11).

(3) The reflexivity and symmetry of $\approx$ are trivial. To show its transitivity, notice that by hypothesis and (1) of Lemma **A.36**, for every (a, b) ,

$$\langle x_n \rangle \rightsquigarrow (a, b) \iff \langle y_n \rangle \rightsquigarrow (a, b) \iff \langle z_n \rangle \rightsquigarrow (a, b),$$

so that by (1), $\langle x_n \rangle \approx \langle z_n \rangle$. ⊢

A.44. Exercise. If $\langle x_n \rangle$ and $\langle y_n \rangle$ are both Cauchy and $x_n \rightarrow x$, then

$$\langle x_n \rangle \approx \langle y_n \rangle \iff y_n \rightarrow x.$$

At this point we could appeal to the existence of *some* quotient B of the set

$$\mathcal{C} = \mathcal{C}(\mathbb{Q}) =_{\text{df}} \{ \langle r_n \rangle \mid \langle r_n \rangle \text{ is Cauchy in the field of rationals} \}$$

by $\approx$ and define the necessary functions and an ordering on B so that it becomes a complete, ordered field. This is one of the classical proofs of the existence of the real numbers, connected with the name of Cantor. Instead of this, we will construct *a specific quotient* of $\mathcal{C}$ by $\approx$ which simplifies the proof a bit and (more significantly) relates this construction with the other classical proof of the existence of the reals, following Dedekind. The basic idea of Dedekind was that a real number x is completely determined by (and hence can be “identified” with) the set

$$(-\infty, x) \cap \mathbb{Q} =_{\text{df}} \{ r \in \mathbb{Q} \mid r < x \}$$

of all rationals preceding it, and that the sets of the form $(-\infty, x) \cap \mathbb{Q}$ can be characterized directly by three simple conditions.

A.45. Definition. A **Dedekind cut** is any set X of rational numbers which satisfies the following three conditions:

1. $X \neq \emptyset, (\mathbb{Q} \setminus X) \neq \emptyset$.
2. $r < q \ \& \ q \in X \implies r \in X$.
3. $q \in X \implies (\exists r)[q < r \ \& \ r \in X]$.

We set

$$\mathcal{D} =_{\text{df}} \{X \subseteq \mathbb{Q} \mid X \text{ is a Dedekind cut}\}.$$

A.46. Exercise. A set $X \subseteq \mathbb{Q}$ is a Dedekind cut if and only if it is non-empty, bounded from above, with no largest member and “downward closed”, i.e., such that $r < q$ & $q \in X \implies r \in X$.

The next theorem is basic for the proof of existence of the real numbers.

A.47. Theorem. For each Cauchy sequence of rationals $\langle x_n \rangle$, let

$$\pi(\langle x_n \rangle) =_{\text{df}} \{a \in \mathbb{Q} \mid (\exists b)[a < b \text{ \& } \langle x_n \rangle \rightsquigarrow (a, b)]\};$$

it follows that each value $\pi(\langle x_n \rangle)$ is a Dedekind cut and that the function

$$\pi : \mathcal{C} \twoheadrightarrow \mathcal{D}$$

is a surjection which determines the equivalence relation $\approx$, so that $\mathcal{D}$ is a quotient of $\mathcal{C}$.

PROOF. That each $\pi(\langle x_n \rangle)$ is a Dedekind cut is quite easy from the definitions and the general properties of the relation $\rightsquigarrow$, and the equivalence

$$\langle x_n \rangle \approx \langle y_n \rangle \iff \pi(\langle x_n \rangle) = \pi(\langle y_n \rangle)$$

is an immediate corollary of **A.43**. The only thing which is not completely obvious is that for every cut X there exists a Cauchy sequence $\langle x_n \rangle \in \mathcal{C}$ in the rationals such that $\pi(\langle x_n \rangle) = X$. For this we construct a nested sequence of closed intervals *in the rationals*

$$[x_0, y_0] \supseteq [x_1, y_1] \supseteq \cdots$$

exactly as in the proof of **A.41**, beginning with some $x_0 \in X$ and some $y_0 \notin X$, so that, in fact, y_0 is an upper bound of X . We argue as in **A.41** that the non-decreasing sequence $\langle x_n \rangle$ is Cauchy, and that, in addition, for all n ,

$$x_n \in X \text{ \& } y_n \notin X,$$

because X is downward closed and has no largest member. Then we compute:

$$\begin{aligned} a \in \pi(\langle x_n \rangle) &\implies (\exists b)[a < b \text{ \& } \langle x_n \rangle \rightsquigarrow (a, b)] \\ &\implies (\exists n)[a < x_n] \\ &\implies a \in X \text{ because } x_n \in X. \end{aligned}$$

To see that $X \subseteq \pi(\langle x_n \rangle)$, notice that if $a \in X$, then there must exist some natural number K such that $a < x_K$, because the opposite supposition $(\forall n)[x_n \leq a]$ implies (easily) that a is the largest point in X , and X does not have a largest point. Thus, for $n \geq K$, $x_K \leq x_n \leq y_K$, hence, $\langle x_n \rangle \rightsquigarrow (x_K, y_K + 1)$ and $a \in \pi(\langle x_n \rangle)$. $\dashv$

A.48. Theorem (Existence of the real numbers). There exists a complete, ordered field.

PROOF. For the domain of the required field we take the set $\mathcal{D}$ of Dedekind cuts and for 0 and 1 we take the obvious:

$$\begin{aligned} 0 &=_{\text{df}} \{r \in \mathbb{Q} \mid r < 0\} = \pi(\langle 0 \rangle), \\ 1 &=_{\text{df}} \{r \in \mathbb{Q} \mid r < 1\} = \pi(\langle 1 \rangle). \end{aligned}$$

In order to define addition and multiplication on $\mathcal{D}$ we need the following two Lemmas, where all the sequences are Cauchy in the rationals.

$$\langle x_n \rangle \approx \langle x'_n \rangle \ \& \ \langle y_n \rangle \approx \langle y'_n \rangle \implies \langle x_n + y_n \rangle \approx \langle x'_n + y'_n \rangle \quad (\text{A-12})$$

$$\langle x_n \rangle \approx \langle x'_n \rangle \ \& \ \langle y_n \rangle \approx \langle y'_n \rangle \implies \langle x_n \cdot y_n \rangle \approx \langle x'_n \cdot y'_n \rangle \quad (\text{A-13})$$

These are the useful interpretations of the classical theorems from the theory of limits,

$$\begin{aligned} \lim_n (x_n + y_n) &= \lim_n x_n + \lim_n y_n, \\ \lim_n (x_n \cdot y_n) &= \lim_n x_n \cdot \lim_n y_n, \end{aligned}$$

in the case at hand, when the limits need not exist since the sequences are in the incomplete ordered field of the rationals. They are not hard to verify after all the preparatory work we have done and we will skip the details. The implications (A-12) and (A-13) assert that $\approx$ is a congruence in $\mathcal{C}$ for the functions

$$\begin{aligned} (\langle x_n \rangle, \langle y_n \rangle) &\mapsto \langle x_n + y_n \rangle, \\ (\langle x_n \rangle, \langle y_n \rangle) &\mapsto \langle x_n \cdot y_n \rangle, \end{aligned}$$

so that by **A.2** there exist functions $+$ and $\cdot$ on the quotient $\mathcal{D}$ which satisfy the identities

$$\begin{aligned} \pi(\langle x_n \rangle) + \pi(\langle y_n \rangle) &= \pi(\langle x_n + y_n \rangle), \\ \pi(\langle x_n \rangle) \cdot \pi(\langle y_n \rangle) &= \pi(\langle x_n \cdot y_n \rangle). \end{aligned}$$

We take these $+$ and $\cdot$ for the operations of addition and multiplication in $\mathcal{D}$.

Next we must show that $(\mathcal{D}, 0, 1, +, \cdot)$ is a field, but this part of the proof is quite trivial, if a bit tiring in its details (which we will skip). The existence of additive inverses, for example, follows from the obvious

$$\pi(\langle x_n \rangle) + \pi(\langle -x_n \rangle) = \pi(\langle x_n + (-x_n) \rangle) = \pi(\langle 0 \rangle) = 0,$$

where the only “delicate” point is the observation that if $\langle x_n \rangle$ is Cauchy, then so is $\langle -x_n \rangle$. To check the corresponding property for multiplication, given a Cauchy sequence $\langle x_n \rangle \not\approx \langle 0 \rangle$, set

$$y_n =_{\text{df}} \begin{cases} 1/x_n, & \text{if } x_n \neq 0, \\ 1, & \text{if } x_n = 0, \end{cases} \quad (\text{A-14})$$

verify that $\langle y_n \rangle$ is also Cauchy and then compute:

$$\pi(\langle x_n \rangle) \cdot \pi(\langle y_n \rangle) = \pi(\langle x_n \cdot y_n \rangle) = \pi(\langle 1 \rangle) = 1.$$

The basic observation (from (2) of **A.43**) is that

$$\begin{aligned} \langle x_n \rangle \not\approx \langle 0 \rangle &\implies (\exists \delta > 0)[\langle x_n \rangle \not\prec (-\delta, \delta)] \\ &\implies (\exists \delta > 0, K)(\forall n \geq K)|x_n| \geq \delta \end{aligned}$$

with which we begin the proof that $\langle y_n \rangle$ is Cauchy, but some epsilonics are unavoidable. The related result from the theory of limits is the assertion

$$\lim_n x_n \neq 0 \implies \lim_n \left(\frac{1}{x_n} \right) = \frac{1}{\lim_n x_n},$$

traditionally known as the first hard theorem in Calculus, when it is taught rigorously.

Next we define on $\mathcal{D}$ the relation

$$X \leq Y \iff_{\text{df}} X \subseteq Y,$$

which is certainly a partial ordering; it is also linear, because for any two Dedekind cuts X and Y , directly from the definition,

$$\begin{aligned} r \in (Y \setminus X) &\implies (\forall q \in X)[q < r] \& (\forall q < r)[q \in Y] \\ &\implies X \subseteq Y, \end{aligned}$$

and, of course, $X \neq Y \implies (\exists r)[[r \in (X \setminus Y)] \vee [r \in (Y \setminus X)]]$. Appealing once more to the definition of Dedekind cuts, we can also show easily that

$$I \subseteq \mathcal{D} \implies \bigcup I = \mathbb{Q} \vee \bigcup I \in \mathcal{D}. \quad (\text{A-15})$$

(For example, if r were the largest point in the union $\bigcup I$, then $r \in X$ for some $X \in I$, and then r would also be the largest point in X , since $X \subseteq I$ —but X has no largest member.) From (A-15) we infer that every set $I \subseteq \mathcal{D}$ which is bounded above has a least upper bound, because

$$(\forall X \in I)[X \leq Z] \implies \bigcup I \subseteq Z \subsetneq \mathbb{Q} \implies \bigcup I \in \mathcal{D},$$

and the union $\bigcup I$ is obviously the least upper bound of I in the relation $\leq = \subseteq$.

It remains to verify that for all $X, Y, Z \in \mathcal{D}$

$$X < Y \implies X + Z < Y + Z, \quad (\text{A-16})$$

$$[Z > 0 \& X < Y] \implies Z \cdot X < Z \cdot Y, \quad (\text{A-17})$$

since these implications imply then immediately their versions with $\leq$. Considering the more difficult (A-17) as an example, choose first Cauchy sequences such that

$$\pi(\langle z_n \rangle) = Z, \quad \pi(\langle x_n \rangle) = X, \quad \pi(\langle y_n \rangle) = Y,$$

and verify (easily) from the definitions (and **A.43**) that there exist rationals

$$x^0 < x^1 < y^0 < y^1$$

satisfying

$$\langle x_n \rangle \rightsquigarrow (x^0, x^1), \quad \langle y_n \rangle \rightsquigarrow (y^0, y^1),$$

and that for each $\varepsilon > 0$ we can find some z^0 and z^1 such that

$$0 < z^0 < z^1, \quad (z^1 - z^0) < \varepsilon, \quad \langle z_n \rangle \rightsquigarrow (z^0, z^1).$$

It follows that

$$\langle z_n x_n \rangle \rightsquigarrow (z^0 x^0, z^1 x^1), \quad \langle z_n y_n \rangle \rightsquigarrow (z^0 y^0, z^1 y^1),$$

and the desired conclusion $Z \cdot X < Z \cdot Y$ will follow quite easily, if we could choose z^0, z^1 so that

$$z^1 x^1 < z^0 y^0,$$

or equivalently,

$$x^1(z^1 - z^0) < z^0(y^0 - x^1). \quad (\text{A-18})$$

Now (A-18) is obvious if $x^1 \leq 0$, because in that case $x^1(z^1 - z^0) \leq 0$ and $z^0(y^0 - x^1) > 0$. If $x^1 > 0$, we find first some $\delta > 0$ such that $\langle z_n \rangle \rightsquigarrow (\delta, \infty)$ and then z^0, z^1 such that

$$0 < \delta < z^0 < z^1, \quad \langle z_n \rangle \rightsquigarrow (z^0, z^1), \quad (z^1 - z^0) < \frac{\delta(y^0 - x^1)}{x^1}$$

which imply (A-18):

$$x^1(z^1 - z^0) < x^1 \frac{\delta(y^0 - x^1)}{x^1} \leq z^0(y^0 - x^1).$$

Verification of (A-16) is substantially simpler and completes the proof of the theorem. $\dashv$

A.49. Exercise. Prove that for all Dedekind cuts X, Y and Z :

$$X \cdot (Y + Z) = X \cdot Y + X \cdot Z.$$

(Use the formal definitions of $+$ and $\cdot$ given in the proof of A.48.)

A.50. Exercise. Prove (A-12) and (A-13).

A.51. Exercise. Show that if $\langle x_n \rangle$ is Cauchy in an ordered field and $\langle x_n \rangle \not\approx \langle 0 \rangle$, then the sequence $\langle y_n \rangle$ defined by (A-14) is also Cauchy and $\langle x_n y_n \rangle \approx \langle 1 \rangle$.

A.52. Theorem (Uniqueness of the real numbers). For any two complete, ordered fields F_1 and F_2 , there exists exactly one bijection

$$\pi^* : F_1 \xrightarrow{\sim} F_2$$

which is an isomorphism, i.e.,

1. $\pi^*(0) = 0, \quad \pi^*(1) = 1,$
2. $\pi^*(x + y) = \pi^*(x) + \pi^*(y), \quad \pi^*(xy) = \pi^*(x)\pi^*(y),$
3. $x \leq y \iff \pi^*(x) \leq \pi^*(y).$

PROOF. By the uniqueness of the rationals A.14, there exists (exactly one) isomorphism

$$\pi : \mathbb{Q}_1 \xrightarrow{\sim} \mathbb{Q}_2,$$

where $\mathbb{Q}_1, \mathbb{Q}_2$ are the sets of rationals in the two fields F_1, F_2 , and the problem is to extend this π to the whole of F_1 .

Lemma 1. For each $x \in F_1$, there exists a sequence $\langle x_n \rangle$ of rationals in F_1 , such that $\lim_n x_n = x$.

Proof. Using the density of the rationals (Exercise A.27), we can find for each $n \in \mathbb{N}$ a rational $x_n \in \mathbb{Q}_1$ such that $|x - x_n| < 1/(n+1)$, and then (easily, using problem A.26) $\lim_n x_n = x$. $\dashv$ (Lemma 1)

Lemma 2. For each sequence $\langle x_n \rangle$ of rationals in F_1 ,

$$(\exists x \in F_1)[\lim_n = x] \implies (\exists x^* \in F_2)[\lim_n \pi(x_n) = x^*].$$

Proof. We know that

$$u < v \iff \pi(u) < \pi(v) \quad (u, v \in \mathbb{Q}_1)$$

because π is an isomorphism, so that for all $a, b \in \mathbb{Q}_1$, $a < b$,

$$\langle x_n \rangle \rightsquigarrow (a, b) \iff \langle \pi(x_n) \rangle \rightsquigarrow (\pi(a), \pi(b)). \quad (\text{A-19})$$

If $\langle x_n \rangle$ converges, then it is Cauchy, so that $\langle \pi(x_n) \rangle$ is also Cauchy by (A-19) and A.36 (using A.26 once more), and therefore $\langle \pi(x_n) \rangle$ converges because F_2 is complete. (Lemma 2)

We can use the same simple idea to verify the third basic fact we need, whose proof we will omit:

Lemma 3. For any two Cauchy sequences in the rationals of F_1 ,

$$[\lim_n x_n = \lim_n y_n] \implies \lim_n \pi(x_n) = \lim_n \pi(y_n).$$

The Lemmas guarantee that we can define unambiguously for each $x \in F_1$,

$$\pi^*(x) =_{\text{df}} \lim_n \pi(x_n), \quad \text{where } \lim_n x_n = x, \text{ with } x_n \in \mathbb{Q}_1. \quad (\text{A-20})$$

Since for each rational $x \in \mathbb{Q}_1$, $\lim_n x = x$, we have

$$\pi^*(x) = \lim_n \pi(x) = \pi(x),$$

so that π^* is an extension of π . It remains to verify that π^* is an isomorphism of F_1 with F_2 .

Suppose first that

$$x = \lim_n x_n, \quad y = \lim_n y_n, \quad x_n, y_n \in \mathbb{Q}_1, \quad x < y.$$

This implies immediately (by A.36) that there exist rationals a, b, c and d satisfying

$$\langle x_n \rangle \rightsquigarrow (a, b), \quad \langle y_n \rangle \rightsquigarrow (c, d), \quad b < c,$$

and hence by (A-19)

$$\langle \pi(x_n) \rangle \rightsquigarrow (\pi(a), \pi(b)), \quad \langle \pi(y_n) \rangle \rightsquigarrow (\pi(c), \pi(d)).$$

It follows that

$$\pi^*(x) = \lim_n \pi(x_n) < \lim_n \pi(y_n) = \pi^*(y)$$

because $\pi(b) < \pi(c)$, and this completes the proof that π^* respects the strict relation $<$:

$$x < y \implies \pi^*(x) < \pi^*(y).$$

Directly from this, π^* is an injection and it respects the ordering,

$$x \leq y \iff \pi^*(x) \leq \pi^*(y).$$

The rest is trivial (if tiresome) and follows mostly from the *limit theorems* of the Calculus, which hold in every complete, ordered field—and can be proved easily with the tools we have developed. As an example:

$$\begin{aligned}\pi^*(x + y) &=_{\text{df}} \lim_n \pi(x_n + y_n), \quad \text{where } x_n \rightarrow x, y_n \rightarrow y, x_n, y_n \in \mathbb{Q}_1, \\ &= \lim_n [\pi(x_n) + \pi(y_n)] \\ &= \lim_n \pi(x_n) + \lim_n \pi(y_n) \\ &= \pi^*(x) + \pi^*(y).\end{aligned}$$

The crucial step in this computation is the identity

$$\lim_n [\pi(x_n) + \pi(y_n)] = \lim_n \pi(x_n) + \lim_n \pi(y_n).$$

We skip the details. —

A.53. Exercise. *Work out the details of the proof of*

$$\pi^*(x + y) = \pi^*(x) + \pi^*(y).$$

A.54. The real numbers. As we did for the natural numbers and the rationals, we now fix some complete, ordered field

$$(\mathbb{R}, 0, 1, +, \cdot, \leq), \tag{A-21}$$

whose members we will call **real numbers**. We emphasize once more that fixing some $\mathbb{R}$ is only a convenience and the specific choice of $\mathbb{R}$ is of no importance: the fundamental mathematical fact for the development of analysis is that a complete, ordered field exists and that any two complete, ordered fields are isomorphic.

A.55. Exercise. *Prove Corollary 2.14 in Chapter 2 from the axioms.*

The open sets of real numbers defined in A.29 form a topology by the easy Exercise A.30, so we have notions of Borel sets of reals and Borel measurable functions from $\mathbb{R}$ to other topological spaces and vice versa, by Definitions 10.25 and 10.39. The notion of Borel isomorphism was defined in Definition 10.40. The next theorem makes it possible to transfer results about Baire space to the reals, and it is the main tool for analyzing the set theoretic properties of $\mathbb{R}$. We will omit its proof, which is quite simple, using Problems x10.12 and x10.13.

A.56. Theorem. *As a topological space, $\mathbb{R}$ is Borel isomorphic with the Baire space $\mathcal{N}$, and, in particular, $\mathbb{R} =_c \mathcal{N}$.*

With the true (von Neumann) cardinals developed in Chapter 12 from ZFC, this gives the congenial equation $|\mathbb{R}| = \mathfrak{c} = 2^{\aleph_0}$.

Problems for Appendix A

***xA.1.** Let F be the set of all real, rational functions, i.e., all (partial) real functions which can be represented as quotients of polynomials with coefficients in $\mathbb{R}$, and prove that it is a field with the obvious algebraic operations.

Show that the relation

$$f \leq g \iff_{\text{df}} (\exists x)(\forall y \geq x)[f(y) \leq g(y)]$$

is a linear ordering of F , and with it F is a non-archimedean ordered field.

HINT. The field $\mathbb{Q}_F$ comprises the constant functions with rational value, and the identity $(x \mapsto x)$ is above all of them.

***xA.2.** Prove that there exists an ordered field which is not complete, but in which every Cauchy sequence has a limit. **HINT.** Show that every ordered field has a *Cauchy completion*.

xA.3. Every open set of reals is a countable union of disjoint open intervals.

xA.4. Every closed interval of real numbers $[a, b]$ is compact, in the topological sense, **10.43**.

xA.5. Every closed set of real numbers is a countable union of compact sets.

xA.6. Every closed set of real numbers F can be written uniquely as the disjoint union of a perfect and a countable set.

***xA.7.** Prove Theorem **A.56**.

APPENDIX B

AXIOMS AND UNIVERSES

The serious study of *models* of axiomatic set theories depends heavily on methods from *mathematical logic* which are outside the scope of these Notes.³² Here we will consider only SET UNIVERSES, generalizations of the Zermelo and the ZFDC-universes of Chapter 11, which are very special models and can be studied by standard mathematical techniques, as we study fields or topological spaces. First we will prove that THE ZERMELO UNIVERSES of Chapter 11 ARE MODELS OF ZDC and THE ZFDC-UNIVERSES ARE MODELS OF ZFDC; this will give us a better understanding of these universes, and it will also yield some simple CONSISTENCY AND INDEPENDENCE RESULTS for the corresponding theories. In the main part of this chapter we will construct some new set universes with quite different properties, including the ANTIFOUNDED UNIVERSE of Aczel which contains a rich variety of ill founded sets. We will glean some consistency results from these models too, but consistency results are not our main concern: our primary interest is to explore and understand several natural, intuitive notions of SET and compare them with the standard conception of PURE, GROUNDED SET discussed in 12.34.

We begin with a result about the least Zermelo universe $\mathcal{Z}$ which is somewhat surprising, given how much we promoted $\mathcal{Z}$ in 11.25 as a rich collection of sets which contains all objects of interest of classical mathematics.

B.1. Theorem. *The set V_ω of all pure, grounded, hereditarily finite sets is not a member of $\mathcal{Z}$; in fact, there is no set $A \in \mathcal{Z}$ such that*

$$\emptyset \in A \ \& \ (\forall X)[X \in A \implies \mathcal{P}(X) \in A]. \quad (\text{B-1})$$

PROOF. For each $x \in \mathcal{Z}$, we let

$$\text{level}(x) = \text{the least } n \text{ such that } x \in \mathcal{Z}_n, \quad (\text{B-2})$$

³²For those who do know logic, we remark here that the most natural way to formalize the theories we have studied is in a *many sorted predicate logic with identity*, with separate variables for objects, definite conditions and definite operations of every arity, and relation symbols Set and $\in$. Notice that we did not assume in 3.18 any extensionality principles for conditions or operations, and we have never appealed to such principles. This means that to get (a notational variant of) the classical *Gödel-Bernays Theory* from the ZFC of 11.31 we must add *extensionality for conditions*. On the other hand, precisely because we did not include any extensionality axioms for conditions, we can view the present ZFC as a notational variant of the classical *Zermelo-Fraenkel Theory*, by interpreting “conditions” to be just “formulas with set parameters”.

so that members of $\mathbb{N}_0$ have level 0, but

$$\text{level}(x) > 0 \implies \text{level}(\{x\}) = \text{level}(x) + 1;$$

because if $x \in \mathcal{Z}_n$, then $\{x\} \in \mathcal{Z}_{n+1}$, and if $\{x\} \in \mathcal{Z}_n$ with $n > 0$, then $\{x\} \subseteq \mathcal{Z}_{n-1}$ by (11-20), which implies $x \in \mathcal{Z}_{n-1}$. Define now by recursion the sets

$$A_0 = \{\emptyset, \{\emptyset\}\}, \quad A_{n+1} = \{A_n\}.$$

Clearly, each $A_n \in V_\omega$, $\text{level}(A_0) = 1$, since $A_0 \subseteq \mathbb{N}_0$ but $A_0 \notin \mathbb{N}_0$, and by induction, $\text{level}(A_n) = n + 1$.

Suppose now that there is some $A \in \mathcal{Z}$ which satisfies (B-1), and notice that $A_0 \in \mathcal{P}(\mathcal{P}(\emptyset)) \in A$, and then by induction, for each n , $A_n \in A$. If $A \in \mathcal{Z}_m$, then each $A_n \in \mathcal{Z}_m$ because $\mathcal{Z}_m$ is transitive, and hence $\text{level}(A_n) \leq m$ for each n , which violates what we just proved. $\neg$

A similar argument shows that (with Kuratowski tupling) $\bigcup_{n=2}^{\infty} \{\mathbb{N}_0\}^n \notin \mathcal{Z}$, Problem **xB.2**, and it is quite easy to extend it to show that $\mathcal{Z}$ misses many more very simple sets, but the fact that it lacks V_ω is undoubtedly the most startling of the lot. The construction of V_ω is so direct, it seems to follow so naturally from our most basic intuitions about sets, that it is really hard to believe that we developed all this set theory in Chapters 3 – 10 and Appendix A from axioms which do not guarantee its existence. One may try to write off this feature of the Zermelo axioms as a small oversight of Zermelo and strengthen the axioms in some minor way to ensure the existence of V_ω , but this is the wrong way to go. On the one hand, we know the natural extension of **ZDC+AC**, it is the addition of the Replacement Axiom which can be justified by arguments only marginally different from those used to justify the Separation Axiom. And on the other hand, the importance of **ZDC+AC** lies precisely in its two, contrasting features: that it can prove so much about classical mathematics (which is its real domain), while it can be interpreted in such simple, easy to comprehend models like $\mathcal{Z}$. Whatever doubts may have lingered about the soundness of set theory from the paradoxes should be at least moderated by the strength of **ZDC+AC** and its ease of interpretation.

We still need to make precise the sense in which $\mathcal{Z}$ is a “model” of **ZDC**. Perhaps the simplest way to introduce the key, new idea we need, is to try and reinterpret **B.1** as an independence result.

B.2. Theorem? *We cannot prove in **ZDC** the proposition that some set A exists, which contains the empty set and is closed under the powerset operation.*

PROOF. Spelled out symbolically for precision, the proposition in question is

$$\phi \iff_{\text{df}} (\exists A)[\emptyset \in A \ \& \ (\forall X)[X \in A \implies \mathcal{P}(X) \in A]].$$

Suppose, towards a contradiction, that ϕ is a theorem of **ZDC**. Since the least Zermelo universe $\mathcal{Z}$ has all the closure properties demanded by the axioms of **ZDC**, any proof of ϕ from these axioms could be translated into a proof of the interpretation of ϕ in $\mathcal{Z}$, which is

$$\phi^{(\mathcal{Z})} \iff (\exists A \in \mathcal{Z})[\emptyset \in A \ \& \ (\forall X \in \mathcal{Z})[X \in A \implies \mathcal{P}(X) \in A]].$$

As a consequence, $\phi^{(\mathcal{Z})}$ is true, so there exists a set $A \in \mathcal{Z}$ satisfying

$$\emptyset \in A \ \& \ (\forall X \in \mathcal{Z})[X \in A \implies \mathcal{P}(X) \in A],$$

which by the transitivity of $\mathcal{Z}$ is equivalent to

$$\emptyset \in A \ \& \ (\forall X)[X \in A \implies \mathcal{P}(X) \in A]; \quad (\text{B-3})$$

but by **B.1**, no $A \in \mathcal{Z}$ satisfies (B-3). $\dashv$

The argument is unfamiliar, unless you know a lot of logic, and in any case it is incomplete, only a sketch. What we need to elucidate is the meaning of “interpreting a proposition in $\mathcal{Z}$ ”, the move from ϕ to $\phi^{(\mathcal{Z})}$ above, and it may help to consider an example. Suppose we have located a traditional mathematician (perhaps an old-fashioned analyst) who disclaims any interest in general set theory beyond its applications to classical mathematics, he has studied Chapters 3 – 10 and Appendix A and he is convinced that all the objects he cares about live in $\mathcal{Z}$. In an effort to simplify his world, he declares that henceforth by “object” he will mean “member of $\mathcal{Z}$ ”: that is his universe. Suppose further that this person now utters the Powerset Axiom and claims that he believes it. *What does he mean?* Spelled out in terms of the primitive notions of sethood and membership, the powerset axiom reads as follows:

(IV): *For each set A , there exists a set B , such that for every X ,*

$$X \in B \iff \text{Set}(X) \ \& \ (\forall t)[t \in X \implies t \in A]. \quad (\text{B-4})$$

Replacing “set” by “member of $\mathcal{Z}$ ” in this, we get something rather different.

(IV)^($\mathcal{Z}$): *For each set $A \in \mathcal{Z}$, there exists some set $B \in \mathcal{Z}$, such that for every $X \in \mathcal{Z}$,*

$$X \in B \iff \text{Set}(X) \ \& \ (\forall t \in \mathcal{Z})[t \in X \implies t \in A]. \quad (\text{B-5})$$

This is what our friend really means when he tries to tell us that every set has a powerset, and it differs enough from the Powerset Axiom that its truth is not immediately apparent. To prove it, for each $A \in \mathcal{Z}$, we naturally take $B = \mathcal{P}(A)$, which is also in $\mathcal{Z}$ and satisfies (B-4), for every X . Notice next that for each $X \in \mathcal{Z}$,

$$(\forall t \in \mathcal{Z})[t \in X \implies t \in A] \iff (\forall t)[t \in X \implies t \in A],$$

easily, by the transitivity of $\mathcal{Z}$. This reduces (B-5) to

$$X \in B \iff \text{Set}(X) \ \& \ (\forall t)[t \in X \implies t \in A], \quad (\text{B-6})$$

which is true for every X and hence (in particular) for every $X \in \mathcal{Z}$.

As a matter of fact, all the axioms of **ZDC** yield true propositions when we replace in them “object” by “member of $\mathcal{Z}$ ”. It follows that all the theorems derived from the axioms of **ZDC** by logic alone also yield true propositions when we understand them as assertions about $\mathcal{Z}$ in the same way, so our stipulated classical friend can safely work in **ZDC** and be assured that he is proving statements which are true of his world. This not entirely trivial proposition expresses the fact that the universe $\mathcal{Z}$ is a “model” of **ZDC**.

B.3. A **set universe** $\mathcal{M}$ is any triple M, S, E , of a class (which may be a set) M , a subclass $S \subseteq M$, and a binary definite condition E such that $E(x, y) \implies x, y \in M$ and for each $x \in M$,

$$b_{\mathcal{M}}(x) =_{\text{df}} \{t \mid t E x\} \text{ is a set,} \quad (\text{B-7})$$

which means that for some set X and all t ,

$$t \in X \iff E(t, x).$$

We write synonymously

$$t E x \iff E(t, x)$$

for the condition E , which interprets the $\in$ condition in $\mathcal{M}$, and we call $b_{\mathcal{M}}(x)$ the **body** of each $x \in M$. The **$\mathcal{M}$ -objects** are the members of M and the **$\mathcal{M}$ -sets** are the members of S . An n -ary definite condition is an **$\mathcal{M}$ -condition** if it only holds of objects in M , i.e.,

$$P(x_1, \dots, x_n) \implies x_1, \dots, x_n \in M;$$

and a definite n -ary operation F is an **$\mathcal{M}$ -operation** if it assigns $\mathcal{M}$ -objects to $\mathcal{M}$ -objects, i.e.,

$$x_1, \dots, x_n \in M \implies F(x_1, \dots, x_n) \in M.$$

A universe $\mathcal{M}$ is **natural** if the class M is transitive and S, E are the standard sethood and membership conditions, i.e.,

$$\begin{aligned} X \in M &\implies X \subseteq M, \\ x \in S &\iff x \in M \ \& \ \text{Set}(x), \\ x E y &\iff x, y \in M \ \& \ x \in y. \end{aligned}$$

A natural universe $\mathcal{M}$ is completely determined by the transitive class M of its objects and we will identify it with that class, i.e., when we refer to *the universe* M for a transitive class M , we will mean the natural universe with objects the members of M . Notice that in a natural universe the body of each object is the set of its members,

$$b_M(x) = \{t \mid t \in x\} \quad (x \in M); \quad (\text{B-8})$$

this means that $b_M(x) = x$, if x is a set, but $b_M(x) = \emptyset$ if x is an atom.

The **relativization to** a universe $\mathcal{M}$ of a proposition θ is the proposition $\theta^{(\mathcal{M})}$ constructed by replacing “object” by “ $\mathcal{M}$ -object”, “set” by “ $\mathcal{M}$ -set”, “ $x \in y$ ” by “ $x E y$ ”, “condition” by “ $\mathcal{M}$ -condition” and “operation” by “ $\mathcal{M}$ -operation”, consistently wherever these expressions occur in θ . If $\theta^{(\mathcal{M})}$ is true, we say that θ is **true in the universe** $\mathcal{M}$ or (synonymously) that $\mathcal{M}$ **satisfies or models** θ .

A universe $\mathcal{M}$ is a **model** of an axiomatic system T if every axiom of T is true in $\mathcal{M}$, i.e., if the relativization $\theta^{(\mathcal{M})}$ of every axiom θ of T is a true proposition. If this holds, we call $\mathcal{M}$ a **universe of** T , or simply a **T -universe**.

B.4. Propositions and relativizations. By *proposition* we mean any ordinary, definite mathematical statement or assertion, just like the axioms, hypotheses and theorems we have considered so far. This is not completely precise, everyday mathematical English not being a perfectly specified language. The basic idea is that for all objects x and y , the expressions $x = y$, $x \in y$ and $\text{Set}(x)$ are propositions; for each definite condition P and objects $x_1, \dots, x_n$, the expression $P(x_1, \dots, x_n)$ is a proposition; and that propositions may be combined by the basic operations of logic, $\neg$, $\vee$, $\exists$ and the like. *Relativizations*³³ to a set universe $\mathcal{M}$ are computed as one might expect, e.g.,

$$\begin{aligned}
 (x = y)^{(\mathcal{M})} &: x = y, \\
 \text{Set}(x)^{(\mathcal{M})} &: x \in S, \\
 (x \in y)^{(\mathcal{M})} &: x E y, \\
 P(x_1, \dots, x_n)^{(\mathcal{M})} &: P(x_1, \dots, x_n), \\
 (\neg \theta)^{(\mathcal{M})} &: \neg(\theta^{(\mathcal{M})}), \\
 (\phi \& \psi)^{(\mathcal{M})} &: \phi^{(\mathcal{M})} \& \psi^{(\mathcal{M})}, \\
 ((\forall x)\phi)^{(\mathcal{M})} &: (\forall x \in M)\phi^{(\mathcal{M})}, \\
 ((\forall P)\phi)^{(\mathcal{M})} &: (\forall P)[(\forall x, y)[P(x, y) \implies x, y \in M] \implies \phi^{(\mathcal{M})}],
 \end{aligned}$$

where P varies over binary, definite conditions, etc. In the proofs which follow, we will take care to spell out laboriously the relativizations of all the propositions that concern us. This will produce many examples which illustrate the notion, and it will also ensure that the specific results we claim will be precise and rigorously established even if the general notion of proposition and the relativization process are not precisely delimited. On the other hand, much of the significance of the results comes from the following general principle. It says simply that logical consequences of true (in $\mathcal{M}$) propositions are true (in $\mathcal{M}$) for any universe $\mathcal{M}$, and, of course, it holds for arbitrary models of arbitrary axiomatic theories, not just set universes.

B.5. Principle of Soundness of Logical Inference. *If a proposition θ is a theorem of an axiomatic system $\mathbf{T}$ (i.e., it can be proved by logic alone from the axioms of $\mathbf{T}$), then every universe of $\mathbf{T}$ satisfies θ .*

B.6. Universes vs. general models. According to the discussion in 8.22, to define a model of an axiomatic set theory we must specify a domain of objects,

³³To those who know formal logic, it might appear that the completely precise (syntactical) relativization operation on *formulas* is much easier to understand than this relativization operation, which is applied directly to propositions of “mathematical English”. But to apply the formal relativization process, we must first “formalize” the ordinary language propositions in which we are ultimately interested, and a moment’s thought will convince you that the formalization process is *exactly* as “vague” as the present operation of relativization. It may be argued that we know how to formalize a certain proposition precisely if we can interpret it in some arbitrary structure, i.e., precisely if we can understand its informal relativizations.

define on it the conditions of membership and sethood and also specify which conditions and operations on the domain will be considered definite. Set universes are very special models in two ways.

(1) When we view a set universe $\mathcal{M}$ as a model for an axiomatic set theory, we take its *definite conditions* to be all the definite conditions of our basic domain $\mathcal{W}$, and we take for its *definite operations* all the $\mathcal{M}$ -operations, i.e., the definite operations of $\mathcal{W}$ which take $\mathcal{M}$ -objects to $\mathcal{M}$ -objects.³⁴ It is routine to verify that all the axioms for definite conditions and operations listed in 3.18 hold with this interpretation, Problem **xB.1**: thus, to prove that a set universe $\mathcal{M}$ is a model of (say) **ZFDC**, it is enough to prove the relativizations to $\mathcal{M}$ of axioms (I) – (VIII).

(2) Because we assume that the body $b_{\mathcal{M}}(x)$ of each $x \in M$ is a set, $\mathcal{M}$ -sets cannot be “larger” than the sets of the intended universe $\mathcal{W}$: for example, there cannot be an $\mathcal{M}$ -set x such that for all t , $t \in x$.

Natural universes are even more special, of course, as they only restrict the domain of objects to some transitive class—and this makes it especially simple to understand the meaning of the relativization operation for them. From the mathematical point of view, natural universes are *subuniverses* of $\mathcal{W}$ and they inherit their structure from $\mathcal{W}$, much like subgroups, subposets, topological subspaces and the like are specified by a subset of some given space and inherit the relevant structure from it. The additional subtlety here is that we need to interpret (relativize) in these subuniverses propositions which are logically quite complex, more complex than the typical identities or inclusions which come up in Algebra or Topology.

We begin with the verification of the axioms we have been studying in natural universes which we have already introduced, where the notions are most familiar.

B.7. Lemma. *Every transitive class M satisfies the Axiom of Extensionality.*

PROOF. The relativization to M of the Extensionality Axiom reads as follows:

(I)^(M) Extensionality: *For all sets $A, B \in M$,*

$$A = B \iff (\forall x \in M)[x \in A \iff x \in B].$$

If $A = B$, then A and B have the same members, so they certainly have the same members in M . To prove the converse, we must show that if $A \neq B$, then there must exist some $x \in M$, such that either $x \in A \setminus B$ or $x \in B \setminus A$; but if $A \neq B$, then there certainly exists some

$$x \in (A \setminus B) \cup (B \setminus A) \subseteq A \cup B,$$

$A, B \subseteq M$ because M is transitive, and hence this x is also in M . ←

³⁴More pedantically, the definite conditions of $\mathcal{M}$ are the restrictions to M of the definite conditions in the intended universe $\mathcal{W}$, and similarly for the operations.

B.8. Lemma. *Every Zermelo universe M is a natural universe of axioms (I) – (VI).*

PROOF. We consider in turn each of the axioms (I) – (VI) other than (I) just shown and (IV) for which we gave the argument in the example above. The reader is advised to compare the relativization of each axiom to M which we must prove with the original statement of the axiom in Chapter 3.

(II)^(M) Emptyset and Pairset: (a) *There is a special object $y \in M$ which is a set but has no members in M .* (b) *For all $x, y \in M$, there is a set $z \in M$ such that*

$$(\forall t \in M)[t \in z \iff [t = x \vee t = y]]. \quad (\text{B-9})$$

(a) $\emptyset \in M$ by hypothesis, it has no members whatsoever, so it certainly has no members in M . (b) If $x, y \in M$, then $z = \{x, y\} \in M$ by hypothesis, and it obviously satisfies (B-9), since it satisfies the stronger

$$(\forall t)[t \in z \iff [t = x \vee t = y]].$$

(III)^(M) Separation Axiom: *For each set $A \in M$ and each unary, definite condition P , there exists a set $B \in M$ which satisfies the equivalence*

$$(\forall x \in M)[x \in B \iff x \in A \ \& \ P(x)]. \quad (\text{B-10})$$

Suppose $A \in M$. By the Axiom of Separation, there exists some B which satisfies

$$(\forall x)[x \in B \iff x \in A \ \& \ P(x)]. \quad (\text{B-11})$$

Now $B \subseteq A$, so $B \in \mathcal{P}(A) \in M$, so $B \in M$ because M is transitive, and (B-11) implies the weaker (B-10).

(V)^(M) Unionset Axiom: *For each set $\mathcal{E} \in M$, there exists a set $B \in M$ which satisfies the equivalence*

$$(\forall t \in M)[t \in B \iff (\exists X \in M)[X \in \mathcal{E} \ \& \ t \in X]]. \quad (\text{B-12})$$

Again, we naturally take $B = \bigcup \mathcal{E}$, which is in M by hypothesis and satisfies the equivalence

$$(\forall t)[t \in B \iff (\exists X)[X \in \mathcal{E} \ \& \ t \in X]]. \quad (\text{B-13})$$

Using once more the transitivity of M , immediately, for every t ,

$$(\exists X)[X \in \mathcal{E} \ \& \ t \in X] \iff (\exists X \in M)[X \in \mathcal{E} \ \& \ t \in X],$$

so that (B-12) reduces to

$$(\forall t \in M)[t \in B \iff (\exists X)[X \in \mathcal{E} \ \& \ t \in X]],$$

which is implied by the stronger (B-13).

(VI)^(M) Axiom of Infinity: *There exists a set $I \in M$ such that*

$$\emptyset \in I \ \& \ (\forall x \in M)[x \in I \implies \{x\} \in I].$$

This is quite simple, taking $I = \mathbb{N}_0 \in M$. ⊣

B.9. Theorem. (1) Every Zermelo universe is a natural universe of **ZDC**, and every **ZFDC**-universe is a natural universe of **ZFDC**.

(2) **(AC)** Every Zermelo universe is a natural universe of **ZDC+AC**, and every **ZFDC**-universe is a natural universe of **ZFDC+AC**.

In particular, $\mathcal{Z}$ and every $M(I)$ such that $\mathbb{N}_0 \subseteq I$ are natural universes of **ZDC**, or universes of **ZAC**, granting **AC**.

(3) (von Neumann) **(AC)** The von Neumann universe $\mathcal{V}$ is a natural universe of **ZFC**.

PROOF. The relativizations of **DC** and **AC** were proved in **11.23** and the relativization of the Axiom of Replacement is exactly the defining condition of a **ZFDC**-universe in **11.33**. For Part (3), we have already shown that $\mathcal{V}$ is a **ZFDC**-universe in **11.34**, so it only remains to prove the relativizations to $\mathcal{V}$ of the Principles of Purity **3.25** and Foundation **11.29**. The first of these is

Purity^($\mathcal{V}$): For every $x \in \mathcal{V}$, $\text{Set}(x)$,

and it is true simply because every object in $\mathcal{V}$ is a set, and the interpretation of “sethood” in $\mathcal{V}$ is the standard one. For the second, it is easiest to relativize the elementary version of the Foundation Principle in **11.30**.

Foundation^($\mathcal{V}$): For every set $X \in \mathcal{V}$, there exists some $m \in X$ such that $m \cap X$ is empty in $\mathcal{V}$, i.e.,

$$(\forall t \in \mathcal{V})[t \notin m \vee t \notin X].$$

The negation of this would give us an $X \in \mathcal{V}$ and some $a \in X$ such that

$$(\forall m \in X)(\exists t \in X)[t \in m],$$

which by **DC** (as in the proof of **11.30**) implies that X is ill founded, contradicting the assumption $X \in \mathcal{V}$. $\dashv$

The Soundness of Logical Inference **B.5** combines well with **B.9** to yield many simple but interesting independence results about **ZDC** and **ZDC+AC**: to prove that a proposition θ cannot be a theorem of **ZDC**, it is enough to find some $I \supseteq \mathbb{N}_0$ such that $\theta^{(M(I))}$ is false. This is exactly the way we proved **B.2**, a bit clumsily without the precise notions. We have included in the problems several examples of this kind.

By the same reasoning, Part (3) of Theorem **B.9** implies that *we cannot refute in **ZFAC** the Principles of Purity or Foundation*, because $\mathcal{V}$ is a model of **ZFAC** which satisfies these principles. It should also be obvious that we cannot prove these principles in **ZFAC**, but to establish this rigorously we need to construct universes of **ZFAC** which are not natural. The basic tool for such constructions is the next simple notion.

B.10. Definition. A **Rieger universe** is any set universe $\mathcal{M} = M, S, E$ such that for every set $Y \subseteq M$, there exists exactly one $\mathcal{M}$ -set X satisfying $Y = \mathbf{b}_{\mathcal{M}}(X)$. For each $Y \subseteq M$ we set

$$\rho_{\mathcal{M}}(Y) =_{\text{df}} \text{the unique } X \in S \text{ such that } \mathbf{b}_{\mathcal{M}}(X) = Y, \quad (\text{B-14})$$

so that $\rho_{\mathcal{M}}$ is a definite operation and immediately from its definition, for every $Y \subseteq M$,

$$X = \rho_{\mathcal{M}}(Y) \iff X \in S \& \mathbf{b}_{\mathcal{M}}(X) = Y, \quad (\text{B-15})$$

$$t E \rho_{\mathcal{M}}(Y) \iff t \in Y. \quad (\text{B-16})$$

B.11. Rieger's Theorem. *Every Rieger universe is a universe of ZFDC.*

PROOF. Fix $\mathcal{M} = M, S, E$ with the Rieger property and let $\mathbf{b}(x) = \mathbf{b}_{\mathcal{M}}(x)$, skipping the subscript since $\mathcal{M}$ is the only universe around. We verify in turn the relativizations of all the axioms of ZFDC.

(I)^(M) Extensionality Axiom: *For all $A, B \in M$, if $S(A)$ and $S(B)$, then*

$$A = B \iff (\forall x \in M)[x E A \iff x E B].$$

If $A = B$, then surely, for all x , $x E A \iff x E B$. Conversely, if for all $x \in M$, $x E A \iff x E B$, then $\mathbf{b}(A) = \mathbf{b}(B)$; by the Rieger property, there is exactly one $C \in S$ such that $\mathbf{b}(A) = \mathbf{b}(C)$; so we must have $C = A$, and similarly $C = B$, hence $A = B$.

(II)^(M) Emptyset and Pairset: (a) *There is a special object $y \in S$ such that $(\forall t \in M) \neg t E y$.* (b) *For all $x, y \in M$, there is some $z \in S$ such that*

$$(\forall t \in M)[t E z \iff t = x \vee t = y].$$

For (a), choose y so that $y \in S$ and $\mathbf{b}(y) = \emptyset$, and for (b) choose $z \in S$ so that $\mathbf{b}(z) = \{x, y\}$, both times by applying directly the Rieger property. In the case of (b), for example, we compute:

$$t E z \iff t \in \mathbf{b}(z) \iff [t = x \vee t = y],$$

which is the required conclusion.

(IV)^(M) Powerset Axiom: *For each set $A \in M$, there exists some $B \in S$, such that for every $X \in M$,*

$$X E B \iff X \in S \& (\forall t \in M)[t E X \implies t E A].$$

Given $A \in M$, choose $B \in S$ by the Rieger property so that

$$\mathbf{b}(B) = \{\rho(Y) \mid Y \subseteq \mathbf{b}(A)\}, \quad (\text{B-17})$$

where $\rho(Y) = \rho_{\mathcal{M}}(Y)$ is the Rieger operation associated with $\mathcal{M}$ by (B-14), and compute:

$$\begin{aligned} X E B &\iff X \in \mathbf{b}(B) \\ &\iff (\exists Y)[Y \subseteq \mathbf{b}(A) \& X = \rho(Y)] \\ &\iff (\exists Y)[Y \subseteq \mathbf{b}(A) \& [X \in S \& \mathbf{b}(X) = Y]] \quad \text{by (B-15)} \\ &\iff X \in S \& \mathbf{b}(X) \subseteq \mathbf{b}(A) \\ &\iff X \in S \& (\forall t \in M)[t E X \implies t E A]. \end{aligned}$$

Verifications of the remaining axioms in $\mathcal{M}$ are similar, following the same ideas as in the proof of Theorem B.9, and we leave them for the exercises. $\dashv$

B.12. Exercise. *Prove that a set universe $\mathcal{M} = M, S, E$ is a Rieger universe if and only if (1) it satisfies the Axiom of Extensionality, and (2) for every $Y \subseteq M$, there exists some $X \in S$ such that $\mathbf{b}_{\mathcal{M}}(X) = Y$.*

B.13. Exercise. *Every Rieger universe is a model of the Axioms of Separation (III) and Replacement (VIII).*

B.14. Exercise. (AC) *Every Rieger universe is a model of ZFAC.*

B.15. Relativization of “faithfully modeled” notions. The Choice Principles DC and AC are formulated in terms of the notions of “function” which was defined in Chapter 4 using an “arbitrary but fixed” ordered pair operation 4.1, and “system of natural numbers”, which was also “arbitrary but fixed” in 5.9. It is not completely obvious how to relativize propositions involving such “faithfully modeled” notions, since (for example) a given universe $\mathcal{M}$ may not be closed under the chosen ordered pair operation. We avoided the problem in 11.23 and B.9 by assuming that the fixed ordered pair is the Kuratowski pair under which every universe satisfying (I) – (VI) is closed, but there may be some lingering vagueness on how to deal with this problem in general. There is an easy solution for Rieger universes, which we outline in Problems xB.13 and xB.14. In discussing Rieger universes from now on, we will assume tacitly that we have fixed an ordered pair operation, a system of natural numbers, etc. for each of them, and we will relativize propositions which involve these faithfully modeled notions in terms of these fixed operations. Problems xB.13 and xB.14 make it clear that which particular definitions are chosen is irrelevant for the results.

B.16. Proposition. *There exists a Rieger universe $\mathcal{M}_{\text{at}}$ in which every set is equinumerous with a set of atoms; in particular, we cannot prove in ZFDC that every set is pure.*

PROOF. The idea is to code every set A by the pair $(0, A)$ in $\mathcal{M}_{\text{at}}$, and to declare that every object which does not code a set in this way is an atom. We set

$$\begin{aligned} x \in M_{\text{at}} &\iff_{\text{df}} x = x, \text{ so } M_{\text{at}} = \mathcal{W}, \\ x \in S_{\text{at}} &\iff_{\text{df}} (\exists A)[\text{Set}(A) \ \& \ x = (0, A)], \\ x E_{\text{at}}(0, A) &\iff_{\text{df}} x \in A, \end{aligned}$$

and proceed to verify that $\mathcal{M}_{\text{at}} = M_{\text{at}}, S_{\text{at}}, E_{\text{at}}$ is a set universe, that it has the Rieger property and that it satisfies the proposition “every set is equinumerous with a set of atoms”.

Skipping the subscript $\mathcal{M}_{\text{at}}$, clearly

$$\mathbf{b}(x) = \begin{cases} \emptyset, & \text{if for all sets } A, \ x \neq (0, A), \\ A, & \text{if for some (necessarily unique) set } A, \ x = (0, A), \end{cases}$$

so each $\mathbf{b}(x)$ is a set. For the Rieger property, we notice first that for each set $A \subseteq M_{\text{at}}$, $(0, A) \in S_{\text{at}}$ and $\mathbf{b}((0, A)) = A$; if $x \in S$ and $\mathbf{b}(x) = A$, then

$x = (0, B)$ and $\mathbf{b}(x) = B$ for some set B , by the definition, so we have $A = B$ and $x = (0, A)$, as required. The Rieger operation is very simple in this case,

$$\rho(Y) = (0, Y).$$

The relativization to $\mathcal{M}_{\text{at}}$ of the proposition “every set is equinumerous with a set of atoms” is the assertion that *for every $\mathcal{M}_{\text{at}}$ -set $(0, A)$, there is a bijection in $\mathcal{M}_{\text{at}}$ between $(0, A)$ and some $\mathcal{M}_{\text{at}}$ -set of atoms $(0, B)$* . We take

$$B =_{\text{df}} \{(1, t) \mid t \in A\};$$

now $(0, B) \in S_{\text{at}}$ and, by the definition,

$$x E_{\text{at}}(0, B) \implies x \in B \implies (\forall y)[x \neq (0, y)] \implies x \notin S_{\text{at}},$$

i.e., each $\mathcal{M}_{\text{at}}$ -member of $(0, B)$ is an atom in $\mathcal{M}_{\text{at}}$. If $(x, y)_{\text{at}}$ is the ordered pair operation of $\mathcal{M}_{\text{at}}$ and

$$f =_{\text{df}} \rho\{(t, (1, t))_{\text{at}} \mid t \in A\},$$

then $\mathcal{M}_{\text{at}}$ recognizes f as a bijection between the $\mathcal{M}_{\text{at}}$ -sets $(0, A)$ and $(0, B)$. $\dashv$

By the anthropomorphic “ $\mathcal{M}_{\text{at}}$ recognizes $f \dots$ ” we simply mean that if $A' = (0, A)$ and $B' = (0, B)$ are the objects in $\mathcal{M}_{\text{at}}$ which code the sets A and B , respectively, and if we set

$$\begin{aligned} \theta \iff_{\text{df}} f \subseteq A \times B \ \& \ (\forall x \in A')(\exists y \in B')(x, y) \in f \\ \& \ (\forall y \in B')(\exists! x \in A')[(x, y) \in f], \end{aligned}$$

then the relativization $\theta^{(\mathcal{M}_{\text{at}})}$ of the proposition θ to $\mathcal{M}_{\text{at}}$ is true. This relativization can be computed in principle, but it is quite messy. It is best to develop a machinery for arguing about relativizations without actually writing them out, and for this the following, traditional “model theoretic” notation is very useful. For each set universe $\mathcal{M}$ and each proposition θ ,

$$\mathcal{M} \models \theta \iff_{\text{df}} \theta \text{ is true in } \mathcal{M} \iff \theta^{(\mathcal{M})}. \quad (\text{B-18})$$

We read $\mathcal{M} \models \theta$ “ $\mathcal{M}$ models θ ”, but also “ $\mathcal{M}$ thinks that θ ”, “ $\mathcal{M}$ believes that θ ”, etc. as befits the occasion. For example, for each pair of classes M, S and each binary condition E , let

$$\begin{aligned} \text{Setuniv}(M, S, E) \iff_{\text{df}} (\forall u, v)[u E v \implies u, v \in M] \\ \& \ (\forall t)[t \in S \implies t \in M] \\ \& \ (\forall x)[x \in M \implies \\ (\exists X)[\text{Set}(X) \ \& \ (\forall t)[t \in X \iff t E x]] \end{aligned}$$

be the fairly complex proposition which asserts that M, S, E comprise a set universe. Consider also

$$\begin{aligned} \text{Rieger}(M, S, E) \iff_{\text{df}} (\forall Y)[(\text{Set}(Y) \ \& \ Y \subseteq M) \\ \implies (\exists! X \in S)[Y = \mathbf{b}_{\mathcal{M}}(X)]] \\ \iff (\forall Y)[(\text{Set}(Y) \ \& \ Y \subseteq M) \\ \implies (\exists! X \in S)(\forall t \in M)[t \in Y \iff t E X]], \end{aligned}$$

which asserts that $\mathcal{M}$ is a Rieger universe. These are propositions about M , S and E , which may be true or false; whether they are true or not, it makes sense to interpret them (relativize them) in some universe $\mathcal{M}'$ and ask if $\mathcal{M}'$ *thinks that* M , S , E comprise a Rieger universe! There is an obvious question here, which has a simple and useful answer.

B.17. Theorem. *Suppose $\mathcal{M}_1 = M_1, S_1, E_1$ is a Rieger universe, $S_2 \subseteq M_2 \subseteq M_1$ are classes and E_2 is a binary condition such that $x E_2 y \implies x, y \in M_2$; if*

$$\mathcal{M}_1 \models \left[\text{Setuniv}(M_2, S_2, E_2) \& \text{Rieger}(M_2, S_2, E_2) \right],$$

then $\mathcal{M}_2 = M_2, S_2, E_2$ is also a Rieger universe.

PROOF. To show first that $\mathcal{M}_2$ is a set universe, we must verify that for every $x \in M_2$,

$$(\exists Y \in \text{Set})(\forall t)[t \in Y \iff t \in M_2 \& t E_2 x]. \quad (\text{B-19})$$

Fix some $x \in M_2$. The proposition (B-19) is true in $\mathcal{M}_1$ since

$$\mathcal{M}_1 \models \text{Setuniv}(M_2, S_2, E_2),$$

which means that some $Y_1 \in S_1$ exists such that for all $t \in M_1$,

$$t E_1 Y_1 \iff t \in M_2 \& t E_2 x; \quad (\text{B-20})$$

and since $\mathcal{M}_1$ is a set universe, there exists some set Y such that for all t ,

$$t \in Y \iff t \in M_1 \& t E_1 Y_1; \quad (\text{B-21})$$

now (B-20) and (B-21) together imply what (B-19) demands for the given $x \in M_2$ and some Y .

To show that $\mathcal{M}_2$ is a Rieger universe, we must show first that if $Y \subseteq M_2$, then there exists some $X_2 \in S_2$ such that for every $t \in M_2$,

$$t \in Y \iff t E_2 X_2, \quad (\text{B-22})$$

and then verify that this X_2 is unique. Since $Y \subseteq M_2 \subseteq M_1$ and $\mathcal{M}_1$ is a Rieger universe, there exists some $X_1 \in S_1$ such that for all $t \in M_1$,

$$t \in Y \iff t E_1 X_1; \quad (\text{B-23})$$

working in $\mathcal{M}_1$, we apply the Rieger property for $\mathcal{M}_2$ to the $\mathcal{M}_1$ -set X_1 , to get some $X_2 \in S_2$ such that

$$\mathcal{M}_1 \models (\forall t)[t \in M_2 \implies [t \in X_1 \iff t E_2 X_2]]; \quad (\text{B-24})$$

and computing the relativization, this means that

$$(\forall t \in M_1)[t \in M_2 \implies [t E_1 X_1 \iff t E_2 X_2]]. \quad (\text{B-25})$$

Compute now, for any $t \in M_2 \subseteq M_1$:

$$\begin{aligned} t \in Y &\iff t E_1 X_1 && \text{by (B-23),} \\ &\iff t E_2 X_2 && \text{by (B-25),} \end{aligned}$$

which proves (B-22). The fact that at most one $X_2 \in S_2$ can satisfy (B-22) for every $t \in M_2$ is proved similarly. $\dashv$

REMARK: Notice that in relativizing (B-19) in this proof, we left the clause $t \in M_2$ alone. In computing relativizations, “primitive” propositions of the form $P(x_1, \dots, x_n)$ (which express that a definite condition P holds of the objects $x_1, \dots, x_n$) are their own relativizations.

This simple theorem makes it possible to construct universes within universes within universes, each time using the properties of the model just constructed. Consider, for example, the next Corollary of **B.16**.

B.18. Proposition. *There exists a ZFDC-universe $\mathcal{M}$ which has exactly one atom.*

PROOF. Suppose c is an atom and let

$$M_c =_{\text{df}} \{x \mid (\forall t \in \text{TC}(x))[\neg \text{Set}(t) \implies t = c]\} \quad (\text{B-26})$$

be the class of objects *supported by* $\{c\}$ in the sense of Problem **11.38**. It is quite easy to verify that M_c is transitive and for each set X ,

$$X \subseteq M_c \implies X \in M_c,$$

so the natural universe M_c has the Rieger property and by **B.11** it is a model of **ZFDC**; and it is quite clear that it has exactly one atom, c .

So far so good, as long as there exists at least one atom, which may or may not be true in our intended domain $\mathcal{W}$. But there are lots of atoms in the Rieger universe $\mathcal{M}_{\text{at}}$ of **B.16**, so what we need to do is *to interpret the proof of the preceding paragraph in the universe $\mathcal{M}_{\text{at}}$* . This argument runs as follows.

Let

$$\begin{aligned} \phi(M, S, E, c) \iff_{\text{df}} & \text{Setuniv}(M, S, E) \ \& \ \text{Rieger}(M, S, E) \\ & \& \ \text{Atom}(c) \ \& \ (\forall t \in M)[t \notin S \implies t = c] \end{aligned} \quad (\text{B-27})$$

be the proposition which asserts of M, S, E, c that they have the properties we are interested in, and let

$$\theta \iff_{\text{df}} (\exists M)(\exists S)(\exists E)(\exists c)\phi(M, S, E, c) \quad (\text{B-28})$$

be the proposition which asserts that some M, S, E, c with these properties exist. We have proved θ from the hypothesis that some atom exists, and other than that we have only used the axioms of **ZFDC**—what else is there! Hence this θ is true in every universe of **ZFDC** which has an atom, in particular $\mathcal{M}_{\text{at}}$, i.e.,

$$\mathcal{M}_{\text{at}} \models \theta.$$

This means that for some $\mathcal{M}_{\text{at}}$ -classes M, S , some binary $\mathcal{M}_{\text{at}}$ -condition E and some $\mathcal{M}_{\text{at}}$ -object c ,

$$\mathcal{M}_{\text{at}} \models \phi(M, S, E, c),$$

and in particular

$$\mathcal{M}_{\text{at}} \models \left[\text{Setuniv}(M, S, E) \ \& \ \text{Rieger}(M, S, E) \right],$$

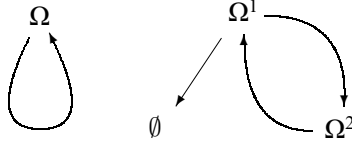


FIGURE B.1. Two decorated, ill-founded graphs.

so by **B.17**, $\mathcal{M} = M, S, E$ is a Rieger universe. In addition

$$\mathcal{M}_{\text{at}} \models [\text{Atom}(c) \ \& \ (\forall t \in M)[t \notin S \implies t = c]],$$

which means precisely that

$$\mathcal{M} \models \text{Atom}(c) \ \& \ (\forall t)[\neg \text{Set}(t) \implies t = c],$$

the required conclusion that $\mathcal{M}$ believes that exactly one atom exists. $\dashv$

It is not hard to manufacture Rieger universes with various types of ill founded sets, by a combination of the techniques in **B.16** and **B.17**. Some of the problems are about such results. Here we will concentrate on the construction of Aczel’s *Antifounded Universe* $\mathcal{A}$, which has a rich variety of ill founded sets with well understood structure.

The idea for $\mathcal{A}$ comes from the Mostowski Collapsing Lemma **11.36**, which gives a “structural” characterization of pure, grounded sets. Recall that by **11.35**, a decoration of a graph G is any surjection $d : G \rightarrow d[G]$ such that

$$d(x) = \{d(y) \mid y \leftarrow x\} \quad (x \in G), \quad (\text{B-29})$$

where $\rightarrow$ is the edge relation on G and $\leftarrow$ is its inverse,

$$y \leftarrow x \iff y \text{ is a child of } x \iff x \rightarrow y.$$

Each grounded graph G admits a unique decoration d_G , and the pure, grounded sets are all the values $d_G(x)$ of these decorations. Can we also “decorate” the nodes of ill founded graphs to get pure, ill founded sets which are related to ill founded graphs in the same way that pure, grounded sets are related to grounded graphs?

B.19. Antifoundation Principle, AFA. *Every graph admits a unique decoration.*

In Figure **B.1** we have labeled the nodes of two ill founded graphs by the values of their unique decorations, assuming that such exist. By the definition of decoration,

$$\Omega = \{\Omega\}, \quad \Omega^1 = \{\emptyset, \Omega^2\}, \quad \Omega^2 = \{\Omega^1\}, \quad (\text{B-30})$$

i.e., Ω , Ω^1 and Ω^2 are the “ultimately frustrating gifts” we discussed in **11.32**. We can refer to “the” frustrating gifts, because, in fact, the equations in (B-30)—or the graphs in Figure **B.1**—characterize these sets under **AFA**, as follows.

B.20. Proposition. (AFA) (1) *There is exactly one set Ω which is its own singleton.* (2) *There is exactly one pair of sets Ω^1, Ω^2 such that $\Omega^1 = \{\emptyset, \Omega^2\}$ and $\Omega^2 = \{\Omega^1\}$.*

PROOF. (1) If $X = \{X\}$ and $Y = \{Y\}$, then we can use either X or Y to decorate the single node graph in Figure B.1; but this graph has only one decoration by AFA, so $X = Y$. The proof of (2) is similar. $\dashv$

This “uniqueness” part of the Antifoundation Principle we just applied makes it possible to specify and analyze the structure of ill founded sets with diverse properties, and is the main advantage of the antifounded universe $\mathcal{A}$ over other models which contain ill founded sets. We now proceed to its construction.

B.21. Definition. A **pointed graph** is a pair (G, p_G) of a graph and a node in it, in full detail, a structured set $(G, \rightarrow_G, p_G)$ where $p_G \in G$ and $\rightarrow_G$ is a binary relation on the field G . The designated node p_G is the **point** of the pointed graph.

B.22. Pictures. A pointed graph (G, p) is a **picture** of a set A , if there exists a decoration $d : G \rightarrow d[G]$ of G such that $d_G(p) = A$. The **canonical picture** of a pure set A is the pointed graph $(\text{TC}(A), \ni, A)$, where $\text{TC}(A)$ is the transitive closure of A and $\ni$ is the restriction of the inverse membership condition to $\text{TC}(A)$. This is a picture of A , because the identity function $d(x) = x$ is obviously a decoration of it, $A \in \text{TC}(A)$ and $d(A) = A$.

B.23. A bijection $\pi : G \rightarrowtail H$ between two graphs is an **isomorphism** if it respects the edge relations,

$$x \rightarrow_G y \iff \pi(x) \rightarrow_H \pi(y) \quad (x, y \in G);$$

and an **isomorphism between two pointed graphs** (G, p) and (H, q) is a graph isomorphism $\pi : G \rightarrow H$ such that $\pi(p) = q$. We call G **isomorphic** with H if there exists an isomorphism $\pi : G \rightarrowtail H$ of the appropriate kind.

It is easy to construct non-isomorphic pointed graphs which picture the same set, even grounded ones, e.g., see Figure B.2 where we have labelled the nodes with the values of the unique decorations. On the other hand, we would expect that if a pointed graph (G, p) admits a unique decoration d_G , then the set $A = d_G[G]$ captures some important invariant of (G, p) . The next fundamental definition identifies that invariant.

B.24. Definition. A relation $R \subseteq G \times H$ is a **bisimulation** between two pointed graphs $(G, \rightarrow_G, p_G)$ and $(H, \rightarrow_H, p_H)$, if it relates the points, i.e., $p_G R p_H$ and also satisfies the implication

$$\begin{aligned} x R y \implies & (\forall u \leftarrow_G x)(\exists v \leftarrow_H y)u R v \\ & \& (\forall v \leftarrow_H y)(\exists u \leftarrow_G x)u R v. \end{aligned} \tag{B-31}$$

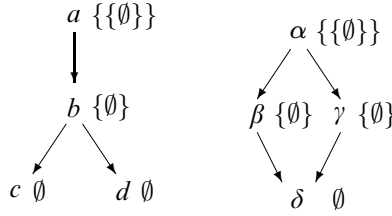


FIGURE B.2. Non-isomorphic, bisimilar, grounded graphs.

Two pointed graphs G , H are **bisimilar** if some bisimulation between them exists,

$$G =_{\text{bs}} H \iff_{\text{df}} (\exists R \subseteq G \times H)[R \text{ is a bisimulation}]. \quad (\text{B-32})$$

As usual with structured sets, we will often refer to “a pointed graph G ”, skipping the explicit reference to the edge relation $\rightarrow_G$ or the point when it is obvious or irrelevant—we already did this in (B-32).

B.25. Exercise. *Isomorphic pointed graphs are bisimilar, and so are the non-isomorphic, grounded, pointed graphs G and H in Figure B.2.*

B.26. Exercise. *If a is a minimal node in a graph G and b is a minimal node in H , then $\{(a, b)\}$ is a bisimulation of the pointed graphs (G, a) and (H, b) .*

B.27. Exercise. *Let L be the “single loop” graph on a singleton $\{a\}$, with the one edge pair (a, a) , and on the set of integers $\mathbb{N}$ define the successor edge relation*

$$n \rightarrow_s m \iff_{\text{df}} n + 1 = m.$$

Show that the relation $\{(a, i) \mid i \in \mathbb{N}\}$ is a bisimulation of (L, a) with $(\mathbb{N}, n)$, for every n .

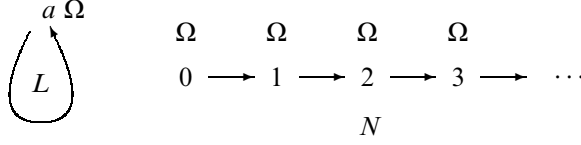
B.28. Lemma. *The condition $=_{\text{bs}}$ is an equivalence condition on the class of all pointed graphs.*

PROOF. Of the three properties of an equivalence condition (defined in 12.42), only the transitivity of $=_{\text{bs}}$ is not immediate. To prove that, suppose G_1 , G_2 and G_3 are pointed graphs, R_1 is a bisimulation of G_1 with G_2 and R_2 is a bisimulation of G_2 with G_3 , and let

$$x R z \iff_{\text{df}} (\exists y)[x R_1 y \ \& \ y R_2 z] \quad (\text{B-33})$$

be the “product relation” of R_1 and R_2 . It is clear that R relates the points p_1 and p_3 of G_1 and G_3 , because $p_1 R_1 p_2$ and $p_2 R_2 p_3$ hold. Suppose that $x R z$, so there exists some $y \in G_2$ such that $x R_1 y$ and $y R_2 z$. If $u \leftarrow_1 x$, then by (B-31) for R_1 , there exists some $v \leftarrow_2 y$ such that $u R_1 v$; and then by (B-31) for R_2 , there exists some $w \leftarrow_3 z$ such that $v R_2 w$, which together with $u R_1 v$ establish $u R w$. This is half of (B-31), and the other half is equally easy. $\dashv$

With these definitions, we can now prove that for a grounded graph G and any $p \in G$, the properties of (G, p) coded by the value $d_G(p)$ of its unique decoration are exactly those preserved under bisimulation of pointed graphs.

FIGURE B.3. Many pictures of $\Omega : (L, a)$ and each $(\mathbb{N}, n)$.

B.29. Theorem. For all grounded graphs G and H with associated decorations d_G and d_H , and for all $p \in G$ and $q \in H$,

$$d_G(p) = d_H(q) \iff (G, p) =_{\text{bs}} (H, q). \quad (\text{B-34})$$

PROOF. We verify first that the relation

$$R = \{(x, y) \in G \times H \mid d_G(x) = d_H(y)\} \quad (\text{B-35})$$

satisfies (B-31), as follows:

$$\begin{aligned} x R y &\implies \{d_G(u) \mid u \leftarrow x\} = \{d_H(v) \mid v \leftarrow y\} \text{ by the def. of decoration} \\ &\implies (\forall u \leftarrow x)(\exists v \leftarrow y)[d_G(u) = d_H(v)] \\ &\quad \& (\forall v \leftarrow y)(\exists u \leftarrow x)[d_G(u) = d_H(v)] \\ &\implies (\forall u \leftarrow x)(\exists v \leftarrow y)u R v \& (\forall v \leftarrow y)(\exists u \leftarrow x)u R v. \end{aligned}$$

Hence, if $p \in G$, $q \in H$, and $d_G(p) = d_H(q)$, then the relation R of (B-35) establishes that $(G, p) =_{\text{bs}} (H, q)$.

For the converse, suppose towards a contradiction that p is minimal in G such that there exists some $q \in H$ and a bisimulation R of the pointed graphs (G, p) and (H, q) , but $d_G(p) \neq d_H(q)$. Now

$$(\forall u \leftarrow p)(\exists v \leftarrow q)u R v,$$

hence,

$$(\forall u \leftarrow p)(\exists v \leftarrow q)[d_G(u) = d_H(v)]$$

by the choice of p , and, similarly, $(\forall v \leftarrow q)(\exists u \leftarrow p)[d_G(u) = d_H(v)]$, which proves $d_G(p) = d_H(q)$, contradicting the choice of p . $\dashv$

It is a crucial property of **AFA** that it yields the same characterization of bisimulation for all graphs, by quite a different argument.

B.30. Theorem (Aczel). (**AFA**) For all graphs G and H with associated decorations d_G and d_H , and for all $p \in G$ and $q \in H$,

$$d_G(p) = d_H(q) \iff (G, p) =_{\text{bs}} (H, q). \quad (\text{B-36})$$

PROOF. The left-to-right implication in (B-36) is proved exactly as in B.29, as that part of the argument did not depend on the given graphs being grounded.

Suppose now that the edge relations of G and H are $\rightarrow_G$ and $\rightarrow_H$ and $R \subseteq G \times H$ is a bisimulation of G with H . We can turn R into a pointed graph, with point the pair (p_G, p_H) and edge relation the product of $\rightarrow_G$ and $\rightarrow_H$:

$$(p, q) \rightarrow (u, v) \iff_{\text{df}} p \rightarrow_G u \& q \rightarrow_H v.$$

If d_G is the unique decoration of the graph G (forgetting the point) given by **AFA**, define on R the function

$$d_G^R(p, q) =_{\text{df}} d_G(p),$$

and compute:

$$\begin{aligned} x \in d_G^R(p, q) &\iff x \in d_G(p) \\ &\iff (\exists u \leftarrow p)[x = d_G(u)] \\ &\iff (\exists u \leftarrow p)(\exists v \leftarrow q)[u R v \ \& \ x = d_G(u)] \quad (\text{B-37}) \\ &\iff (\exists (u, v) \leftarrow (p, q))[x = d_G^R(u, v)], \end{aligned}$$

where the key equivalence (B-37) holds because R is a bisimulation and $p R q$, and hence for each $u \leftarrow p$, there exists some $v \leftarrow q$ satisfying $u R v$. Thus, the function d_G^R is a decoration of R , and the corresponding extension

$$d_H^R(p, q) =_{\text{df}} d_H(q)$$

of the decoration d_H of H is also a decoration of R , by the same argument. By **AFA** then, for all $(p, q) \in R$,

$$d_G(p) = d_G^R(p, q) = d_H^R(p, q) = d_H(q),$$

which completes the proof. $\dashv$

This characterization under **AFA** of the properties of (G, p) which are coded into the value $d_G(p)$ of its unique decoration, suggests a method for the construction of $\mathcal{A}$.

B.31. The Antifounded Universe. Let

$$\mathcal{A}_0 =_{\text{df}} \{(G, \rightarrow_G, p_G) \in \mathcal{V} \mid \rightarrow_G \subseteq G \times G \ \& \ p_G \in G\} \quad (\text{B-38})$$

be the class of all pointed graphs on pure, grounded sets, and on $\mathcal{A}_0$ define the binary definite condition

$$\begin{aligned} (G, p_G) \varepsilon_0 (H, p_H) \\ \iff_{\text{df}} (\exists q \in H)[q \leftarrow p_H \ \& \ (G, p_G) =_{\text{bs}} (H, q)], \quad (\text{B-39}) \end{aligned}$$

skipping the edge relations in the notation.

First we note that ε_0 respects bisimulation:

$$G_1 \varepsilon_0 H_1 \ \& \ G_1 =_{\text{bs}} G_2 \ \& \ H_1 =_{\text{bs}} H_2 \implies G_2 \varepsilon_0 H_2. \quad (\text{B-40})$$

To prove this, suppose $\rightarrow_1, p_1$ are the edge relation and the designated node of H_1 , and similarly with $\rightarrow_2, p_2$ for H_2 . The hypothesis of (B-40) gives us some $q_1 \leftarrow p_1$ such that

$$G_2 =_{\text{bs}} G_1 =_{\text{bs}} (H_1, q_1), \quad (\text{B-41})$$

and a bisimulation R of (H_1, p_1) with (H_2, p_2) . By the basic property of bisimulations, there must exist some $q_2 \leftarrow H_2$ such that $q_1 R q_2$; this means that R is a bisimulation of (H_1, q_1) with (H_2, q_2) , and then (B-41) with the transitivity of $=_{\text{bs}}$ gives $G_2 =_{\text{bs}} (H_2, q_2)$, hence $G_2 \varepsilon_0 H_2$.

Now each $G \in \mathcal{A}_0$ is a pure, grounded *set* (a triple in $\mathcal{V}$), even if it is ill founded *as a graph*, and the bisimulation condition $=_{\text{bs}}$ is an equivalence condition on $\mathcal{A}_0$ by **B.28**. By Problem **x12.45**, there exists a definite operation α which is determining for $=_{\text{bs}}$, i.e., for $G, H \in \mathcal{A}_0$

$$G =_{\text{bs}} H \iff \alpha(G) = \alpha(H).$$

The domain of the antifounded universe is the quotient class of $\mathcal{A}_0$ by $=_{\text{bs}}$,

$$\mathcal{A} =_{\text{df}} \{\alpha(G) \mid G \in \mathcal{A}_0\}. \quad (\text{B-42})$$

We define on $\mathcal{A}$ the membership condition

$$x \varepsilon y \iff_{\text{df}} (\exists G, H)[x = \alpha(G) \ \& \ y = \alpha(H) \ \& \ G \varepsilon_0 H], \quad (\text{B-43})$$

unambiguously by (B-40), and finally we take the *Pure Antifounded Universe* to be the triple $\mathcal{A}, \mathcal{A}, \varepsilon$. We will refer to it by the name of its domain $\mathcal{A}$, which is also the collection of its sets—there are no atoms in $\mathcal{A}$.

B.32. Theorem (Aczel). (AC) $\mathcal{A}$ is a Rieger universe, which further satisfies the Antifoundation Principle **AFA**, the Axiom of Choice **AC** and the Principle of Purity.

PROOF. The key property of $\mathcal{A}$ is that for each graph $H \in \mathcal{V}$ with edge relation $\rightarrow_H$ and each node $p \in H$,

$$\mathbf{b}_{\mathcal{A}}(\alpha(H, p)) = \{\alpha(H, q) \mid q \leftarrow_H p\}, \quad (\text{B-44})$$

which follows from the following trivial computation:

$$\begin{aligned} x \in \mathbf{b}_{\mathcal{A}}(\alpha(H, p)) &\iff (\exists G \in \mathcal{A}_0)(\exists q \leftarrow_H p)[x = \alpha(G) \ \& \ G =_{\text{bs}} (H, q)] \\ &\iff (\exists q \leftarrow_H p)[x = \alpha(H, q)]. \end{aligned}$$

This implies, in particular, that each $\mathbf{b}_{\mathcal{A}}(x)$ is a set, so $\mathcal{A}$ is a set universe.

For the Rieger property, suppose $Y \subseteq \mathcal{A}$ and (using **AC**) choose for each $y \in Y$ a pointed graph $G_y \in \mathcal{A}_0$, such that (1) $\alpha(G_y) = y$. By replacing each G_y by an isomorphic copy if necessary, we can also ensure that (2) $y \neq z \implies G_y \cap G_z = \emptyset$, and (3) for all $y \in Y$, $\emptyset \notin G_y$. Let

$$\begin{aligned} H &=_{\text{df}} \bigcup \{G_y \mid y \in Y\} \cup \{\emptyset\}, \\ u \rightarrow_H v &\iff_{\text{df}} (\exists y \in Y)[u \rightarrow_y v \vee (u = \emptyset \ \& \ v = p_y)], \end{aligned}$$

where $\rightarrow_y$ and p_y are the edge relation and the point of G_y . The pointed graph H with edge relation $\rightarrow_H$ and point $\emptyset$ is obviously in $\mathcal{A}_0$, and for each $y \in Y$,

$$(H, p_y) =_{\text{bs}} G_y \quad (\text{B-45})$$

by the trivial (identity) bisimulation

$$\{(u, v) \in H \times G_y \mid u = v\};$$

thus, by (B-44), and the definition,

$$\begin{aligned} \mathbf{b}_{\mathcal{A}}(\alpha(H, \emptyset)) &= \{\alpha(H, q) \mid q \leftarrow_H \emptyset\} \\ &= \{\alpha(H, p_y) \mid y \in Y\} \\ &= \{\alpha(G_y) \mid y \in Y\} \quad \text{by (B-45)} \\ &= Y. \end{aligned}$$

To prove the uniqueness of $\alpha(H)$, suppose H' is any pointed graph in $\mathcal{A}_0$ with edge relation $\rightarrow'$ and point q' such that

$$G \varepsilon_0 H' \iff \alpha(G) \in Y.$$

By (B-44) again,

$$\begin{aligned} \mathbf{b}_{\mathcal{A}}(\alpha(H', q')) &= \{\alpha(H', q) \mid q \leftarrow_{H'} q'\} \\ &= \{\alpha(H, p_y) \mid y \in Y\} \quad \text{by hyp.} \end{aligned}$$

Thus, for each $y \in Y$,

$$y = \alpha(G_y) = \alpha(H, p_y) \varepsilon \alpha(H'),$$

and we can choose (by **AC**) some $q_y \leftarrow q'$ and a bisimulation S_y of (H, p_y) with (H', q_y) ; and conversely, by the same argument, for each $q \leftarrow q'$ we can choose some $y_q \in Y$ and some bisimulation T_q of (H, q_y) with (H', q) . It is now easy to verify that the union

$$R = \bigcup \{S_y \mid p_y \leftarrow_H \emptyset\} \cup \{T_q \mid q \leftarrow_{H'} q'\} \cup \{(\emptyset, q')\}$$

is a bisimulation which establishes that $H =_{\text{bs}} H'$, i.e., $\alpha(H) = \alpha(H')$.

Finally, to verify **AFA** for $\mathcal{A}$, suppose G is a graph in $\mathcal{A}$ with edge relation $\rightarrow_G \in \mathcal{A}$. To prove that G admits a decoration in $\mathcal{A}$, it is enough to define an $\mathcal{A}$ -operation δ such that

$$\mathcal{A} \models (\forall p \in G)[\delta(p) = \{\delta(q) \mid q \leftarrow_G p\}], \quad (\text{B-46})$$

since $\mathcal{A}$ is a **ZFDC**-universe, so it “knows” from (B-46) that the restriction of δ to G is a function, which is then a decoration of G . Let

$$H =_{\text{df}} \mathbf{b}_{\mathcal{A}}(G) \in \mathcal{V},$$

and make H into a graph in $\mathcal{V}$ with the edge relation

$$x \rightarrow_H y \iff_{\text{df}} \mathcal{A} \models x \rightarrow_G y \quad (x, y \in H). \quad (\text{B-47})$$

For each $p \in H$, set

$$\delta(p) =_{\text{df}} \alpha(H, p) \in \mathcal{A} \quad (p \in H) \quad (\text{B-48})$$

and compute:

$$\begin{aligned} \mathbf{b}_{\mathcal{A}}(\delta(p)) &= \{\alpha(H, q) \mid q \leftarrow_H p\} && \text{by (B-44)} \\ &= \{\alpha(H, q) \mid \mathcal{A} \models q \leftarrow_G p\} && \text{by (B-47)} \\ &= \{\delta(q) \mid \mathcal{A} \models q \leftarrow_G p\} && \text{by (B-48)}. \end{aligned}$$

Put another way, for each $p \in G$,

$$x \in \delta(p) \iff \mathcal{A} \models (\exists q \leftarrow p)[x = \delta(q)],$$

which is equivalent to (B-46).

It remains to show that G admits at most one decoration in $\mathcal{A}$, and for this it suffices (as above) to show that if δ' is any $\mathcal{A}$ -operation such that

$$\mathcal{A} \models (\forall p \in G)[\delta'(p) = \{\delta'(q) \mid q \leftarrow p\}], \quad (\text{B-49})$$

then $\delta'(p) = \delta(p)$, for every $p \in H$. Given such a δ' , choose (by **AC**) a pointed graph H'_p with point r_p for each $p \in H$, such that

$$\delta'(p) = \alpha(H'_p) \quad (p \in H),$$

and make sure as in the proof of the Rieger property that these graphs are all pairwise disjoint. If $H' = \bigcup \{H'_p \mid p \in H\}$ is the union of all the graphs, then

$$\delta'(p) = \alpha(H', r_p) \quad (p \in H),$$

since (trivially) the identity relation $\{(q, q) \mid q \in H\}$ is a bisimulation of (H'_p, r_p) with (H, r_p) . We now claim that *the relation*

$$R =_{\text{df}} \{(p, s) \in H \times H' \mid \alpha(H', s) = \delta'(p)\}$$

is a bisimulation of (H, p) with (H', r_p) for each $p \in H$. This will complete the proof, because for $p \in H$, $\alpha(H', r_p) = \delta'(p)$, hence $p R r_p$, and hence

$$\delta(p) = \alpha(H, p) = \alpha(H', r_p) = \delta'(p).$$

To show the somewhat less trivial half of the italicized statement, let $\rightarrow'$ be the edge relation of H' , assume $p R s$ and compute:

$$\begin{aligned} t' \leftarrow s &\implies \alpha(H', t) \varepsilon \alpha(H', s) \\ &\implies \alpha(H', t) \varepsilon \delta'(p) && \text{because } p R s \\ &\implies \text{for some } q \leftarrow p, \alpha(H', t) = \delta'(q) && \text{by (B-49)} \\ &\implies \text{for some } q \leftarrow p, q R t. \end{aligned}$$

The Axiom of Choice for $\mathcal{A}$ follows from **B.14**, and the Principle of Purity is trivial. $\dashv$

Problems for Appendix B

xB.1. Prove that for each set universe $\mathcal{M} = M, S, E$, the axioms for definite conditions and operations listed in **3.18** become true, if we replace in them “condition” by “ $\mathcal{M}$ -condition”, “operation” by “ $\mathcal{M}$ -operation”, $\in$ by E , Set by S and $(\forall y)$ by $(\forall y \in M)$.

***xB.2.** Suppose pairs and Cartesian products are defined by the Kuratowski operation of **4.2**. Show that $\bigcup_{n=2}^{\infty} \{\mathbb{N}_0\}^n \notin \mathcal{Z}$ and infer that the following

proposition is not a theorem of **ZDC**: for each set A , there exists a function $f : \mathbb{N} \rightarrow f[A]$ such that

$$f(0) = A \times A, \quad f(n+1) = f(n) \times A.$$

***xB.3.** Construct a definite operation $(x, y)'$ with the following properties. (1) $(x, y)'$ is an ordered pair operation, i.e., it satisfies (OP1) and (OP2) of **4.1**. (2) If $X, Y \in \mathcal{Z}$, then their Cartesian product $X \times Y$ is also in $\mathcal{Z}$. (3) If $\bigcup_{n=2}^{\infty} A^n$ is defined using this pair, then for each $A \in \mathcal{Z}$, $\bigcup_{n=2}^{\infty} A^n \in \mathcal{Z}$.

***xB.4.** Show that the implication (OP1) $\implies$ (OP2) in **4.1** is not a theorem of **ZDC** for any definite operation (x, y) .

***xB.5.** Verify that if I is a transitive set, then

$$A \in M(I) \implies \text{TC}(A) \in M(I).$$

xB.6. For each I , define $K_n(I)$ by the recursion

$$K_0(I) = I, \quad K_{n+1}(I) = K_n(I) \cup \mathcal{P}(K_n(I)). \quad (\text{B-50})$$

Show that

$$M_n(I) \subseteq K_n(\text{TC}_{n+1}(I)) \subseteq M(I),$$

where $\text{TC}_n(I)$ and $M_n(I)$ are defined by (11-16) and (11-18), respectively.

***xB.7.** Find some $I \supset \mathbb{N}_0$ such that $\text{TC}(I) \notin M(I)$. Infer that **ZDC** cannot prove that “every set has a transitive closure”.

***xB.8.** The implication (C1) $\implies$ (C3) in **4.20** cannot be proved for an arbitrary, definite operation $|A|$ in **ZDC**. (Cf. Problem **x11.6**.)

***xB.9.** The equivalence in Problem **x11.7** is not a theorem of **ZDC**.

***xB.10.** Assume that the full Axiom of Choice and the Generalized Continuum Hypothesis is true, so for all cardinals κ , $2^\kappa =_c \kappa^+$. Prove that

$$(n \mapsto \aleph_n) \subseteq \mathcal{Z}, \quad (n \mapsto \aleph_n) \notin \mathcal{Z}.$$

Infer that **ZDC+AC** cannot prove the existence of an infinite, increasing sequence of infinite cardinals, i.e., the proposition

$$\theta : (\exists f : \mathbb{N} \rightarrow f[\mathbb{N}])(\forall n \in \mathbb{N})[\mathbb{N} \leq_c f(n) <_c f(n+1)].$$

***xB.11.** Show that **ZDC** cannot prove the proposition “the well ordered set $\mathbb{N}$ of integers is similar with an ordinal”. **HINT.** Use Problem **x12.27**. The less trivial part of the problem is how to compute (or avoid computing) the relativization of this fairly complex proposition.

***xB.12.** (**AC**) Show that **ZFC** cannot prove that strongly inaccessible cardinals exist. **HINT.** Go by contradiction and interpret the meaning of the alleged theorem in $\mathcal{V}_\kappa$, where κ is the least strongly inaccessible cardinal.

xB.13. An **ordered pair operation** in a Rieger universe $\mathcal{M}$ is any binary $\mathcal{M}$ -operation C such that for all $x, y, x', y' \in M$,

$$C(x, y) = C(x', y') \iff x = x' \ \& \ y = y'. \quad (\text{B-51})$$

Cartesian products and function spaces relative to C are defined by

$$\begin{aligned} A \times_C B &=_{\text{df}} \rho\{C(x, y) \mid x \in A, y \in B\}, \\ (A \rightarrow_C B) &=_{\text{df}} \rho\{f \in M \mid (\forall t \in A)[t \in A \times_C B] \\ &\quad \& (\forall x \in A)(\exists! y \in B)[C(x, y) \in f]\}, \end{aligned}$$

where $\rho(Y)$ is the Rieger operation of $\mathcal{M}$ defined in (B-14). Verify that these definitions make sense (i.e., ρ is applied to appropriate arguments) and hence $A \times_C B$ and $A \rightarrow_C B$ are $\mathcal{M}$ -operations.

***xB.14.** Define triples, structured sets and systems of natural numbers in an arbitrary Rieger universe $\mathcal{M}$, relative to an arbitrary ordered pair operation $C(x, y)$ in $\mathcal{M}$. Formulate the Choice Principles **DC**, **AC_N** and **AC** using these notions and prove that every Rieger universe $\mathcal{M}$ satisfies **DC**, and if **AC** is also true, then $\mathcal{M}$ also satisfies **AC**.

xB.15. Show that the Rieger universe $\mathcal{M}_{\text{at}}$ of **B.16** has an ordered pair operation C such that for all x and y , the “pair” $C(x, y)$ is an atom.

***xB.16. (AC)** Define a Rieger universe $\mathcal{M}$ which satisfies the following two propositions.

(a) There exists a binary, definite condition $\leq_{\text{at}}$ which well orders the class of atoms, in the sense that (1) for all atoms a, b, c ,

$$a \leq a, \quad [a \leq b \ \& \ b \leq c] \implies a \leq c, \quad [a \leq b \ \& \ b \leq a] \implies a = c,$$

(2) for every two atoms a, b , either $a \leq b$ or $b \leq a$, and (3) every non-empty set of atoms has a $\leq$ -least member.

(b) Every set X is equinumerous with an $\leq$ -initial segment of atoms, i.e., for some atom b , $X =_c \{a \mid \text{Atom}(a) \ \& \ a < b\}$. **HINT.** Make the ordinals atoms in some Rieger universe.

***xB.17.** Define a Rieger universe which has at least two, distinct self-singletons, i.e., sets a and b such that $a \neq b$, $a = \{a\}$ and $b = \{b\}$. **HINT.** Start with a universe which has two atoms and imitate the coding construction in **B.16**.

***xB.18.** Define a Rieger universe which contains an infinite sequence of distinct sets $x_0, x_1, \dots$, such that for each i , $x_i = \{x_{i+1}\}$.

xB.19. Given a graph G and a node $p \in G$, let

$$G \restriction p = \{x \in G \mid x = p \vee p \Rightarrow x\}$$

consist of p and all the nodes on a path below it. Consider $G \restriction p$ as a subgraph of G , with the restriction of the edge relation $\rightarrow_G$ to it, and prove that

$$(G, p) =_{\text{bs}} (G \restriction p, p).$$

B.33. Definition. A **partial bisimulation** between two graphs G and H is any relation $R \subseteq G \times H$ which is a bisimulation of the pointed graphs (G, p) and (H, q) for every $(p, q) \in R$; it is a **total bisimulation** if in addition

$$(\forall p \in G)(\exists q \in H)p R q \ \& \ (\forall q \in H)(\exists p \in G)p R q.$$

Two graphs are **bisimilar** if there exists a total bisimulation between them.

xB.20. For all pairs of graphs G, H , there exists a largest (under $\subseteq$) partial bisimulation R between G and H , and $G =_{\text{bs}} H$ if and only if this largest bisimulation is total.

xB.21. Two graphs G, H are bisimilar if and only if every pointed graph (G, p) with $p \in G$ is bisimilar with some (H, q) , $q \in H$, and conversely, every (H, q) is bisimilar with some (G, p) .

xB.22. (AFA) Prove that there exist distinct, pure sets x, y and z such that

$$x \ni y \ni z \ni x,$$

and draw a picture of them.

***xB.23. (AFA)** Prove that there are only two, transitive, pure singletons. How many transitive, pure doubletons are there? Draw pictures of them.

xB.24. (AFA) With the Kuratowski pair, prove that there exists a pure set x such that

$$x = (\emptyset, x),$$

and draw a picture of it.

xB.25. (AFA) With the Kuratowski pair, prove that there exists a pure set x such that

$$x = \{(n, x) \mid n \in \mathbb{N}\},$$

and draw a picture of it.

SOLUTIONS TO THE EXERCISES IN CHAPTERS 1 – 12

2.5. The identity function ($x \mapsto x$) shows that $A \leq_c A$. And if $f : A \rightarrow B$ and $g : B \rightarrow C$, then the composition $h(x) = g(f(x))$ is an injection of A into C , so that $A \leq_c C$.

2.8. The assumptions give $A \leq_c \mathbb{N}$ and $B \leq_c A$, so $B \leq_c \mathbb{N}$.

2.9. If A is empty, then $B = f[\emptyset] = \emptyset$. If A is not empty, then there exists a surjection $\pi : \mathbb{N} \twoheadrightarrow A$, and then the composition $h(i) = f(\pi(i))$ is a surjection of $\mathbb{N}$ onto B , so B is countable.

2.20. Suppose $f : A \rightarrow B$ is a bijection, and let $\pi : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ be the image map,

$$\pi(X) = f[X] \quad (X \subseteq A).$$

To prove that π is an injection, note that if $x \in X \setminus Y$, then $f(x) \in f[X]$ but it is not possible that $f(x) \in f[Y]$; because if $f(x) = f(y)$ for some $y \in Y$, then $y = x$ since f is an injection, and so $x = y \in Y$, which is not true. Thus

$$x \in X \setminus Y \implies f(x) \in f[X] \setminus f[Y],$$

and by the symmetric argument,

$$y \in Y \setminus X \implies f(y) \in f[Y] \setminus f[X].$$

The two implications together show that

$$X \neq Y \implies f[X] \neq f[Y],$$

so that the image map is an injection.

To see that the image map is also a surjection, given $Y \subseteq B$, let $X = f^{-1}[Y]$, so that, immediately $f[X] \subseteq Y$; but each $y \in Y$ is the value $f(x)$ of some $x \in A$ since f is a surjection, and then $x = f^{-1}(y) \in X$, so that $f[X] = Y$.

2.23. Given bijections $f : A_1 \rightarrow A_2$ and $g : B_1 \rightarrow B_2$, we let for each $p : A_1 \rightarrow B_1$,

$$\pi(p) = q \text{ where, for each } x \in A_2, \quad q(x) = g(p(f^{-1}(x))).$$

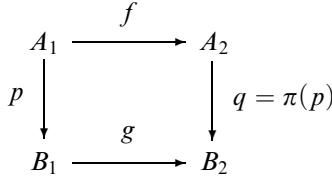


FIGURE 1. Diagram for Exercise 2.23.

In short: q is the unique function which makes the diagram in Figure 1 *commutative*, i.e., such that

$$g(p(x)) = q(f(x)) \quad (x \in A_1).$$

We include the details of the argument for completeness, but they can all be read from the diagram.

To check that if $p : A_1 \rightarrow B_1$, then $\pi(p) = q : A_2 \rightarrow B_2$, compute:

$$\begin{aligned}
 x \in A_2 &\implies f^{-1}(x) \in A_1 \\
 &\implies p(f^{-1}(x)) \in B_1 \implies q(x) = g(p(f^{-1}(x))) \in B_2.
 \end{aligned}$$

The proof that π is a bijection is very similar to that of the corresponding Exercise 2.20 for powersets, only a bit simpler.

First, if $p_1, p_2 : A_1 \rightarrow B_1$ and $p_1 \neq p_2$, then there is some $y \in A_1$ such that $p_1(y) \neq p_2(y)$. If $x = f(y)$, then

$$p_1(f^{-1}(x)) = p_1(y) \neq p_2(y) = p_2(f^{-1}(x))$$

using the fact that f^{-1} is an injection; and so applying g to both sides and using the fact that g is an injection, we get that

$$\pi(p_1)(x) \neq \pi(p_2)(x),$$

so that $\pi(p_1) \neq \pi(p_2)$. Thus π is an injection. To check that π is also a surjection, define for each $q : A_2 \rightarrow B_2$ the function

$$p(y) = g^{-1}(q(f(y))) \quad (y \in A_1),$$

so that $p : A_1 \rightarrow B_1$, and for each $x \in A_2$,

$$\pi(p)(x) = g(p(f^{-1}(x))) = g(g^{-1}(q(f(f^{-1}(x))))) = q(x),$$

so that $\pi(p) = q$ and π is a surjection.

3.9. We must show that these two sets do not have the same members, and this is true because $\emptyset \in \{\emptyset\}$ while $\emptyset \notin \emptyset$.

3.13. The empty set is a subset of every set, so $\emptyset \in \mathcal{P}(\emptyset)$; and if $X \subseteq \emptyset$, then X must be empty, since any member of X would have to be a member of $\emptyset$ —and there are none such. For the second claim, it is again immediate for every set A that $\emptyset, A \subseteq A$, and so $\{\emptyset, \{\emptyset\}\} \subseteq \mathcal{P}(\{\emptyset\})$; and for the other direction, if $X \subseteq \{\emptyset\}$, then X can have only one member, $\emptyset$, and so $X = \emptyset$ or $X = \{\emptyset\}$.

3.14. Using the powerset and subset axioms, we set

$$B = \{X \in \mathcal{P}(A) \mid (\exists t \in A)[X = \{t\}]\}.$$

3.16. The emptyset has no members, and so $\bigcup \emptyset = \emptyset$; and the only member of the singleton $\{\emptyset\}$ is $\emptyset$, which has no members, so again, $\bigcup \{\emptyset\} = \emptyset$.

3.20. The unary condition P defined by

$$P(x) \iff x \in A$$

is coextensive with A , and so by (3-10), $\{x \mid x \in A\} = A$. Similarly, if

$$Q(X) \iff \text{Set}(X) \ \& \ X \subseteq A,$$

then $Q =_e \mathcal{P}(A)$, and so by (3-10), $\{X \mid \text{Set}(X) \ \& \ X \subseteq A\} = \mathcal{P}(A)$.

3.21. That Set is not a set is part of Theorem 3.11. For $\mathcal{W}$ we argue by contradiction: if it were a set, so would Set be a set by the Axiom of Subsets, since $\text{Set} = \{x \in \mathcal{W} \mid \text{Set}(x)\}$.

3.22. If $A = \{w \mid w \text{ is a singleton}\}$ were a set, then its unionset $\bigcup A$ would also be a set; but every object x is a member of its singleton $\{x\}$, and so $\bigcup A = \mathcal{W}$, which is not a set by the preceding Exercise.

3.23. This is primarily an exercise in terminology, and we can prove it by a simple round-robin argument. We assume that A is a class.

If A is a set, then $A \in \{A\}$ and the singleton $\{A\}$ is a set, and hence a class, so A belongs to a class.

If A belongs to some class B , then it must be an object (atom or set), since only objects are put into classes by (3-8); and since A is a class by the hypothesis, it is not an atom, and so it must be a set. It follows that $A \subseteq A$, and so A is a subset of a set.

Finally, suppose that A is a class and $A \subseteq X$, for some set X . If A is a set, we are done; otherwise, by (3-10), $A = P$ for some definite condition P , but $A \subseteq X$, so that only members of X can satisfy P and $A = \{x \in X \mid P(x)\}$; and this makes A a set by the Subset Axiom, contradicting the case hypothesis.

4.3. If $\text{Pair}(z)$, then $z = (x, y)$ for some x, y , and by the definitions,

$$x = \text{First}(z), y = \text{Second}(z);$$

and if $z = (\text{First}(z), \text{Second}(z))$, then $z = (x, y)$ with these x, y , and so $\text{Pair}(x, y)$ holds.

4.4. If $(x, y, z) = (x, (y, z)) = (x', (y', z')) = (x', y', z')$, then by the basic property of pairs we have $x = x'$ and $(y, z) = (y', z')$; and so applying the same property once more, we have $y = y'$ and $z = z'$. The other direction is trivial.

4.6. $A \uplus \emptyset = \{(\text{blue}, x) \mid x \in A\}$, and $(\text{blue}, x) \in A \uplus B$ for every $x \in A$ and every B . For the second claim, notice that for any A, B , the members of $A \uplus B$

are ordered pairs; but if we take $A = B = \{\emptyset\}$, then (with the Kuratowski pair), $\emptyset$ is not an ordered pair, and so we do not have $\{\emptyset\} \subseteq \{\emptyset\} \uplus \{\emptyset\}$.

4.9. There is nothing much to prove for (2) either, since $P_{A,B}$ is a set by the Subset Axiom (III) and it is a subset of $A \times B$ by its definition—so it is a relation on A, B .

4.11. The claim is trivial for the first two examples. For the third one, the reflexivity $x \sim_{A/B} x$ and symmetry $x \sim_{A/B} y \implies y \sim_{A/B} x$ are again trivial. For the transitivity, observe first that

$$[x \sim_{A/B} y \ \& \ y = z] \implies x \sim_{A/B} z;$$

this is because the hypothesis tells us that either $x = y$, and then with $y = z$ we have $x = z$, so that $x \sim_{A/B} z$; or $x, y \in B$, and so with $y = z$ we have $x, z \in B$, i.e., $x \sim_{A/B} z$ again. Suppose now that $x \sim_{A/B} y$ and $y \sim_{A/B} z$, and assume that all three x, y, z are distinct, since each of the assumptions $x = y, y = z$ or $y = z$ implies immediately that $x \sim_{A/B} z$ by the observation just made; but if $x \neq y$ and $y \neq z$, the first hypothesis $x \sim_{A/B} y$ gives $x \in B \ \& \ y \in B$, and the second hypothesis $y \sim_{A/B} z$ gives $y \in B \ \& \ z \in B$, so that $x \in B \ \& \ z \in B$, which implies $x \sim_{A/B} z$.

4.13. If $=_A$ is the restriction of the identity relation on A , then each $x \in A$ is equivalent only to itself and so $[x/_A] = \{x\}$. For the universal relation $\sim_A$ in which $x \sim_A y$ for all $x, y \in A$, clearly $[x/_\sim_A] = A$, for any $x \in A$. Finally, for the more complex $\sim_{A/B}$, we have $[x/_\sim_{A/B}] = \{x\}$ if $x \in A \setminus B$, and $[x/_\sim_{A/B}] = B$ if $x \in B$.

4.15. If f is a function, then there exist sets A, B such that $f \subseteq A \times B$, and so

$$\text{Domain}(f) = \{x \in A \mid (\exists y)[(x, y) \in f]\}$$

and $\text{Domain}(f)$ is a set by the Subset Axiom (III). The same argument shows that $\text{Image}(f)$ is a set (a subset of B), and the last implication is trivial.

4.17. The proof is exactly that in Exercise 2.20, so the point is to read that argument carefully and see exactly what axioms are needed to justify it. Recall that the required equinumerosity between $\mathcal{P}(A)$ and $\mathcal{P}(B)$ was the image function

$$\pi(X) = f[X] \quad (X \subseteq A),$$

so we must check what axioms are needed to construct this function as a set of ordered pairs. First, the values are sets by the Subset Axiom (III), since

$$f[X] = \{y \in B \mid (\exists x \in X)[(x, y) \in f]\};$$

but this part of the argument uses an assumed ordered pair operation, and so we are implicitly appealing to all the axioms required to construct such an operation, i.e., those used in the proof of Lemma 4.2. Looking carefully at that argument, it uses the Pair Axiom (II) to define the Kuratowski ordered pair; the Extensionality Axiom (I) to prove that this operation satisfies (OP1); and

to prove (OP2), first the Pair and Unionset axioms (II) and (V) to construct $A \cup B$ and then the Powerset axiom (IV). In short, to make precise and to show that if $f : A \rightarrow B$, then the image $f[X]$ of any $X \subseteq A$ is a set, we need all the axioms except for the Axiom of Infinity (VI).

To finish the proof we define π as a set of ordered pairs:

$$\pi = \{(X, Y) \in \mathcal{P}(A) \times \mathcal{P}(B) \mid (\forall y)[y \in Y \iff (\exists x \in X)[(x, y) \in f]]\}.$$

This is a set by the Subset Axiom (III) again, once we have the basic properties of the ordered pair operation as above.

4.18. The detailed proofs for this are almost identical to the argument in Exercise 4.17 and it is not worth repeating them. We confine ourselves to defining the functions required to show these equinumerosities, assuming that $f : A \rightarrow B$ and $g : B \rightarrow C$.

To show that $A \times B =_c A' \times B'$, we set

$$\pi((i, x)) = \begin{cases} (\text{blue}, f(x)), & \text{if } i = \text{blue}, \\ (\text{white}, g(x)), & \text{otherwise, i.e., if } i = \text{white}. \end{cases}$$

To show that $A \times B =_c A' \times B'$ we set

$$\pi((x, y)) = (f(x), g(y)).$$

Finally, to show that $(A \rightarrow B) =_c (A' \rightarrow B')$, we set for $p : A \rightarrow B$,

$$\pi(p) = \{(x, y) \in A' \times B' \mid y = g(p(f^{-1}(x)))\},$$

following the proof in Exercise 2.23.

4.22. Using successively (C1), the hypothesis and (C1) again,

$$A =_c |A| = |B| =_c B.$$

4.23. Given the two bijections $f : A \rightarrow |A|$ and $g : B \rightarrow |B|$, we define $h : A \cup B \rightarrow A \uplus B$ by

$$h(x) = \begin{cases} (\text{blue}, f(x)), & \text{if } x \in A, \\ (\text{white}, g(x)), & \text{otherwise (i.e., if } x \in B \setminus A). \end{cases}$$

If $A \cap B = \emptyset$, then the two cases in the definition of h are mutually exclusive, and so h is a bijection which witnesses

$$A \cup B =_c A \uplus B =_c |A| + |B|.$$

4.24. These identities follow immediately from the definitions, the basic (C1) and Exercise 4.18: e.g.,

$$\kappa_1 + \kappa_2 =_c \kappa_1 \uplus \kappa_2 =_c \lambda_1 \uplus \lambda_2 \text{ (by Exercise 4.18)} =_c \lambda_1 + \lambda_2.$$

4.26. If κ is a cardinal number, then $\kappa = |A|$ for some set A , and by (C1), $\kappa =_c A$; but then (C2) (which holds of a strong cardinal assignment) implies that $|\kappa| = |A|$, i.e., $|\kappa| = \kappa$. For the non-trivial direction of the second claim: if $\kappa =_c \lambda$, then $|\kappa| = |\lambda|$ by (C2), and so $\kappa = \lambda$ by the first claim.

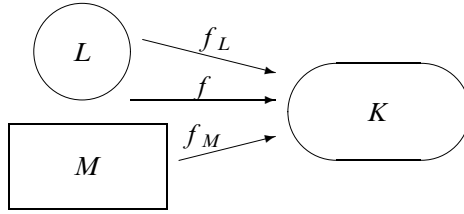


FIGURE 2. Diagram for Exercise 4.28.

4.27. By the definition, each member of $K \uplus (L \uplus M)$ is in exactly one of these forms, where k, l, m are uniquely determined members of K, L, M respectively:

$$x = (\underline{\text{blue}}, k), \text{ or } x = (\underline{\text{white}}, (\underline{\text{blue}}, l)), \text{ or } x = (\underline{\text{white}}, (\underline{\text{white}}, m)).$$

In the same way, each member of $(K \uplus L) \uplus M$ is in exactly one of the following forms, where k, l, m are uniquely determined members of K, L, M respectively:

$$y = (\underline{\text{blue}}, (\underline{\text{blue}}, k)), \text{ or } y = (\underline{\text{blue}}, (\underline{\text{white}}, l)), \text{ or } y = (\underline{\text{white}}, m).$$

The required bijection $f : K \uplus (L \uplus M) \xrightarrow{\sim} (K \uplus L) \uplus M$ is defined by matching the corresponding elements in the obvious way:

$$\begin{aligned} f(\underline{\text{blue}}, k) &= (\underline{\text{blue}}, (\underline{\text{blue}}, k)), \\ f(\underline{\text{white}}, (\underline{\text{blue}}, l)) &= (\underline{\text{blue}}, (\underline{\text{white}}, l)), \\ f(\underline{\text{white}}, (\underline{\text{white}}, m)) &= (\underline{\text{white}}, m). \end{aligned}$$

4.28. If $L \cap M = \emptyset$, then $(L \cup M) \times K$ splits into the two disjoint sets $L \times K$ and $M \times K$, and so each function $f \subseteq (L \cup M) \times K$ splits into two functions

$$f_L = f \cap (L \times K) : L \rightarrow K, \quad f_M = f \cap (M \times K) : M \rightarrow K.$$

This defines a mapping $\pi : ((L \cup M) \rightarrow K) \rightarrow (L \rightarrow K) \times (M \rightarrow K)$ given by

$$\pi(f) = (f_L, f_M),$$

and it is very easy to check that π is, in fact, a bijection.

To get the cardinal identity (4-17) from this, use $\lambda \uplus \mu = L \cup M$ where $L = \{\underline{\text{blue}}\} \times \lambda$ and $M = \{\underline{\text{white}}\} \times \mu$, so that $L \cap M = \emptyset$ and $\lambda =_c L$, $\mu =_c M$, and compute:

$$\begin{aligned} \kappa^{(\lambda+\mu)} &= (L \cup M) \rightarrow K && (\text{def., (C1) and 4.18}) \\ &= (L \rightarrow K) \times (M \rightarrow K) \\ &= \kappa^\lambda \cdot \kappa^\mu && (\text{def., (C1) and 4.18}). \end{aligned}$$

5.18. We prove by induction on n , that each function

$$f_n(s) = n + s$$

is an injection. This is trivial at the basis, since $f_0(s) = s$, and for the induction step,

$$f_{n+1}(s) = (n + 1) + s = (n + s) + 1 = f_n(s) + 1,$$

and so f_{n+1} is the composition of f_n and the successor function; both of these are injections, and so f_{n+1} is an injection.

5.25. For the first claim, suppose that there is a $k \in [n, n)$, so that there exist t and s satisfying

$$n + t = k, \quad k + s = n, \quad k \neq n;$$

it follows that $n + t + s = n$, which by three applications of Exercise **5.18** implies that $t = 0$ and $s = 0$; and that gives $k = n$ contradicting the hypothesis.

The second claim is essentially a restatement of Lemma **5.22**. First we compute:

$$\begin{aligned} k < Sm &\iff k \leq Sm \ \& \ k \neq Sm && (\text{def.}), \\ &\iff (k \leq m \vee k = Sm) \ \& \ k \neq Sm && (\text{by Lemma 5.22}), \\ &\iff k \leq m \\ &\iff (k \leq m \ \& \ k \neq m) \vee k = m \\ &\iff k < m \vee k = m, \end{aligned}$$

where for the third equivalence we have used the fact that

$$k \leq m \implies k \neq Sm;$$

this holds because its negation implies that $Sm \leq m$ which is absurd (if $Sm + t = m$, then $m + St = m$, so $St = 0$ by Exercise **5.18**, which is false). It follows that for all k and $n \leq m$,

$$\begin{aligned} k \in [n, Sm) &\iff n \leq k \ \& \ k < Sm \iff n \leq k \ \& \ (k < m \vee k = m) \\ &\iff (n \leq k \ \& \ k < m) \vee (n \leq k \ \& \ k = m) \iff k \in ([n, m) \cup \{m\}), \end{aligned}$$

the last inference using the hypothesis $n \leq m$.

6.3. For each $x \in P$, we have $x \leq M$ since M is the maximum; and if M' is any upper bound of P , then $M \leq M'$ because $M \in P$.

6.4. (1) $\{e, f\}$ has no upper bound. (2) $\{a, c\}$ has upper bounds $(b, d, e$ and $f)$ but no least upper bound. (3) $\{b, d\}$ has a least upper bound (e) but no maximum.

6.5. Every member of P is an upper bound of $\emptyset$, because it satisfies (formally) the condition for being an upper bound,

$$(\forall x)[x \in \emptyset \implies x \leq M];$$

it follows that M is *the least upper bound* of $\emptyset$ if and only if it is the least element of P .

6.6. The conditions

$$X \subseteq X, [X \subseteq Y \ \& \ Y \subseteq Z] \implies X \subseteq Z, [X \subseteq Y \ \& \ Y \subseteq X] \implies X = Y$$

are all trivial. If $\mathcal{E} \subseteq \mathcal{P}(A)$ is any set of subsets of A , then the unionset $\bigcup \mathcal{E}$ is an upper bound of $\mathcal{E}$, because $X \in \mathcal{E} \implies X \subseteq \bigcup \mathcal{E}$; and it is the least upper bound, because for any M' , if $X \subseteq M'$ for every $X \in \mathcal{E}$, then $\bigcup \mathcal{E} \subseteq M'$.

6.9. If E is empty, then $(A \rightarrow E) = (A \rightarrow E) = \{\emptyset\}$ and there is nothing to prove, so assume that $e_0 \in E$. If $f : A \rightarrow E$ and $X = \text{Domain}(f)$, we set

$$f'(x) = \begin{cases} f(x), & \text{if } f(x) \downarrow, \\ e_0, & \text{otherwise,} \end{cases}$$

so that $f' : A \rightarrow E$ and $f' \upharpoonright X = f$; and if $f : A \rightarrow E$ and $X \subseteq A$, then $f \upharpoonright X : A \rightarrow E$ with $\text{Domain}(f \upharpoonright X) = X$, directly from the definitions.

6.11. The empty set satisfies (formally) the condition for being a chain,

$$(\forall x, y)[(x \in \emptyset \ \& \ y \in \emptyset) \implies x \leq y \vee y \leq x].$$

6.12. The only chains in a flat poset P are singletons (including $\{\perp\}$) and doubletons of the form $\{\perp, a\}$, and every one of these sets has a maximum, which is also its supremum.

A discrete poset with at least two elements a, b , cannot be inductive, because it has no least element; and so the only inductive, discrete posets are the singletons.

6.13. If $u, v \in \{x_n \mid n \in \mathbb{N}\}$, then there exist n, m such that $u = x_n$ and $v = x_m$; if $n \leq m$, then $u = x_n \leq x_m = v$, and, similarly, if $m \leq n$, then $v \leq u$.

6.15. With the usual ordering, $\mathbb{N}$ is a chain and it has no upper bound; and the poset of Figure 6.2 does not have a least element.

6.16. We claim that if $X \subseteq P = E^* \cup (\mathbb{N} \rightarrow E)$ and X is a chain, then the union set $\bigcup X$ is a function with domain some subset of $\mathbb{N}$: because if $(x, y), (x, y') \in \bigcup X$, then there exist $p, q \in X$ such that $(x, y) \in p$ and $(x, y') \in q$; and if $p \subseteq q$, we have $(x, y), (x, y') \in q$ so that $y = y'$, while if $q \subseteq p$ we have $(x, y), (x, y') \in p$, so that, again, $y = y'$. Moreover, the domain of $\bigcup X$ is either all of $\mathbb{N}$ or a finite, initial segment of $\mathbb{N}$, because it is closed under $<$:

$$\begin{aligned} x < x' \in \text{Domain}(\bigcup X) &\implies \text{for some } p \in X, x' \in \text{Domain}(p) \\ &\implies x \in \text{Domain}(p) \\ &\implies x \in \text{Domain}(\bigcup X). \end{aligned}$$

Thus $\bigcup X \in P$, and it is clear that it is the least upper bound of X .

6.17. Suppose $S \subseteq (A \rightarrow B)$ is a chain of *partial injections* and let $p^* = \bigcup S$ be the unionset of S . Since p^* is obviously the least upper bound of S under $\subseteq$, it is enough to show that p is a partial injection; i.e., we must show that

$$(x, y), (x, y') \in p^* \implies y = y', \quad (\text{a})$$

$$(x, y), (x', y) \in p^* \implies x = x'. \quad (\text{b})$$

For (a): from the hypotheses, there are $p, p' \in S$ such that $(x, y) \in p$ and $(x, y') \in p'$; if $p \leq p'$, then $(x, y) \in p'$, and since p' is a partial function and it also contains (x, y') , we must have $y = y'$; and if $p' \leq p$, then the analogous symmetric reasoning applies and leads to the same conclusion.

The argument for (b) is similar.

6.20. If $x_0 \leq_P x_1 \leq_P \dots$, then the set $X = \{x_n \mid n \in \mathbb{N}\}$ of values of the sequence $\{x_n\}_{n \in \mathbb{N}}$ and its image $\pi[X] = \{\pi(x_n) \mid n \in \mathbb{N}\}$ under a monotone π are both non-empty, countable chains, an observation that was needed in order to define limits of monotone sequences in the first place,

$$\lim_n x_n = \sup X, \quad \lim_n \pi(x_n) = \sup \pi[X];$$

it follows that if π is countably continuous, then

$$\pi(\lim_n x_n) = \pi(\sup(X)) = \sup \pi[X] = \lim_n \pi(x_n),$$

as required for one direction of the Exercise. For the other direction, suppose that $S = \{s_0, s_1, \dots\}$ is a non-empty, countable chain in P , and define the function (sequence) $x : \mathbb{N} \rightarrow S$ by the recursion

$$x_0 = s_0, \quad x_{n+1} = \max_P(x_n, s_{n+1}).$$

The definition makes sense, since each x_n is a member of the chain S and hence comparable with s_{n+1} . By a trivial induction on n ,

$$x_n \in S, \quad s_n \leq_P x_n, \quad x_n \leq_P x_{n+1},$$

from which we get immediately that

$$\sup S = \sup\{x_n \mid n \in \mathbb{N}\} = \lim_n x_n;$$

and the same holds for the images by any monotone π ,

$$\sup \pi[S] = \sup\{\pi(x_n) \mid n \in \mathbb{N}\} = \lim_n \pi(x_n).$$

Thus, if π respects limits of monotone sequences, we have the required

$$\sup \pi[S] = \lim_n \pi(x_n) = \pi(\lim_n x_n) = \pi[\sup S].$$

6.24. Directly from the definition,

$$\pi(f)(n) = \pi(f_0)(n),$$

where f_0 is the restriction of f to the finite set $\{0, \dots, n\}$. For the second part, if $f(n) = 2n$, then

$$\pi(f)(2) = f(0) + f(1) + f(2) = 0 + 2 + 4 = 6.$$

6.26. If $f : X \rightarrow Y$ is continuous and $F \subseteq Y$ is closed, then $G = Y \setminus F$ is open, and $F = Y \setminus G$, so

$$f^{-1}[F] = f^{-1}[Y \setminus G] = f^{-1}[Y] \setminus f^{-1}[G] = X \setminus f^{-1}[G],$$

so $f^{-1}[F]$ is closed. The symmetric argument shows that if $f^{-1}[F]$ is closed for every closed $F \subseteq Y$, then $f^{-1}[G]$ is open for every open $G \subseteq Y$, and so f is continuous.

6.32. Directly from the definition,

$$\pi(f)(n) = \pi(f_0)(n),$$

where $f_0 = \{(n, 0)\}$ if $n \in A$, and if $n \notin A$, then

$$f_0 = \{(n, h(w)) \mid (n+1, w) \in f\}.$$

This f_0 is finite, in fact it is defined on exactly one number if $n \in A$ or $n \notin A$ & $f(n+1) \downarrow$, and it is the empty partial function otherwise; and so by **6.23**, π is a continuous map.

7.3. Suppose $\leq_C$ is a wellordering of C and $f : A \rightarrow C$ is an injection, and let

$$x \leq_A y \iff f(x) \leq_C f(y) \quad (x, y \in A).$$

We prove that the relation $\leq_A$ is a wellordering of A .

Proof that $\leq_A$ is a linear ordering is trivial, basically by inspection. For example,

$$\begin{aligned} x \leq_A y \text{ \& } y \leq_A z &\implies f(x) \leq_C f(y) \text{ \& } f(y) \leq_C f(z) \\ &\implies f(x) \leq_C f(z) \quad (\text{because } \leq_C \text{ is transitive}) \\ &\implies x \leq_A z \quad (\text{by def.}), \end{aligned}$$

so that $\leq_A$ is transitive.

To check that $\leq_A$ wellorders A , suppose $X \subseteq A$ and $X \neq \emptyset$, and let $f[X]$ be the image of X ; now $f[X]$ is a non-empty subset of C , and so it has a least element $f(m)$; and then m is the $\leq_A$ -least element of X , because for every $x \in X$, $f(m) \leq_C f(x)$, and so $m \leq_A x$, by the definition.

7.4. If $(C, \leq_C)$ is a well ordered set and $f : C \rightarrow A$, define “an inverse” function $g : A \rightarrow C$ by

$$g(x) = \min_C \{y \mid f(y) = x\}.$$

This is an injection; because if $x \neq x'$, then

$$\{y \mid f(y) = x\} \cap \{y \mid f(y) = x'\} = \emptyset,$$

and $g(x)$ is in the first of these two sets while $g(x')$ is in the second.

We now appeal to Exercise **7.3** using this injection.

7.6. This is trivial: the restriction $\leq_I$ of $\leq_U$ to I inherits the properties of reflexivity, transitivity and antisymmetry, and every non-empty $X \subseteq I$ has a least member in $I \cap U$, which is its least member in I .

7.8. By the definition, $\mathbf{seg}(0) = \{x \in U \mid x <_U 0\}$, and so $\mathbf{seg}(0) = \emptyset$ since 0 is the least element.

Again by the definition,

$$S(x) = \min\{y \in U \mid x <_U y\},$$

so that there is no $z \in U$ such that $x <_U z <_U S(x)$; this means that

$$y <_U S(x) \iff y <_U x \vee y = x,$$

i.e., $\mathbf{seg}(S(x)) = \mathbf{seg}(x) \cup \{x\}$.

7.10. If V is the set of all initial segments of U , then

$$V = V_p \cup \{U\},$$

where the “proper” initial segments in V_p are those of the form $\mathbf{seg}_U(x)$ with $x \in U$. Moreover, the mapping

$$f(I) = \text{the unique } x \in U \text{ such that } I = \mathbf{seg}_U(x)$$

is obviously order preserving from V_p into U (in fact it is a similarity) and so V_p is well ordered. Now V is obtained by adding a point (U) at the top of V_p , and so it remains well ordered—for a detailed proof of this, see Exercise 7.18 below.

7.12. Recall from Definition 6.18 that a mapping $\pi : P \rightarrow Q$ is monotone if

$$x \leq y \implies \pi(x) \leq \pi(y),$$

and so, it is order-preserving if, *in addition*, the converse holds,

$$\pi(x) \leq \pi(y) \implies x \leq y;$$

hence, order-preserving mappings are monotone. For a counterexample to the converse, take any constant mapping $\pi(t) = c$, with P chosen so that it has (at least) two members $x \neq y$; constant mappings are trivially monotone, but this one cannot be order preserving, since $\pi(x) = \pi(y) = c$, which would imply (if π were order preserving) that $x \leq y$ and $y \leq x$, i.e., $x = y$.

7.13. Suppose (towards a contradiction) that P and Q are linear, $f : P \rightarrow Q$ is order-preserving, $x <_P y$ but $f(x) \not\leq_Q f(y)$; it follows by the linearity of Q that $f(y) \leq_Q f(x)$; but this implies $y \leq_P x$, since f is order-preserving; and this contradicts the hypothesis $x <_P y$.

For the converse, suppose (with P and Q linear again) that f is strictly monotone but not order-preserving. Easily, $x \leq_P y \implies f(x) \leq_Q f(y)$, taking cases on whether $x = y$ or $x <_P y$, and so, since f is not order-preserving, there must exist $x, y \in P$ such that $f(x) \leq_Q f(y)$ but $x \not\leq_P y$; but then $y <_P x$ by the linearity of P ; and so $f(y) <_Q f(x)$, since f is strictly monotone, contradicting the other assumption, $f(x) \leq_Q f(y)$.

7.14. The identity mapping $\pi(x) = x$ on P is a similarity, so $P =_o P$.

To verify the symmetry property, suppose $\pi : P \rightarrowtail Q$ is a similarity, and let $\pi^{-1} : Q \rightarrowtail P$ be the inverse bijection. Now, if $x, y \in Q$, then

$$\pi^{-1}(x) \leq_P \pi^{-1}(y) \iff \pi(\pi^{-1}(x)) \leq_Q \pi(\pi^{-1}(y))$$

since π is order-preserving, and the order-preserving property for π^{-1} follows because $\pi(\pi^{-1}(z)) = z$ for every $z \in Q$.

For the transitivity property, we first observe that *the composition of order-preserving mappings is order-preserving*:

$$x \leq_P y \iff \pi(x) \leq_Q \pi(y) \iff \rho(\pi(x)) \leq_R \rho(\pi(y)).$$

Since the composition of bijections is a bijection, it follows that *the composition of similarities is a similarity*, and so $=_o$ is transitive.

7.17. If $\pi : P \rightarrowtail Q$ is a similarity of P with Q , then its extension by one value $p \cup \{(t_P, t_Q)\}$ is easily a similarity of $\text{Succ}(P)$ with $\text{Succ}(Q)$.

7.18. It is easy to show that $\leq_{\text{Succ}(U)}$ is a linear order of $\text{Field}(U) \cup \{t_U\}$. To check the wellordering property, suppose $\emptyset \neq X \subseteq \text{Field}(U) \cup \{t_U\}$; if $X = \{t_U\}$, then t_U is the least member of X in $\text{Succ}(U)$, and if $X \cap \text{Field}(U) \neq \emptyset$, then the $\leq_U$ -least member of this set is the $\leq_{\text{Succ}(U)}$ -least member of X .

7.28. Let $\sigma = \rho\pi$ be the composition of π and ρ , so that

$$\sigma(x) = \rho(\pi(x)) \quad (x \in U).$$

To prove that $\sigma : U \rightarrow W$ is order preserving, we compute, using the fact that π and ρ are order preserving:

$$\begin{aligned} x \leq_U y &\iff \pi(x) \leq_V \pi(y) \\ &\iff \rho(\pi(x)) \leq_W \rho(\pi(y)) \\ &\iff \sigma(x) \leq_W \sigma(y). \end{aligned}$$

To show that $\sigma[U]$ is an initial segment of W , suppose $w \in \sigma[U] = \rho[\pi[U]]$, so that $w = \rho(v)$ for some $v \in \pi[U]$, and $w' <_W w$. Since ρ is an initial similarity, there must exist some $v' \in V$ such that $w' = \rho(v')$; and since $\rho(v') <_W \rho(v)$ and ρ is order preserving, we must have $v' <_V v$. Since $v \in \pi[U]$, we have at this point some $u \in U$ such that

$$v = \pi(u) \text{ and } v' <_V v.$$

But π is also an initial similarity, so that (as above) there must exist some $u' <_U u$ such that $v' = \pi(u')$, from which we get

$$\sigma(u') = \rho(\pi(u')) = \rho(v') = w',$$

which completes the proof.

8.3. There is no choice set S for a family $\mathcal{E}$ which contains $\emptyset$, since $S \cap \emptyset$ is not singleton. For the second example, if S is a choice set for $\mathcal{E}$, then $S \cap \{a\} = \{a\}$ and also $S \cap \{b\} = \{b\}$, so that $\{a, b\} \subseteq S$ and $S \cap \{a, b\}$ is not a singleton.

8.7. Suppose $(\forall x)(\exists y)P(x, y)$, and let ε be a choice function for B , so that

$$[X \subseteq B \ \& \ X \neq \emptyset] \implies \varepsilon(X) \in X.$$

Define $f : A \rightarrow B$ by

$$f(x) = \varepsilon(\{y \mid P(x, y)\});$$

this makes sense since for each $x \in A$, $\{y \mid P(x, y)\} \neq \emptyset$ by the hypothesis. Moreover, $f(x) \in \{y \mid P(x, y)\}$ for any $x \in A$, since ε is a choice function for B , and so $P(x, f(x))$, as required.

8.15. Suppose $a \in A$, $P \subseteq A \times A$, and $(\forall x \in A)(\exists y \in A)P(x, y)$. Let

$$P^*(u, x) \iff u \in A^* \ \& \ x \in A \\ \& \ [u = \emptyset \ \& \ x = a] \vee [\text{lh}(u) > 0 \ \& \ P(u(\text{lh}(u) - 1), x)].$$

If $u = \emptyset$, then $P^*(u, a)$, and if $u = \langle u_0, \dots, u_n \rangle$ with $n \geq 0$ and $P(u_n, x)$, then $P^*(u, x)$; thus the hypothesis on P guarantees that $(\forall u)(\exists x)P^*(u, x)$, and the assumed version of **DC** supplies a function $f : \mathbb{N} \rightarrow A$ such that $(\forall n)P^*(\bar{f}(n), f(n))$; which in particular implies that $f(0) = a$, when we apply it to $u = \emptyset$. Moreover, if $n > 0$,

$$P^*(\bar{f}(n), f(n)) \iff P(f(n-1), f(n)),$$

so that $(\forall n > 0)P(f(n-1), f(n))$, which is the same as $(\forall n)P(f(n), f(n+1))$.

8.18. Suppose first that $(P, \leq)$ is linear and grounded, and let $X \subseteq P$ be a non-empty set of points; it follows that there is some $m \in X$ such that for all $x \in X$, $x \not\leq m$, which by the linearity of $\leq$ means that $m \leq x$, and so m is the least member of X . Conversely, if $(P, \leq)$ is a wellordering and X is any non-empty subset of P , then X has a least element m , which is certainly minimal in X .

9.4. If u is a leaf, then $u \sqsubseteq w \implies w = u$, so that $T_u = \{w \in T \mid w \sqsubseteq u\}$; on the other hand, the second summand in (9-2) is empty (because u has no children), and so (9-2) holds. If u is not a leaf, then clearly, for every w ,

$$u \sqsubseteq w \iff (\exists v)[v \text{ is a child of } u \ \& \ v \sqsubseteq w],$$

which again implies (9-2).

9.5. An infinite branch $f : \mathbb{N} \rightarrow \mathbb{N}$ would need to satisfy

$$f(0) > f(1) > \dots,$$

and there are no infinite, descending sequences of natural numbers.

9.12. In full detail, we know that $\kappa <_c 2^\kappa$, and so $\kappa^+ \leq_c 2^\kappa$ by (9-5). Now the **GCH** asserts that no cardinal is properly between an infinite κ and 2^κ , so that $\kappa^+ =_c 2^\kappa$; and, conversely, if $\kappa^+ =_c 2^\kappa$, then there is no cardinal properly between κ and 2^κ , again by (9-5).

9.19. If κ_1 and κ_2 satisfy (1) – (3), then $\kappa_1 =_c A$ for some $A \in \mathcal{E}$ by (2) for κ_1 ; and so $\kappa_2 \leq_c A \leq_c \kappa_1$ by (3) for κ_2 . The symmetric argument gives $\kappa_1 \leq_c \kappa_2$.

9.22. This is immediate, from the theorem and the fact that for every i , $\{i\} <_c 2$, since 2 is a set with two members.

9.24. This is immediate.

10.2. Suppose $u \sqsubseteq v$ and compute:

$$x \in \mathcal{N}_v \implies v \sqsubseteq x \implies u \sqsubseteq x \implies x \in \mathcal{N}_u.$$

The converse is equally obvious.

10.3. The family of neighborhoods is equinumerous with $\mathbb{N}^*$, and $\mathbb{N}^* =_c \mathbb{N}$.

10.5. If G is open and $x \in G$, then by (10-6), there is a u such that $x \in \mathcal{N}_u \subseteq G$; and if each $x \in G$ belongs to some $\mathcal{N}_u \subseteq G$, then (10-6) holds trivially.

10.9. By definition, $\mathcal{N}_u = [T_u]$, where the subtree T_u is defined in (9-1) and it is splitting, since for each $v \in T_u$ and each n , $v \star \langle n \rangle \in T_u$.

10.13. If A is uncountable and $A = \bigcup_n A_n \in \mathbb{N}$, then at least one A_n must be uncountable; and if $A_n \in \Gamma$, then A_n has a non-empty, perfect subset P by the hypothesis, and so also $P \subseteq A$.

10.17. If $\pi : (\mathbb{N} \rightarrow \mathbb{N}) \rightarrow (\mathbb{N} \rightarrow \mathbb{N})$ is continuous and $x \in \mathcal{N}$, then

$$\pi(x) = \sup\{\pi(v) \mid v \in \mathbb{N}^*, v \sqsubseteq x\},$$

and so if $\pi(x) \in \mathcal{N}_u$ (which means $u \sqsubseteq \pi(x)$), there must exist some $v \sqsubseteq x$ such that $u \sqsubseteq \pi(v)$; but this implies that $u \sqsubseteq \pi(y)$ for any $y \in \mathcal{N}$ such that $v \sqsubseteq y$, so that finally,

$$y \in \mathcal{N}_v \implies \pi(y) \in \mathcal{N}_u$$

for this v , which establishes the continuity of $f = \pi \upharpoonright \mathcal{N}$ at x .

10.26. Call (temporarily) a collection of pointsets *good* if it is one of the collections whose intersection defines $\mathcal{B}(X)$, i.e.,

$$\begin{aligned} \mathcal{E} \text{ is good} &\iff \mathcal{G} \subseteq \mathcal{E} \\ &\& (\forall \{A_n\}_n)[(\forall n) A_n \in \mathcal{E} \implies \bigcup_n A_n \in \mathcal{E}] \\ &\& (\forall A \in \mathcal{E})[cA \in \mathcal{E}]. \end{aligned}$$

To verify that $\mathcal{B}(X) = \bigcap \{\mathcal{E} \mid \mathcal{E} \text{ is good}\}$, is a σ -field which contains all the open sets (i.e., good), we must check the following three claims:

(1) *Every open set $G \subseteq X$ is in $\mathcal{B}(X)$* ; this is because $G \in \mathcal{E}$ for every good $\mathcal{E}$, and so $G \in \mathcal{B}(X)$.

(2) *If $A \in \mathcal{B}(X)$, then $cA \in \mathcal{B}(X)$* : if $A \in \mathcal{E}$ for every good $\mathcal{E}$, then $cA \in \mathcal{E}$ for every good $\mathcal{E}$, by the definition of “goodness”, and so $cA \in \mathcal{B}(X)$.

(3) If $A_n \in \mathcal{B}(X)$ for every n , then $\bigcup_n A_n \in \mathcal{B}(X)$. Fix some good collection $\mathcal{E}$. The hypothesis tells us that each $A_n \in \mathcal{E}$, and so by the definition of “goodness”, $\bigcup_n A_n \in \mathcal{E}$; but $\mathcal{E}$ was arbitrary, and so $\bigcup_n A_n$ belongs to every good collection, which means that $\bigcup_n A_n \in \mathcal{B}(X)$.

10.27. By Problem x1.3, $\bigcap_n A_n = c(\bigcup_n cA_n)$.

11.4. By the Replacement Axiom, the image $F[X]$ of any set X is a set, so there exists a w satisfying the condition

$$(\forall y)[y \in w \iff (\exists x \in X)[y = F(x)]];$$

and by the Extensionality Axiom, only one such w can exist, so we can set

$$G(X) = \text{the unique } w \text{ such that } (\forall y)[y \in w \iff (\exists x \in X)[y = F(x)]].$$

11.7.

$$H(w, n) = \begin{cases} a, & \text{if } n = 0, \\ y, & \text{if } n > 0 \text{ \& } (n-1, y) \in w \text{ \& } (\forall t \neq y)[(n-1, t) \notin w], \\ \emptyset, & \text{otherwise.} \end{cases}$$

11.9. In these cases, it is easiest to verify $\bigcup M \subseteq M$ by inspection, after computing the relevant unionsets:

$$\begin{aligned} \bigcup \emptyset &= \emptyset; \\ \bigcup \{\emptyset, \{\emptyset\}\} &= \{\emptyset\}; \\ \bigcup \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} &= \{\emptyset, \{\emptyset\}\}; \\ \bigcup \mathbb{N}_0 &= \mathbb{N}_0. \end{aligned}$$

Finally, if M is a class of atoms, then $\bigcup M = \emptyset$, so $\bigcup M \subseteq M$.

11.11. By the definition, $A \in \text{TC}(A)$ and $\text{TC}(A)$ is transitive, so that $A \subseteq \text{TC}(A)$ and hence $A \cup \{A\} \subseteq \text{TC}(A)$, for any A . For the other direction, suppose A is transitive and $x \in A \cup \{A\}$. If $x \in A$, then $x \subseteq A \subseteq A \cup \{A\}$, and if $x = A$, then again $x \subseteq A \subseteq A \cup \{A\}$; which shows that $A \cup \{A\}$ is transitive, and hence $A \cup \{A\} \subseteq \text{TC}(A)$.

11.13. If there are no atoms, then the transitive closure $\text{TC}(A)$ of every set has no atoms, and so every set is pure. Conversely, if there exists some atom a , then $\text{TC}(\{a\}) = \{\{a\}, a\}$, and so $\{a\}$ is not pure.

11.14. If A is transitive, then $\text{TC}(A) = A \cup \{A\}$ by Exercise 11.11, and $A \cup \{A\}$ is finite or countable exactly when A is finite or countable.

11.16. False: the singleton $\{a\}$ of an atom is transitive with $\text{TC}(\{a\}) = \{\{a\}, a\}$, but it is not a subset of its powerset $\mathcal{P}(\{a\}) = \{\emptyset, \{a\}\}$, all of whose members are (by definition) sets.

11.17. The inclusion $M_n(I) \subseteq M_n(J)$ is proved by a simple induction on n , with the basis $M_0(I) = I \subseteq J = M_0(J)$ supplied by the hypothesis.

11.20. The class $\mathcal{W}$ of all objects is transitive, because if y is an object and $x \in y$, then x is also an object—simply because we have assumed that membership is a condition on pairs of objects; and it is a Zermelo universe, because we have assumed of it all the conditions we demand of a Zermelo universe—that it contains the unordered pair $\{x, y\}$ of any two objects (in it), and the unionset $\bigcup X$ and powerset $\mathcal{P}(X)$ of any set (in it), and that it also contains a set I which satisfies the Axiom of Infinity, from which $\mathbb{N}_0$ can be constructed using the closure properties of $\mathcal{W}$, as in the proof of Theorem 5.4.

For the second claim, notice first that $\mathbb{N}_0 \subseteq M$, since M is transitive and contains $\mathbb{N}_0$, and so $\emptyset \in M$, since $\emptyset \in \mathbb{N}_0$. Moreover, if $A \in M$, then $\mathcal{P}(A) \in M$ and so $\mathcal{P}(A) \subseteq M$, which means that every subset of A is in M .

11.22. The Peano system constructed in the proof of Theorem 5.4 is a member of every Zermelo universe M , because by Proposition 11.21, M is closed under all the operations we used in that proof to construct it.

11.24. By the Axiom of Choice, the hypothesis of (11-24) implies that there exists some $f : A \rightarrow B$ such that for all $x \in A$, $P(x, f(x))$. But the function space $(A \rightarrow B) \in M$, and so $f \in M$ since M is transitive.

11.27. No descending $\in$ -chain can start with x if x has no members, which is why atoms and $\emptyset$ are grounded. For $\mathbb{N}_0$, we use the fact that it is a Peano system with $0 = \emptyset$ and $Sm = \{m\}$, and we prove by induction that for all $m \in \mathbb{N}_0$, there is no infinite, descending $\in$ -chain which starts with m : this is clear if $m = 0$, since $0 = \emptyset$, and if it is true of m , then it is also true of $Sm = \{m\}$, for which the alleged chain would have to start with

$$\{m\} \ni m \ni \cdots$$

immediately yielding a chain which starts with m .

If A is grounded, then so is each $x \in A$: because if $x \ni x_1 \ni \cdots$ were an infinite, descending $\in$ -chain starting with some $x \in A$, then $A \ni x \ni x_1 \ni \cdots$ would be an infinite, descending $\in$ -chain starting with A ; and conversely, if every member of A is grounded, then we cannot have a chain $A \ni x_1 \ni \cdots$, because the tail $x_1 \ni \cdots$ would witness that x_1 is not grounded.

Similarly, if A is grounded, then so is $\mathcal{P}(A)$: because any chain $\mathcal{P}(A) \ni X \ni x_1 \ni \cdots$ would yield a chain $x_1 \ni \cdots$ starting with $x_1 \in A$. And conversely, if $\mathcal{P}(A)$ is grounded, then so is A , because any chain $\mathcal{P}(A) \ni X \ni x \ni x_1 \ni \cdots$ starting with $\mathcal{P}(A)$ would yield a chain $x \ni x_1 \ni \cdots$ starting with some $x \in A$.

Finally, the class of all grounded sets is transitive, because, again, all elements of a grounded set grounded.

12.2. For any function $f : A \rightarrow B$, the image $f[\emptyset]$ of the empty set is empty; and so with $\nu_U : U \rightarrow \text{ord}(U)$, since 0_U has no predecessors,

$$\nu_U(0_U) = \nu_U[\{y \in U \mid y < 0_U\}] = \nu_U[\emptyset] = \emptyset.$$

By the definition of the successor operation on U ,

$$y <_U S(x) \iff y <_U x \vee y = x,$$

and using this, we can compute:

$$\begin{aligned}\mathfrak{v}_U(S(x)) &= \{\mathfrak{v}_U(y) \mid y <_U S(x)\} \\ &= \{\mathfrak{v}_U(y) \mid y <_U x\} \cup \{\mathfrak{v}_U(x)\} \\ &= \mathfrak{v}_U(x) \cup \{\mathfrak{v}_U(x)\}.\end{aligned}$$

12.3. If 0_U is least in U , then $0_U <_U x$, and so $\emptyset = \mathfrak{v}_U(0_U) \in \mathfrak{v}_U(x)$. If $\alpha \in \mathfrak{v}_U(x)$, then $\alpha = \mathfrak{v}_U(y)$, for some $y <_U x$; but then y has a successor $S(y) <_U x$, since x is a limit point, $\mathfrak{v}_U(S(y)) \in \mathfrak{v}_U(x)$, by the definition, and using the preceding Exercise **12.2**,

$$\mathfrak{v}_U(S(y)) = \mathfrak{v}_U(y) \cup \{\mathfrak{v}_U(y)\} = \alpha \cup \{\alpha\} \in \mathfrak{v}_U(x).$$

12.4. Let

$$G = \{X \mid \emptyset \in X \ \& \ (\forall \alpha \in X)[\alpha \cup \{\alpha\} \in X]\}.$$

Since ω_U is a limit point, $\mathfrak{v}_U(\omega_U) \in G$ by Exercise **12.3**—and, in particular, G is a non-empty class. For the converse, suppose towards a contradiction that there is some $X \in G$ such that $\mathfrak{v}_U(\omega_U) \not\subseteq X$, and let y be least in U such that $\mathfrak{v}_U(y) \notin X$. Now y is not the least element of U , since $\mathfrak{v}_U(0_U) = \emptyset \in X$, by the hypothesis on X . And y is not a limit point of U , since $y < \omega_U$; hence $y = S(x)$ for some $x \in U$, $\mathfrak{v}_U(x) \in X$, by the choice of y , and $\mathfrak{v}_U(y) = \mathfrak{v}_U(x) \cup \{\mathfrak{v}_U(x)\} \in X$ by the hypothesis on X , which contradicts the assumption on y .

12.6. If $\pi : U \rightarrow V$ is an initial similarity, then, directly from Lemma **12.5**,

$$\text{ord}(U) = \{\mathfrak{v}_U(x) \mid x \in U\} = \{\mathfrak{v}_V(\pi(x)) \mid x \in U\} \subseteq \text{ord}(V).$$

12.8. If $U = \text{Succ}((\mathbb{N}, \leq_{\mathbb{N}}))$ is the next well ordered set to $(\mathbb{N}, \leq_{\mathbb{N}})$ with t added on top, then $t = \omega_U$, $\mathbb{N} = \text{seg}_U(t)$ and $\text{ord}(\mathbb{N}, \leq_{\mathbb{N}}) = \mathfrak{v}_U(t) = \omega$. The rest follows immediately from Exercise **12.2**.

12.10. More precisely, the claim is that $\text{ord}(\alpha, \leq_{\alpha}) = \alpha$, and the proof is as follows: if $\alpha = \text{ord}(U)$, then $(\alpha, \leq_{\alpha}) =_o U$ by Lemma **12.9**, and so by Exercise **12.6**

$$\text{ord}(\alpha) = \text{ord}(U) = \alpha.$$

12.13. If $U =_o V$, then

$$\text{ord}(U) =_o U =_o V =_o \text{ord}(V)$$

by (12-12), and so $\text{ord}(\text{ord}(U)) =_o \text{ord}(\text{ord}(V))$ by (12-12) again, from which $\text{ord}(U) = \text{ord}(V)$ follows by Exercise **12.10**.

For the second claim: if $U =_o \alpha$ and $U =_o \beta$, then $\alpha =_o \beta$, and so, in the same way,

$$\alpha = \text{ord}(\alpha) = \text{ord}(\beta) = \beta.$$

12.16. If β is the least member of $\mathcal{E}$, then $\beta \leq \alpha$, for every $\alpha \in \mathcal{E}$, so that by Lemma 12.9, $\beta \subseteq \alpha$; thus $\beta \subseteq \bigcap \mathcal{E}$. On the other hand, if $\gamma \in \beta$, then $\gamma < \beta$, so $\gamma < \alpha$ for every $\alpha \in \mathcal{E}$, i.e., $\gamma \in \alpha$ for every $\alpha \in \mathcal{E}$, hence $\gamma \in \bigcap \mathcal{E}$; thus $\gamma \in \beta \implies \gamma \in \mathcal{E}$, i.e., $\beta \subseteq \bigcap \mathcal{E}$.

12.20. The successor poset $\text{Succ}(\alpha)$ is defined by adding a new element $r \notin \alpha$ to the field α and placing it above all the members of α . Define $\pi : \alpha + 1 = \alpha \cup \{\alpha\} \rightarrow \text{Succ}(\alpha)$ by

$$\pi(x) = \begin{cases} x, & \text{if } x \in \alpha, \\ r, & \text{otherwise (i.e., if } x = \alpha). \end{cases}$$

It follows immediately that π is a similarity, so that $\alpha + 1 =_0 \text{Succ}(\alpha)$, and hence $\alpha + 1 = \text{ord}(\alpha + 1) = \text{ord}(\text{Succ}(\alpha))$.

12.21. This is an elaboration of the preceding Exercise 12.20. By the definition of the sum of two posets in 7.37, we have

$$\alpha +_o \beta = (\{0\} \times \alpha \cup \{1\} \times \beta, \leq),$$

where the ordering $\leq$ is defined by

$$(i, x) \leq (j, y) \iff i < j \vee [i = j \ \& \ [x = y \vee x \in y]],$$

using the properties of the ordering on the ordinals. Fix α , and for each β , define the function

$$\pi_\beta : \alpha +_o \beta \rightarrow \text{ON}$$

by

$$\pi(0, x) = x, \quad \pi(1, y) = \alpha + y;$$

we show by ordinal induction on β that π_β is a similarity of $\alpha +_o \beta$ with $\alpha + \beta$, so that

$$\text{ord}(\alpha +_o \beta) = \text{ord}(\alpha + \beta) = \alpha + \beta.$$

At the basis $\beta = 0$, $\alpha +_o 0 = (\{0\} \times \alpha, \leq)$, so that $\pi_0(0, x) = x$ and this is obviously a similarity of $\alpha +_o 0$ with $\alpha + 0 = \alpha$.

At the induction step, if $\beta = \gamma + 1$, then the required result is exactly the preceding Exercise 12.20.

Finally, at the induction step when β is a limit ordinal, the induction hypothesis gives us that for each $\gamma < \beta$, the function π_γ is a similarity of $\alpha +_o \gamma$ with $\alpha + \gamma$. It is obvious from the definition of these functions that

$$\gamma < \delta < \beta \implies \pi_\gamma \subseteq \pi_\delta;$$

and from this we get easily that the union

$$\pi_\beta = \bigcup_{\gamma < \beta} \pi_\gamma$$

is a similarity of $\alpha +_o \beta$ with $\alpha + \beta$, as required.

The results about associativity and (not)-commutativity follow now from the problems in Chapter 7, especially Problem x7.4.

Finally, (12-26) also follows from the main part of the exercise, because if $\pi : \beta \rightarrow \gamma$ is an initial similarity of β onto a proper initial segment of γ , then the function

$$\rho((i, x)) = \begin{cases} (i, x), & \text{if } i = 0, \\ (i, \pi(x)), & \text{if } i = 1 \end{cases}$$

is easily an initial similarity of $\alpha +_o \beta$ onto a proper, initial segment of $\alpha +_o \gamma$.

12.22. By the definition,

$$\alpha \cdot_o \beta = (\alpha \times \beta, \leq),$$

where $\leq$ is the “inverse lexicographic” ordering on the pairs,

$$(x_1, y_1) \leq (x_2, y_2) \iff y_1 < y_2 \vee [y_1 = y_2 \ \& \ x_1 \leq x_2].$$

To illustrate a different method than the one we used in the preceding exercise, we prove for each fixed α by ordinal induction on β , that

$$\alpha \cdot_o \beta =_o \alpha \cdot \beta,$$

which gives the required result by taking the ordinals of both sides.

At the basis $\beta = 0$, we have that $\alpha \cdot_o 0$ is the empty poset, $\alpha \cdot 0 = 0 = \emptyset$, so we have literal equality.

At the successor induction step, $\beta = \gamma + 1 = \gamma \cup \{\gamma\}$, so

$$\alpha \times \beta = (\alpha \times \gamma) \cup (\alpha \times \{\gamma\}).$$

Notice that

$$(\alpha \times \gamma) \cap (\alpha \times \{\gamma\}) = \emptyset,$$

because the pairs (x, γ) in the second part have second member γ , while every pair (x, y) in the first part has second member some $y < \gamma$. Moreover,

$$(\{(x, \gamma) \mid x \in \alpha\}, \leq) =_o \alpha$$

by the trivial similarity $\rho(x, \gamma) = x$, and so, by the definition of poset addition,

$$\alpha \cdot_o \beta =_o \alpha \cdot_o \gamma +_o \alpha;$$

but then the induction hypothesis, the preceding Exercise **12.21** and the definition of ordinal multiplication imply together that

$$\alpha \cdot_o \beta =_o \alpha \cdot \gamma + \alpha = \alpha \cdot \beta$$

which is what we needed to show.

For the limit case in the induction step, the induction hypothesis gives us for each $\gamma < \beta$ a similarity

$$\pi_\gamma : \alpha \cdot_o \gamma \rightarrow \alpha \cdot \gamma,$$

which is an initial similarity of $\alpha \cdot_o \gamma$ into $\alpha \cdot \beta$ because $\alpha \cdot \gamma \leq \alpha \cdot \beta$, so that $\alpha \cdot \gamma$ is an initial segment of $\alpha \cdot \beta$. We can now appeal to that fact that if U, V

are well ordered sets and $U \leq_o V$, then there is exactly one initial similarity $\pi : U \rightarrow V$ by Problem **x7.13**. In the present case, this means that

$$\delta < \gamma < \beta \implies \pi_\beta \upharpoonright \gamma = \pi_\gamma,$$

i.e., these initial similarities “cohere”, so that their union is an initial similarity of $\alpha \cdot_o \beta$ into $\alpha \cdot \beta$,

$$\cup \{\pi_\gamma \mid \gamma < \beta\} = \pi : \alpha \cdot_o \beta \rightarrow \alpha \cdot \beta.$$

But this π is onto $\alpha \cdot \beta$ since

$$\alpha \cdot \beta = \bigcup_{\gamma < \beta} \alpha \cdot \gamma$$

by the definition of ordinal multiplication, and so π is a similarity and we have the desired $\alpha \cdot_o \beta =_o \alpha \cdot \beta$.

The results about associativity and (not)-commutativity follow now from Problems **x7.9** and **x7.8**.

Finally, if $\pi : \beta \rightarrow \gamma$ is a proper initial similarity, then so is (easily) the function $\rho : \alpha \cdot_o \beta \rightarrow \alpha \cdot_o \gamma$ defined by

$$\rho(x, y) = (x, \pi(y)),$$

and it establishes the required $\alpha \cdot_o \beta <_o \alpha \cdot_o \gamma$.

Note. Another way to do the main part of this exercise is to show directly that for all α, β , the map

$$\pi(x, y) = x \cdot y \quad ((x, y) \in \alpha \times \beta)$$

is a similarity of $\alpha \cdot_o \beta$ with $\alpha \cdot \beta$. The proof (by ordinal recursion on β) is very similar to the argument we just gave, but it involves a somewhat more detailed “chasing” of similarities.

12.23. We prove the claimed identity by ordinal induction on γ , simultaneously for all α and β : i.e., we prove by induction on γ that

$$(\forall \alpha, \beta)[\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma].$$

The basis $\gamma = 0$ is trivial.

For $\gamma = \delta + 1$, compute:

$$\begin{aligned} \alpha \cdot (\beta + (\delta + 1)) &= \alpha \cdot ((\beta + \delta) + 1) \\ &= \alpha \cdot (\beta + \delta) + \alpha \quad (\text{def. of } \cdot) \\ &= \alpha \cdot \beta + \alpha \cdot \delta + \alpha \quad (\text{ind. hyp.}) \\ &= \alpha \cdot \beta + \alpha \cdot (\delta + 1). \end{aligned}$$

Finally, for limit γ , we observe first that

$$\bigcup_{\eta < \gamma} \{\beta + \eta\} \text{ is a limit ordinal,}$$

because for each $\eta < \gamma$ there is a ζ such that $\eta < \zeta < \gamma$, and then $\beta + \eta < \beta + \zeta$ by (12-26). Using this to distinguish cases in the definition of multiplication,

we compute:

$$\begin{aligned}
 \alpha \cdot (\beta + \gamma) &= \alpha \cdot (\bigcup_{\eta < \gamma} \{\beta + \eta\}) \\
 &= \bigcup_{\eta < \gamma} \{\alpha \cdot (\beta + \eta)\} \\
 &= \bigcup_{\eta < \gamma} \{\alpha \cdot \beta + \alpha \cdot \eta\} \quad (\text{ind. hyp.}) \\
 &= \alpha \cdot \beta + \alpha \cdot \gamma \quad (\text{def. of } +),
 \end{aligned}$$

where for the last inference we have used (12-27) to infer that if $\eta < \zeta < \gamma$, then $\alpha \cdot \eta < \alpha \cdot \zeta < \alpha \cdot \gamma$, so that $\alpha \cdot \gamma$ is a limit ordinal and $\alpha \cdot \gamma = \bigcup_{\eta < \gamma} \alpha \cdot \eta$.

12.26. Suppose first $\kappa = (\mu\xi \in \text{ON})[\xi =_c A]$, and assume towards a contradiction that $\kappa =_c \alpha$ for some $\alpha < \kappa$; but then $A =_c \alpha < \kappa$, contradicting the definition of κ . For the converse (and the second claim), if κ is not equinumerous with any $\alpha < \kappa$, then, $\kappa = |\kappa|$, since $\kappa =_c \kappa$, and so $\kappa \in \text{Card}_v$.

12.27. If $A =_c B$, then, for all $\xi \in \text{ON}$,

$$A =_c \xi \iff B =_c \xi,$$

and hence $(\mu\xi \in \text{ON})[A =_c \xi] = (\mu\xi \in \text{ON})[B =_c \xi]$. The third property of strong cardinal assignments is trivial when we assume the Axiom of Replacement: because $\{|X| \mid X \in \mathcal{E}\}$ is the image of $\mathcal{E}$ under the definite operation $X \mapsto |X|$, and so it is a set.

12.30. If **AC** holds, then every set is equinumerous with an ordinal λ , and then $|A|$ is a von Neumann cardinal, by definition. Conversely, if $|A| = \aleph_\alpha$, then $A =_c \aleph_\alpha$, and so A is well orderable.

12.31. Granting **AC**, the Generalized Continuum Hypothesis is the claim that for every infinite cardinal number κ ,

$$|\mathcal{P}(\kappa)| = 2^\kappa = \kappa^+;$$

which becomes exactly the claimed identity since the cardinals are exactly the alephs and $\aleph_\alpha^+ = \aleph_{\alpha+1}$.

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